

FINAL TRIM AND STABILITY BOOKLET OF ARCADIA SUMERA (YARD NO. GMS-005)

AS AMENDED ON PAGES 2, 25 - 42, 58 - 61

SEE LETTER : E-115640-155303



VALID FOR LOADING CONDITIONS APPROVED IN THE STABILITY BOOKLET

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CONTENTS

General

Page No

1.General Particulars	03
2.Stability Criteria	04
3.Notes to the Master	08
4.Draft & Trim Calculation	09
5.Worked Out Example	12
6.Definitions, Symbols and Sign Conventions	20
7.Metric Conversion Table	21

Ship's Data

1. Angle of Down flooding	22
2. Capacity Plan and Table	23
# 3. Sounding Tables	25
4. Table of Hydrostatic Particulars	43
5. Definition of KN	49
6. Volume used For KN Computation	50
7. Displacement Vs KN tables	51
8. Deadweight Table	57
# 9. Table of Maximum Permissible KG_{SHIP}	58
10. Table of Freeboards	62
# NOT PART OF THIS APPROVAL	

Summary of Conditions

64

Loading Conditions

1. Cond#0- Lightship	66
2. Cond#1- Lightship + Crane + Ballast	70
FOR 150T CRAWLER CRANE	
3. Cond#2- Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Aft	75
4. Cond#3- Lightship + Load at Stbd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Port	86
5. Cond#4- Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Fwd	97
6. Cond#5- Lightship + Crane + Ballast + Deck Cargo	108
7. Cond#6- Lightship + Max. Deck Cargo	113
FOR 250T CRAWLER CRANE	
8. Cond#7- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @18m Radius Aft	117
9. Cond#8- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @18m Radius Stbd	128
10. Cond#9- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @18m Radius Fwd	139

Appendix

I. Lightship Particulars	150
II. Memorandum of Freeboards	151
III. Draft Survey Report Approval Letter	152

GENERAL PARTICULARS

Name of the Vessel	ARCADIA SUMERA
Client	ARVIND & CO-SHIPPING AGENCIES PVT. LTD
Yard No.	GMS-005
Type of Vessel	PONT00N
Class Notation	HSU-, CRANE PONT00N, INDIAN COASTAL SERVICE DESCRIPTIVE NOTE "FOR CRANE OPERATION"

Main Particulars

LOA	55.000 M
LBP	55.000 M
Breadth (Mld.)	16.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1855.34 T
Lightship	423.35 T
Deadweight	1431.99 T
Gross Tonnage	670
Net Tonnage	201

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15^\circ$
- $GZC \leq 0,6 GZ_{MAX}$
- $A1 \geq 0.4 ATOT$

where:

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC , GZ_{MAX} : Defined in Fig 1

$A1$: Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle equal to the lesser of:

- heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θ_F , corresponding to flooding of unprotected openings

$ATOT$: Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = (Pd - Zz) / \Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z : Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM . The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

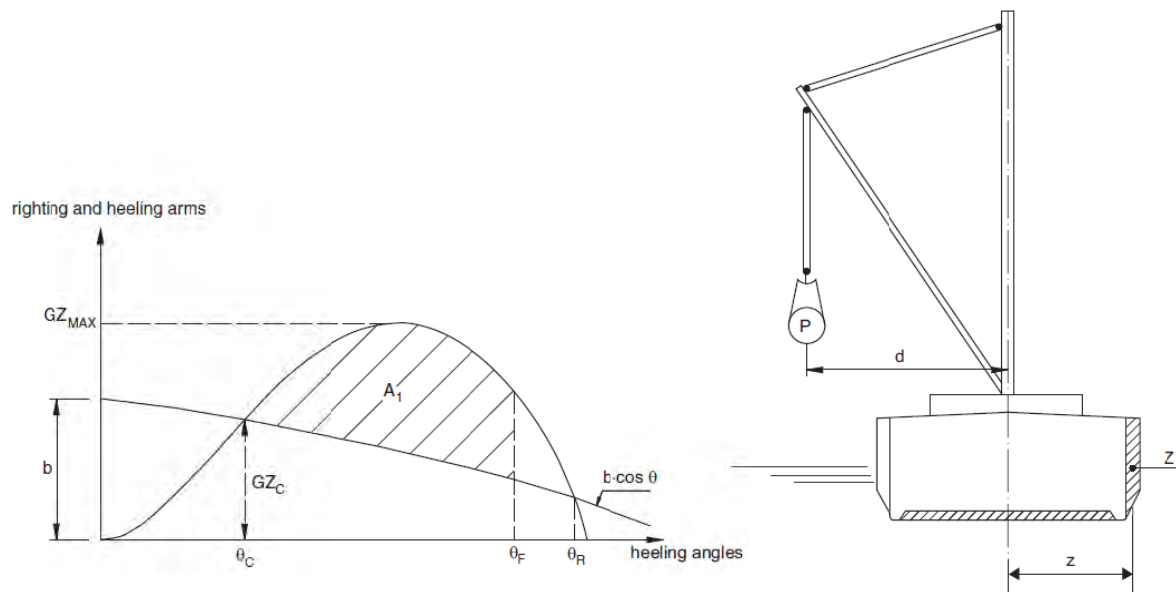


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A_2 / A_1 \geq 1$
- $\theta_2 - \theta_3 \geq 20^\circ$

where:

A_1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_1 to the heeling angle θ_C (see Fig 2)

A_2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_2 (see Fig 2)

A_3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_3 (see Fig 2)

θ_1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ_2 : Heeling angle corresponding to the lesser of θ_R and θ_F

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ_3 : Maximum heeling angle due to roll, at which $A_3 = A_1$, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θ_R : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θ_F : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$$b = Zz \cos\theta/\Delta$$

HEELING AND RIGHTING ARMS

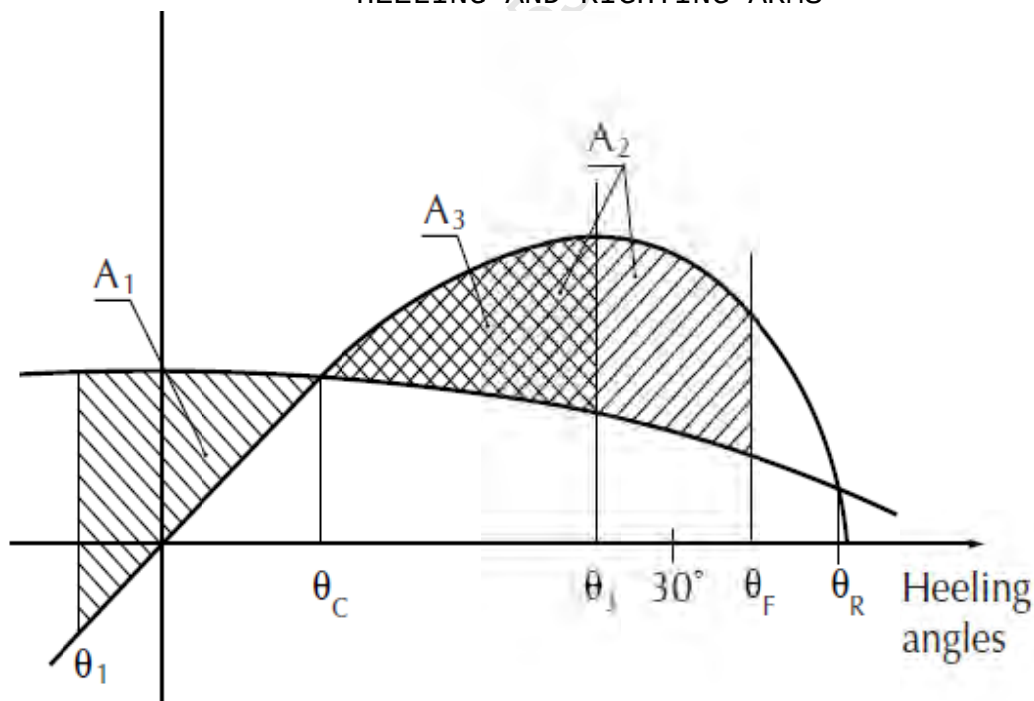


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

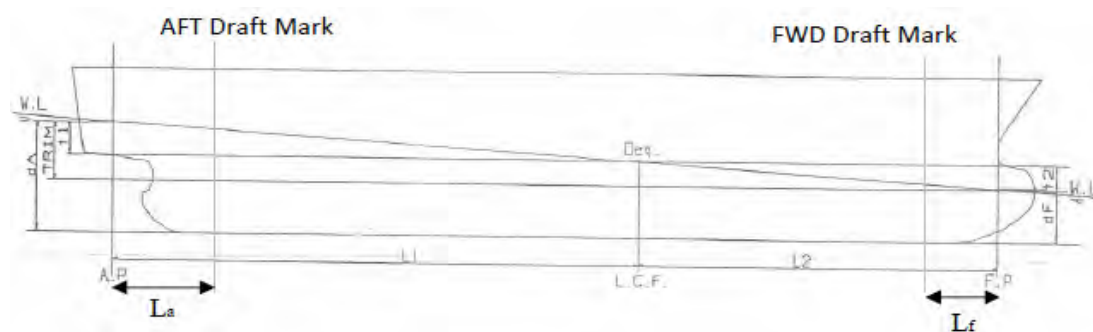
-0.50m, -0.25m, 0.00m, 0.25m, 0.50m

These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$\text{LCG} = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
- c) Displacement, Δ
- d) Moment to change trim, MCT1cm
- e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (\text{LCB} - \text{LCG})}{\text{MCT1cm} * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{\text{LCF} * \text{Trim}}{\text{LBP.}} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

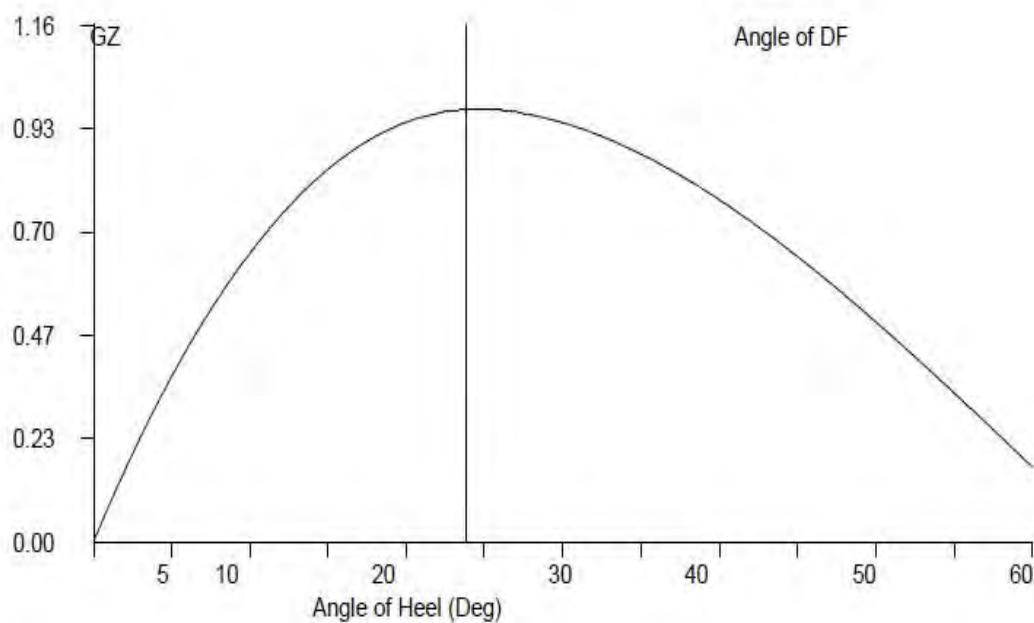
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples**1 Loading the Vessel**

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

**Condition 5: Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m**

Displacement	1855.24	Tonnes
LCG	27.199	M Fwd of AP
VCG (KG)	3.397	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \frac{\text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}}{12}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo**Towing Condition with Max. draft 2.212m**

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
N01_WBTK.P	59.38	1.025	60.86	2.927	178.14	-5.223	-317.87	1.797	109.37	69.64
N01_WBTK.S	59.38	1.025	60.86	2.927	178.14	5.223	317.87	1.797	109.37	69.64
N05_WBTK.C	132.29	1.025	135.60	45.500	6169.80	0.000	0.00	1.500	203.40	94.17
N06_WBTK.P	59.40	1.025	60.88	52.074	3170.27	-5.223	-317.98	1.797	109.40	69.64
N06_WBTK.S	59.40	1.025	60.88	52.074	3170.27	5.223	317.98	1.797	109.40	69.64
TOTAL TANK			379.08	33.942	12866.60	0.000	0.00	1.691	640.93	372.73

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			423.35	27.500	11,642.13	0.000	0.00	3.000	1,270.05
CRANE			150.00	7.500	1,125.00	0.000	0.00	5.200	780.00
DECK CARGO			902.81	27.500	24,827.28	0.000	0.00	4.000	3,611.24
TOTAL FIXED WEIGHT			1,476.16	25.468	37,594.40	0.000	0.00	3.835	5,661.29

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			379.080	33.942	12866.605	0.000	0.000	1.691	640.934
TOTAL FIXED WEIGHT			1476.160	25.468	37594.400	0.000	0.000	3.835	5661.290
TOTAL WEIGHT			1855.240	27.199	50461.005	0.000	0.000	3.397	6302.224

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Free Surface Correction for VCG = FSM/Displacement

	0.201	m
VCGc	3.598	m

From the MaxVCG table,

Displacement	1855.24	T
Allowable Max VCG	5.653	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1855.240	T
----------------	----------	---

Draft ,Tm @ 27.50f =	2.212	m
LCB =	27.500	m
LCF =	27.499	m
MCT =	41.710	T-m/cm
KMT =	11.525	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.134 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perpendicular,Midship and aft Perpendicular Draft

LBP = 55m	
Draft @ 27.500 (MS),Tmid	2.212 m
Draft @ 55.0f (FP),Tfwd	=Tm - Trim/2 2.145 m
Draft@0(AP),Taft	=Tm + Trim/2 2.279 m

Condition 5 : Lightship + Crane + Ballast + Deck Cargo**From HYDROSTATICS**

Hydrostatic Particulars			Drafts & Trims	
MCT	41.710	T-m	Draft (Mean)	2.212 m
LCB	27.500	m	Draft AP (USK)	2.279 m
			Draft FP (USK)	2.145 m
Metacentric Height				
KMt	11.525	m		
KG	3.397	m		
KG0	3.598	m		
GMt	7.927	m	Trim	0.134 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	40.550	T-m	Draft (Mean)	2.210 m
LCB	27.193	m	Draft AP (USK)	2.281 m
			Draft FP (USK)	2.143 m
Metacentric Height				
KMt	11.524	m		
KG	3.410	m		
KG0	3.611	m		
GMt	7.913	m	Trim	0.138 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo**Intact Stability (Primary Criterion)****Minimum Required**

Displacement	1855.24	T
KG0	3.611	M
GMt	7.913	M

0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.011	1.744	2.348	2.577	2.629	2.571
GZ (M)	-0.011	1.112	1.108	0.767	0.304	-0.198

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.2017 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	46.09 m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	2.56 m-rad	0.00 Deg.

Plot GZ values against the angles of heel as above.

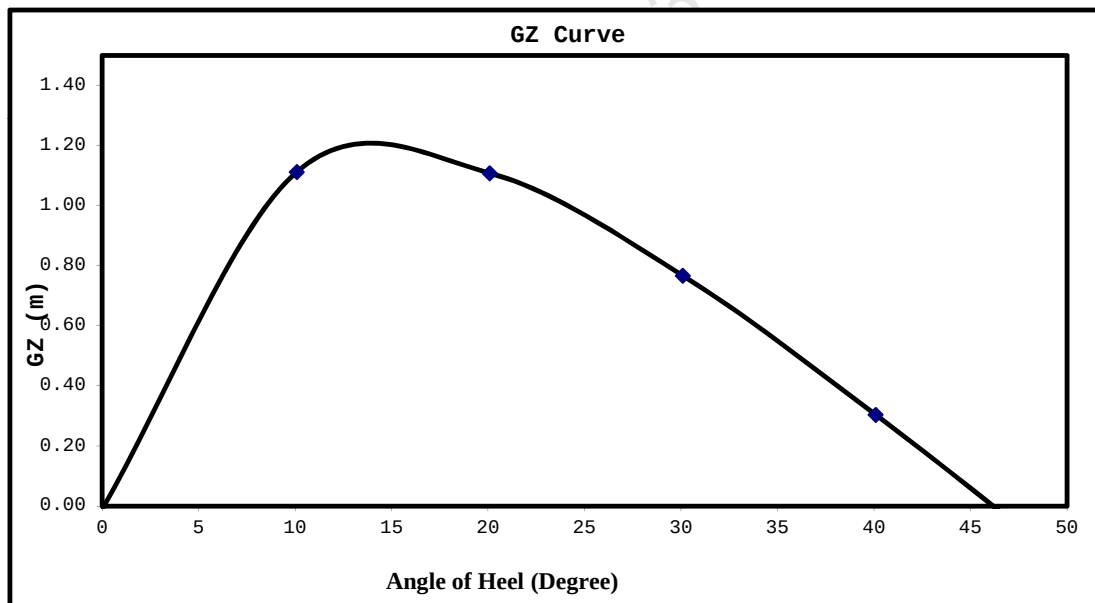
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees

then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	2.212	M
LCF from AP	27.494	M
LCB from AP	27.193	M
MCT	40.560	T-M/Cm
Trim	0.138	M
Draft (AP)	2.281	M
Draft (FP)	2.143	M
KG	3.410	M
KGo	3.598	M
GoMt	7.926	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
N01_WBTK.P		1.025								
N01_WBTK.S		1.025								
N01_WBTK.C		1.025								
N02_WBTK.P		1.025								
N02_WBTK.S		1.025								
N02_WBTK.C		1.025								
N03_WBTK.P		1.025								
N03_WBTK.S		1.025								
N03_WBTK.C		1.025								
N04_WBTK.P		1.025								
N04_WBTK.S		1.025								
N04_WBTK.C		1.025								
N05_WBTK.P		1.025								
N05_WBTK.S		1.025								
N05_WBTK.C		1.025								
N06_WBTK.P		1.025								
N06_WBTK.S		1.025								
N06_WBTK.C		1.025								
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			423.350	27.500	11,642.125	0.000	0.000	3.000	1,270.050
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK									
TOTAL FIXED WEIGHT									
TOTAL WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

	m
VCGc	m

From the MaxVCG table,

Displacement	T
Allowable Max VCG	m

$$\text{VCGc} < \text{Maxvcg}$$

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	T
Draft ,Tm @ 27.50f =	m
LCB =	m
LCF =	m
MCT =	T-m/cm
KMT =	m

The vessels trim is to be determined as follow,

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCTX} \times 100)$$

m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular, Midship and aft Perpendicular Draft

LBP = 55m
Draft @ 27.500 (MS), Tmid m

$$\text{Draft @ 55.0 f (FP), } T_{fwd} = T_m - T_{rim}/2$$

$$\text{Draft}_{00}(\text{AP}), \text{Taft} = T_m + T_{\text{Trim}}/2$$

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.110
Midship	Fr.55
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floataction

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

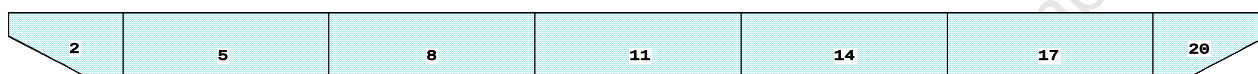
Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

CAPACITY PLAN AND TABLE

Profile View



Plan View

1	4	7	10	13	16	19
3	6	9	12	15	18	21
2	5	8	11	14	17	20

Tanks

1 N01_WBTK.P	4 N02_WBTK.P	8 N03_WBTK.S	12 N01_VOIDTK.C	16 N05_WBTK.P	20 N06_WBTK.S
2 N01_WBTK.S	5 N02_WBTK.S	9 N03_WBTK.C	13 N04_WBTK.P	17 N05_WBTK.S	21 N06_WBTK.C
3 N01_WBTK.C	6 N02_WBTK.C	10 N01_VOIDTK.P	14 N04_WBTK.S	18 N05_WBTK.C	
	7 N03_WBTK.P	11 N01_VOIDTK.S	15 N04_WBTK.C	19 N05_WBTK.P	

10-12-19 15:52:15 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTON

TANK STATUS

Trim: zero, Heel: zero

Part-----	Cu.M.-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	59.4	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	59.4	1.025	60.86	2.927f	5.223s	1.797	69.64*
N01_WBTK.C	54.5	1.025	55.90	2.922f	0.000	1.789	52.32*
N02_WBTK.P	144.4	1.025	148.03	9.500f	5.230p	1.510	125.35*
N02_WBTK.S	144.4	1.025	148.03	9.500f	5.230s	1.510	125.35*
N02_WBTK.C	132.3	1.025	135.60	9.500f	0.000	1.500	94.17*
N03_WBTK.P	144.4	1.025	148.03	18.500f	5.230p	1.510	125.35*
N03_WBTK.S	144.4	1.025	148.03	18.500f	5.230s	1.510	125.35*
N03_WBTK.C	132.3	1.025	135.60	18.500f	0.000	1.500	94.17*
N04_WBTK.P	144.4	1.025	148.03	36.500f	5.230p	1.510	125.35*
N04_WBTK.S	144.4	1.025	148.03	36.500f	5.230s	1.510	125.35*
N04_WBTK.C	132.3	1.025	135.60	36.500f	0.000	1.500	94.17*
N05_WBTK.P	144.4	1.025	148.03	45.500f	5.230p	1.510	125.35*
N05_WBTK.S	144.4	1.025	148.03	45.500f	5.230s	1.510	125.35*
N05_WBTK.C	132.3	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	59.4	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	59.4	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	54.6	1.025	55.92	52.078f	0.000	1.788	52.32*

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N01_WBTK.P, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.010	0.602	0.62	4.390f	5.025p	0.051	1.638	14.768
200	.022	1.332	1.37	4.327f	5.048p	0.105	1.187	18.019
300	.037	2.176	2.23	4.269f	5.070p	0.162	1.041	21.148
400	.053	3.131	3.21	4.213f	5.093p	0.218	1.001	24.521
500	.071	4.198	4.30	4.158f	5.114p	0.278	0.984	27.844
600	.090	5.364	5.50	4.104f	5.133p	0.337	0.978	30.271
700	.112	6.627	6.79	4.051f	5.147p	0.396	0.989	32.832
800	.134	7.986	8.19	3.997f	5.157p	0.457	1.026	35.354
900	.159	9.449	9.69	3.941f	5.165p	0.518	1.072	38.030
1000	.186	11.018	11.29	3.884f	5.171p	0.580	1.124	40.756
1100	.214	12.690	13.01	3.826f	5.177p	0.642	1.175	43.367
1200	.244	14.468	14.83	3.767f	5.182p	0.704	1.230	46.207
1300	.275	16.351	16.76	3.707f	5.186p	0.767	1.285	48.872
1400	.309	18.338	18.80	3.647f	5.189p	0.831	1.340	51.533
1500	.344	20.433	20.94	3.586f	5.193p	0.894	1.399	54.364
1600	.381	22.630	23.20	3.525f	5.195p	0.958	1.456	57.013
1700	.420	24.933	25.56	3.464f	5.198p	1.022	1.515	59.797
1800	.460	27.341	28.02	3.402f	5.200p	1.086	1.573	62.444
1900	.503	29.853	30.60	3.340f	5.202p	1.150	1.632	65.098
2000	.547	32.473	33.29	3.277f	5.204p	1.215	1.693	67.978
2100	.592	35.149	36.03	3.220f	5.206p	1.278	1.755	68.554
2200	.637	37.832	38.78	3.169f	5.209p	1.340	1.472	68.988
2300	.682	40.520	41.53	3.125f	5.211p	1.401	1.379	69.277
2400	.728	43.212	44.29	3.086f	5.213p	1.460	1.296	69.454
2500	.773	45.905	47.05	3.052f	5.215p	1.518	1.223	69.638
2600	.819	48.601	49.82	3.021f	5.217p	1.575	1.155	69.638
2700	.864	51.296	52.58	2.994f	5.219p	1.632	1.095	69.638
2800	.909	53.991	55.34	2.969f	5.221p	1.687	1.040	69.638
2900	.955	56.686	58.10	2.947f	5.222p	1.743	0.990	69.638
3000	1.000	59.376	60.86	2.927f	5.223p	1.797		

Soundings in mm.-----Other distances in METERS.-----

N01_WBTK.P Reference Point: Long.= 2.500f Trans.= 5.250p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

TANK CHARACTERISTICS

No Trim, No Heel

Tank: N01_WBTK.S, Contents: WB at 1.025 Specific Gravity

NOT PART OF THIS APPROVAL

Snding	Load	Volume	Weight	Center of Gravity			GML	FSM m.-MT
		CU. METERS	METRIC TON	LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.010	0.602	0.62	4.390f	5.025s	0.051	1.638	14.768
200	.022	1.332	1.37	4.327f	5.048s	0.105	1.187	18.019
300	.037	2.176	2.23	4.269f	5.070s	0.162	1.041	21.148
400	.053	3.131	3.21	4.213f	5.093s	0.218	1.001	24.521
500	.071	4.198	4.30	4.158f	5.114s	0.278	0.984	27.844
600	.090	5.364	5.50	4.104f	5.133s	0.337	0.978	30.271
700	.112	6.627	6.79	4.051f	5.147s	0.396	0.989	32.832
800	.134	7.986	8.19	3.997f	5.157s	0.457	1.026	35.354
900	.159	9.449	9.69	3.941f	5.165s	0.518	1.072	38.030
1000	.186	11.018	11.29	3.884f	5.171s	0.580	1.124	40.756
1100	.214	12.690	13.01	3.826f	5.177s	0.642	1.175	43.367
1200	.244	14.468	14.83	3.767f	5.182s	0.704	1.230	46.207
1300	.275	16.351	16.76	3.707f	5.186s	0.767	1.285	48.872
1400	.309	18.338	18.80	3.647f	5.189s	0.831	1.340	51.533
1500	.344	20.433	20.94	3.586f	5.193s	0.894	1.399	54.364
1600	.381	22.630	23.20	3.525f	5.195s	0.958	1.456	57.013
1700	.420	24.933	25.56	3.464f	5.198s	1.022	1.515	59.797
1800	.460	27.341	28.02	3.402f	5.200s	1.086	1.573	62.444
1900	.503	29.853	30.60	3.340f	5.202s	1.150	1.632	65.098
2000	.547	32.473	33.29	3.277f	5.204s	1.215	1.693	67.978
2100	.592	35.149	36.03	3.220f	5.206s	1.278	1.575	68.554
2200	.637	37.832	38.78	3.169f	5.209s	1.340	1.472	68.988
2300	.682	40.520	41.53	3.125f	5.211s	1.401	1.379	69.277
2400	.728	43.212	44.29	3.086f	5.213s	1.460	1.296	69.454
2500	.773	45.905	47.05	3.052f	5.215s	1.518	1.223	69.638
2600	.819	48.601	49.82	3.021f	5.217s	1.575	1.155	69.638
2700	.864	51.296	52.58	2.994f	5.219s	1.632	1.095	69.638
2800	.909	53.991	55.34	2.969f	5.221s	1.687	1.040	69.638
2900	.955	56.686	58.10	2.947f	5.222s	1.743	0.990	69.638
3000	1.000	59.376	60.86	2.927f	5.223s	1.797		

Soundings in mm.-----Other distances in METERS.-----

N01_WBTK.S Reference Point: Long.= 2.500f Trans.= 5.250s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N01_WBTK.C, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.011	0.596	0.61	4.390f	0.000	0.051	1.625	13.957
200	.024	1.306	1.34	4.327f	0.000	0.105	1.177	16.277
300	.039	2.116	2.17	4.268f	0.000	0.161	1.030	18.283
400	.055	3.019	3.09	4.211f	0.000	0.216	0.986	20.289
500	.074	4.014	4.11	4.155f	0.000	0.275	0.968	22.174
600	.093	5.095	5.22	4.100f	0.000	0.333	0.967	24.001
700	.115	6.262	6.42	4.046f	0.000	0.392	0.981	25.829
800	.138	7.518	7.71	3.991f	0.000	0.452	1.022	27.832
900	.163	8.869	9.09	3.935f	0.000	0.513	1.071	29.874
1000	.189	10.316	10.57	3.877f	0.000	0.574	1.123	31.915
1100	.217	11.858	12.15	3.818f	0.000	0.636	1.177	33.956
1200	.247	13.496	13.83	3.759f	0.000	0.699	1.232	35.998
1300	.279	15.229	15.61	3.699f	0.000	0.762	1.288	38.037
1400	.313	17.058	17.48	3.638f	0.000	0.825	1.345	40.076
1500	.348	18.983	19.46	3.577f	0.000	0.888	1.403	42.115
1600	.385	21.003	21.53	3.516f	0.000	0.952	1.461	44.155
1700	.424	23.118	23.70	3.454f	0.000	1.016	1.520	46.196
1800	.464	25.330	25.96	3.392f	0.000	1.080	1.579	48.237
1900	.507	27.636	28.33	3.330f	0.000	1.144	1.639	50.279
2000	.551	30.039	30.79	3.267f	0.000	1.208	1.699	52.317
2100	.596	32.489	33.30	3.209f	0.000	1.272	1.571	52.317
2200	.641	34.939	35.81	3.159f	0.000	1.334	1.461	52.317
2300	.686	37.389	38.32	3.116f	0.000	1.394	1.365	52.317
2400	.731	39.839	40.83	3.078f	0.000	1.453	1.281	52.317
2500	.775	42.289	43.35	3.045f	0.000	1.510	1.207	52.317
2600	.820	44.739	45.86	3.015f	0.000	1.567	1.141	52.317
2700	.865	47.189	48.37	2.988f	0.000	1.624	1.082	52.317
2800	.910	49.639	50.88	2.964f	0.000	1.679	1.028	52.317
2900	.955	52.089	53.39	2.942f	0.000	1.734	0.980	52.317
3000	1.000	54.535	55.90	2.922f	0.000	1.789		

Soundings in mm.-----Other distances in METERS.-----

N01_WBTK.C Reference Point: Long.= 2.500f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N02_WBTK.P, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	9.500f	5.025p	0.050	68.169	99.935
200	.062	8.996	9.22	9.500f	5.050p	0.101	34.412	105.933
300	.094	13.627	13.97	9.500f	5.076p	0.151	23.156	112.166
400	.127	18.346	18.80	9.500f	5.101p	0.203	17.524	118.640
500	.160	23.153	23.73	9.500f	5.127p	0.254	14.143	125.349
600	.194	28.004	28.70	9.500f	5.148p	0.305	11.693	125.349
700	.227	32.855	33.68	9.500f	5.163p	0.356	9.966	125.349
800	.261	37.706	38.65	9.500f	5.174p	0.407	8.684	125.349
900	.295	42.557	43.62	9.500f	5.183p	0.457	7.694	125.349
1000	.328	47.408	48.59	9.500f	5.190p	0.508	6.907	125.349
1100	.362	52.259	53.57	9.500f	5.196p	0.558	6.266	125.349
1200	.395	57.110	58.54	9.500f	5.200p	0.608	5.734	125.350
1300	.429	61.962	63.51	9.500f	5.204p	0.659	5.285	125.349
1400	.463	66.813	68.48	9.500f	5.207p	0.709	4.901	125.349
1500	.496	71.664	73.46	9.500f	5.210p	0.759	4.569	125.349
1600	.530	76.515	78.43	9.500f	5.213p	0.809	4.280	125.349
1700	.563	81.366	83.40	9.500f	5.215p	0.859	4.024	125.349
1800	.597	86.217	88.37	9.500f	5.217p	0.909	3.798	125.349
1900	.631	91.068	93.34	9.500f	5.219p	0.959	3.596	125.349
2000	.664	95.919	98.32	9.500f	5.220p	1.010	3.414	125.349
2100	.698	100.770	103.29	9.500f	5.222p	1.060	3.249	125.349
2200	.731	105.621	108.26	9.500f	5.223p	1.110	3.100	125.349
2300	.765	110.472	113.23	9.500f	5.224p	1.160	2.964	125.349
2400	.799	115.323	118.21	9.500f	5.225p	1.210	2.839	125.349
2500	.832	120.175	123.18	9.500f	5.226p	1.260	2.725	125.349
2600	.866	125.026	128.15	9.500f	5.227p	1.310	2.619	125.349
2700	.899	129.877	133.12	9.500f	5.228p	1.360	2.521	125.349
2800	.933	134.728	138.10	9.500f	5.229p	1.410	2.430	125.349
2900	.966	139.579	143.07	9.500f	5.230p	1.460	2.346	125.349
3000	1.000	144.422	148.03	9.500f	5.230p	1.510		

Soundings in mm.-----Other distances in METERS.-----

N02_WBTK.P Reference Point: Long.= 9.500f Trans.= 5.250p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N02_WBTK.S, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	9.500f	5.025s	0.050	68.169	99.935
200	.062	8.996	9.22	9.500f	5.050s	0.101	34.412	105.933
300	.094	13.627	13.97	9.500f	5.076s	0.151	23.156	112.166
400	.127	18.346	18.80	9.500f	5.101s	0.203	17.524	118.640
500	.160	23.153	23.73	9.500f	5.127s	0.254	14.143	125.349
600	.194	28.004	28.70	9.500f	5.148s	0.305	11.693	125.349
700	.227	32.855	33.68	9.500f	5.163s	0.356	9.966	125.349
800	.261	37.706	38.65	9.500f	5.174s	0.407	8.684	125.349
900	.295	42.557	43.62	9.500f	5.183s	0.457	7.694	125.349
1000	.328	47.408	48.59	9.500f	5.190s	0.508	6.907	125.349
1100	.362	52.259	53.57	9.500f	5.196s	0.558	6.266	125.349
1200	.395	57.110	58.54	9.500f	5.200s	0.608	5.734	125.350
1300	.429	61.961	63.51	9.500f	5.204s	0.659	5.285	125.349
1400	.463	66.813	68.48	9.500f	5.207s	0.709	4.901	125.349
1500	.496	71.664	73.46	9.500f	5.210s	0.759	4.569	125.349
1600	.530	76.515	78.43	9.500f	5.213s	0.809	4.280	125.349
1700	.563	81.366	83.40	9.500f	5.215s	0.859	4.024	125.349
1800	.597	86.217	88.37	9.500f	5.217s	0.909	3.798	125.349
1900	.631	91.068	93.34	9.500f	5.219s	0.959	3.596	125.349
2000	.664	95.919	98.32	9.500f	5.220s	1.010	3.414	125.349
2100	.698	100.770	103.29	9.500f	5.222s	1.060	3.249	125.349
2200	.731	105.621	108.26	9.500f	5.223s	1.110	3.100	125.349
2300	.765	110.472	113.23	9.500f	5.224s	1.160	2.964	125.349
2400	.799	115.323	118.21	9.500f	5.225s	1.210	2.839	125.349
2500	.832	120.175	123.18	9.500f	5.226s	1.260	2.725	125.349
2600	.866	125.026	128.15	9.500f	5.227s	1.310	2.619	125.349
2700	.899	129.877	133.12	9.500f	5.228s	1.360	2.521	125.349
2800	.933	134.728	138.10	9.500f	5.229s	1.410	2.430	125.349
2900	.966	139.579	143.07	9.500f	5.230s	1.460	2.346	125.349
3000	1.000	144.422	148.03	9.500f	5.230s	1.510		

Soundings in mm.-----Other distances in METERS.-----

N02_WBTK.S Reference Point: Long.= 9.500f Trans.= 5.250s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N02_WBTK.C, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU.	METERS	METRIC	TON	LCG	TCG	VCG	GML
0	.000	0.000		0.00					
100	.033	4.410		4.52		9.500f	0.000	0.050	67.500
200	.067	8.820		9.04		9.500f	0.000	0.100	33.750
300	.100	13.230		13.56		9.500f	0.000	0.150	22.500
400	.133	17.640		18.08		9.500f	0.000	0.200	16.875
500	.167	22.050		22.60		9.500f	0.000	0.250	13.500
600	.200	26.460		27.12		9.500f	0.000	0.300	11.250
700	.233	30.870		31.64		9.500f	0.000	0.350	9.643
800	.267	35.280		36.16		9.500f	0.000	0.400	8.438
900	.300	39.690		40.68		9.500f	0.000	0.450	7.500
1000	.333	44.100		45.20		9.500f	0.000	0.500	6.750
1100	.367	48.510		49.72		9.500f	0.000	0.550	6.136
1200	.400	52.920		54.24		9.500f	0.000	0.600	5.625
1300	.433	57.330		58.76		9.500f	0.000	0.650	5.192
1400	.467	61.740		63.28		9.500f	0.000	0.700	4.821
1500	.500	66.150		67.80		9.500f	0.000	0.750	4.500
1600	.533	70.560		72.32		9.500f	0.000	0.800	4.219
1700	.567	74.970		76.84		9.500f	0.000	0.850	3.971
1800	.600	79.380		81.36		9.500f	0.000	0.900	3.750
1900	.633	83.790		85.88		9.500f	0.000	0.950	3.553
2000	.667	88.200		90.41		9.500f	0.000	1.000	3.375
2100	.700	92.610		94.93		9.500f	0.000	1.050	3.214
2200	.733	97.020		99.45		9.500f	0.000	1.100	3.068
2300	.767	101.430		103.97		9.500f	0.000	1.150	2.935
2400	.800	105.840		108.49		9.500f	0.000	1.200	2.813
2500	.833	110.250		113.01		9.500f	0.000	1.250	2.700
2600	.867	114.660		117.53		9.500f	0.000	1.300	2.596
2700	.900	119.070		122.05		9.500f	0.000	1.350	2.500
2800	.933	123.480		126.57		9.500f	0.000	1.400	2.411
2900	.967	127.890		131.09		9.500f	0.000	1.450	2.328
3000	1.000	132.293		135.60		9.500f	0.000	1.500	

Soundings in mm.-----Other distances in METERS.-----

N02_WBTK.C Reference Point: Long.= 9.500f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N03_WBTK.P, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	18.500f	5.025p	0.050	68.167	99.934
200	.062	8.996	9.22	18.500f	5.050p	0.101	34.412	105.933
300	.094	13.627	13.97	18.500f	5.076p	0.151	23.156	112.166
400	.127	18.346	18.80	18.500f	5.101p	0.203	17.524	118.640
500	.160	23.153	23.73	18.500f	5.127p	0.254	14.143	125.349
600	.194	28.004	28.70	18.500f	5.148p	0.305	11.693	125.349
700	.227	32.855	33.68	18.500f	5.163p	0.356	9.966	125.349
800	.261	37.706	38.65	18.500f	5.174p	0.407	8.684	125.349
900	.295	42.557	43.62	18.500f	5.183p	0.457	7.694	125.349
1000	.328	47.408	48.59	18.500f	5.190p	0.508	6.907	125.349
1100	.362	52.259	53.57	18.500f	5.196p	0.558	6.266	125.349
1200	.395	57.110	58.54	18.500f	5.200p	0.608	5.734	125.349
1300	.429	61.961	63.51	18.500f	5.204p	0.659	5.285	125.349
1400	.463	66.812	68.48	18.500f	5.207p	0.709	4.901	125.349
1500	.496	71.663	73.46	18.500f	5.210p	0.759	4.569	125.349
1600	.530	76.515	78.43	18.500f	5.213p	0.809	4.280	125.349
1700	.563	81.366	83.40	18.500f	5.215p	0.859	4.024	125.349
1800	.597	86.217	88.37	18.500f	5.217p	0.909	3.798	125.349
1900	.631	91.068	93.34	18.500f	5.219p	0.959	3.596	125.349
2000	.664	95.919	98.32	18.500f	5.220p	1.010	3.414	125.349
2100	.698	100.770	103.29	18.500f	5.222p	1.060	3.249	125.349
2200	.731	105.621	108.26	18.500f	5.223p	1.110	3.100	125.349
2300	.765	110.472	113.23	18.500f	5.224p	1.160	2.964	125.349
2400	.799	115.323	118.21	18.500f	5.225p	1.210	2.839	125.349
2500	.832	120.174	123.18	18.500f	5.226p	1.260	2.725	125.349
2600	.866	125.025	128.15	18.500f	5.227p	1.310	2.619	125.349
2700	.899	129.876	133.12	18.500f	5.228p	1.360	2.521	125.349
2800	.933	134.727	138.10	18.500f	5.229p	1.410	2.430	125.349
2900	.966	139.578	143.07	18.500f	5.230p	1.460	2.346	125.349
3000	1.000	144.422	148.03	18.500f	5.230p	1.510		

Soundings in mm.-----Other distances in METERS.-----

N03_WBTK.P Reference Point: Long.= 18.500f Trans.= 5.250p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.

GHS 16.68C

PONTOON

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: N03_WBTK.S, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	18.500f	5.025s	0.050	68.167	99.934
200	.062	8.996	9.22	18.500f	5.050s	0.101	34.412	105.933
300	.094	13.627	13.97	18.500f	5.076s	0.151	23.156	112.166
400	.127	18.346	18.80	18.500f	5.101s	0.203	17.524	118.640
500	.160	23.153	23.73	18.500f	5.127s	0.254	14.143	125.349
600	.194	28.004	28.70	18.500f	5.148s	0.305	11.693	125.349
700	.227	32.855	33.68	18.500f	5.163s	0.356	9.966	125.349
800	.261	37.706	38.65	18.500f	5.174s	0.407	8.684	125.349
900	.295	42.557	43.62	18.500f	5.183s	0.457	7.694	125.349
1000	.328	47.408	48.59	18.500f	5.190s	0.508	6.907	125.349
1100	.362	52.259	53.57	18.500f	5.196s	0.558	6.266	125.349
1200	.395	57.110	58.54	18.500f	5.200s	0.608	5.734	125.349
1300	.429	61.961	63.51	18.500f	5.204s	0.659	5.285	125.349
1400	.463	66.812	68.48	18.500f	5.207s	0.709	4.901	125.349
1500	.496	71.663	73.46	18.500f	5.210s	0.759	4.569	125.349
1600	.530	76.515	78.43	18.500f	5.213s	0.809	4.280	125.349
1700	.563	81.366	83.40	18.500f	5.215s	0.859	4.024	125.349
1800	.597	86.217	88.37	18.500f	5.217s	0.909	3.798	125.349
1900	.631	91.068	93.34	18.500f	5.219s	0.959	3.596	125.349
2000	.664	95.919	98.32	18.500f	5.220s	1.010	3.414	125.349
2100	.698	100.770	103.29	18.500f	5.222s	1.060	3.249	125.349
2200	.731	105.621	108.26	18.500f	5.223s	1.110	3.100	125.349
2300	.765	110.472	113.23	18.500f	5.224s	1.160	2.964	125.349
2400	.799	115.323	118.21	18.500f	5.225s	1.210	2.839	125.349
2500	.832	120.174	123.18	18.500f	5.226s	1.260	2.725	125.349
2600	.866	125.025	128.15	18.500f	5.227s	1.310	2.619	125.349
2700	.899	129.876	133.12	18.500f	5.228s	1.360	2.521	125.349
2800	.933	134.727	138.10	18.500f	5.229s	1.410	2.430	125.349
2900	.966	139.578	143.07	18.500f	5.230s	1.460	2.346	125.349
3000	1.000	144.422	148.03	18.500f	5.230s	1.510		

Soundings in mm.-----Other distances in METERS.-----

N03_WBTK.S Reference Point: Long.= 18.500f Trans.= 5.250s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N03_WBTK.C, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU.	METERS	METRIC	TON	LCG	TCG	VCG	GML
0	.000	0.000		0.00					
100	.033	4.410		4.52		18.500f	0.000	0.050	67.500
200	.067	8.820		9.04		18.500f	0.000	0.100	33.750
300	.100	13.230		13.56		18.500f	0.000	0.150	22.500
400	.133	17.640		18.08		18.500f	0.000	0.200	16.875
500	.167	22.050		22.60		18.500f	0.000	0.250	13.500
600	.200	26.460		27.12		18.500f	0.000	0.300	11.250
700	.233	30.870		31.64		18.500f	0.000	0.350	9.643
800	.267	35.280		36.16		18.500f	0.000	0.400	8.438
900	.300	39.690		40.68		18.500f	0.000	0.450	7.500
1000	.333	44.100		45.20		18.500f	0.000	0.500	6.750
1100	.367	48.510		49.72		18.500f	0.000	0.550	6.136
1200	.400	52.920		54.24		18.500f	0.000	0.600	5.625
1300	.433	57.330		58.76		18.500f	0.000	0.650	5.192
1400	.467	61.740		63.28		18.500f	0.000	0.700	4.821
1500	.500	66.150		67.80		18.500f	0.000	0.750	4.500
1600	.533	70.560		72.32		18.500f	0.000	0.800	4.219
1700	.567	74.970		76.84		18.500f	0.000	0.850	3.971
1800	.600	79.380		81.36		18.500f	0.000	0.900	3.750
1900	.633	83.790		85.88		18.500f	0.000	0.950	3.553
2000	.667	88.200		90.40		18.500f	0.000	1.000	3.375
2100	.700	92.610		94.92		18.500f	0.000	1.050	3.214
2200	.733	97.020		99.45		18.500f	0.000	1.100	3.068
2300	.767	101.430		103.97		18.500f	0.000	1.150	2.935
2400	.800	105.840		108.49		18.500f	0.000	1.200	2.813
2500	.833	110.250		113.01		18.500f	0.000	1.250	2.700
2600	.867	114.660		117.53		18.500f	0.000	1.300	2.596
2700	.900	119.070		122.05		18.500f	0.000	1.350	2.500
2800	.933	123.480		126.57		18.500f	0.000	1.400	2.411
2900	.967	127.889		131.09		18.500f	0.000	1.450	2.328
3000	1.000	132.292		135.60		18.500f	0.000	1.500	

Soundings in mm.-----Other distances in METERS.-----

N03_WBTK.C Reference Point: Long.= 18.500f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N04_WBTK.P, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	36.500f	5.025p	0.050	68.166	99.934
200	.062	8.996	9.22	36.500f	5.050p	0.101	34.413	105.933
300	.094	13.627	13.97	36.500f	5.076p	0.151	23.156	112.166
400	.127	18.346	18.80	36.500f	5.101p	0.203	17.526	118.640
500	.160	23.153	23.73	36.500f	5.127p	0.254	14.143	125.349
600	.194	28.004	28.70	36.500f	5.148p	0.305	11.693	125.349
700	.227	32.855	33.68	36.500f	5.163p	0.356	9.966	125.349
800	.261	37.706	38.65	36.500f	5.174p	0.407	8.684	125.349
900	.295	42.557	43.62	36.500f	5.183p	0.457	7.694	125.349
1000	.328	47.408	48.59	36.500f	5.190p	0.508	6.907	125.349
1100	.362	52.259	53.57	36.500f	5.196p	0.558	6.266	125.349
1200	.395	57.110	58.54	36.500f	5.200p	0.608	5.734	125.349
1300	.429	61.961	63.51	36.500f	5.204p	0.659	5.285	125.349
1400	.463	66.812	68.48	36.500f	5.207p	0.709	4.901	125.349
1500	.496	71.663	73.46	36.500f	5.210p	0.759	4.569	125.349
1600	.530	76.515	78.43	36.500f	5.213p	0.809	4.280	125.349
1700	.563	81.366	83.40	36.500f	5.215p	0.859	4.024	125.349
1800	.597	86.217	88.37	36.500f	5.217p	0.909	3.798	125.349
1900	.631	91.068	93.34	36.500f	5.219p	0.959	3.596	125.349
2000	.664	95.919	98.32	36.500f	5.220p	1.010	3.414	125.349
2100	.698	100.770	103.29	36.500f	5.222p	1.060	3.249	125.349
2200	.731	105.621	108.26	36.500f	5.223p	1.110	3.100	125.349
2300	.765	110.472	113.23	36.500f	5.224p	1.160	2.964	125.349
2400	.799	115.323	118.21	36.500f	5.225p	1.210	2.839	125.349
2500	.832	120.174	123.18	36.500f	5.226p	1.260	2.725	125.349
2600	.866	125.025	128.15	36.500f	5.227p	1.310	2.619	125.349
2700	.899	129.876	133.12	36.500f	5.228p	1.360	2.521	125.349
2800	.933	134.727	138.10	36.500f	5.229p	1.410	2.430	125.349
2900	.966	139.578	143.07	36.500f	5.230p	1.460	2.346	125.349
3000	1.000	144.422	148.03	36.500f	5.230p	1.510		

Soundings in mm.-----Other distances in METERS.-----

N04_WBTK.P Reference Point: Long.= 36.500f Trans.= 5.250p Vert.= 0.000
(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

TANK CHARACTERISTICS

No Trim, No Heel

Tank: N04_WBTK.S, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	36.500f	5.025s	0.050	68.166	99.934
200	.062	8.996	9.22	36.500f	5.050s	0.101	34.413	105.933
300	.094	13.627	13.97	36.500f	5.076s	0.151	23.156	112.166
400	.127	18.346	18.80	36.500f	5.101s	0.203	17.526	118.640
500	.160	23.153	23.73	36.500f	5.127s	0.254	14.143	125.349
600	.194	28.004	28.70	36.500f	5.148s	0.305	11.693	125.349
700	.227	32.855	33.68	36.500f	5.163s	0.356	9.966	125.349
800	.261	37.706	38.65	36.500f	5.174s	0.407	8.684	125.349
900	.295	42.557	43.62	36.500f	5.183s	0.457	7.694	125.349
1000	.328	47.408	48.59	36.500f	5.190s	0.508	6.907	125.349
1100	.362	52.259	53.57	36.500f	5.196s	0.558	6.266	125.349
1200	.395	57.110	58.54	36.500f	5.200s	0.608	5.734	125.349
1300	.429	61.961	63.51	36.500f	5.204s	0.659	5.285	125.349
1400	.463	66.812	68.48	36.500f	5.207s	0.709	4.901	125.349
1500	.496	71.663	73.46	36.500f	5.210s	0.759	4.569	125.349
1600	.530	76.515	78.43	36.500f	5.213s	0.809	4.280	125.349
1700	.563	81.366	83.40	36.500f	5.215s	0.859	4.024	125.349
1800	.597	86.217	88.37	36.500f	5.217s	0.909	3.798	125.349
1900	.631	91.068	93.34	36.500f	5.219s	0.959	3.596	125.349
2000	.664	95.919	98.32	36.500f	5.220s	1.010	3.414	125.349
2100	.698	100.770	103.29	36.500f	5.222s	1.060	3.249	125.349
2200	.731	105.621	108.26	36.500f	5.223s	1.110	3.100	125.349
2300	.765	110.472	113.23	36.500f	5.224s	1.160	2.964	125.349
2400	.799	115.323	118.21	36.500f	5.225s	1.210	2.839	125.349
2500	.832	120.174	123.18	36.500f	5.226s	1.260	2.725	125.349
2600	.866	125.025	128.15	36.500f	5.227s	1.310	2.619	125.349
2700	.899	129.876	133.12	36.500f	5.228s	1.360	2.521	125.349
2800	.933	134.727	138.10	36.500f	5.229s	1.410	2.430	125.349
2900	.966	139.578	143.07	36.500f	5.230s	1.460	2.346	125.349
3000	1.000	144.422	148.03	36.500f	5.230s	1.510		

Soundings in mm.-----Other distances in METERS.-----

N04_WBTK.S Reference Point: Long.= 36.500f Trans.= 5.250s Vert.= 0.000
(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N04_WBTK.C, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU.	METERS	METRIC	TON	LCG	TCG	VCG	GML
0	.000	0.000		0.00					
100	.033	4.410		4.52		36.500f	0.000	0.050	67.500
200	.067	8.820		9.04		36.500f	0.000	0.100	33.750
300	.100	13.230		13.56		36.500f	0.000	0.150	22.500
400	.133	17.640		18.08		36.500f	0.000	0.200	16.875
500	.167	22.050		22.60		36.500f	0.000	0.250	13.500
600	.200	26.460		27.12		36.500f	0.000	0.300	11.250
700	.233	30.870		31.64		36.500f	0.000	0.350	9.643
800	.267	35.280		36.16		36.500f	0.000	0.400	8.438
900	.300	39.690		40.68		36.500f	0.000	0.450	7.500
1000	.333	44.100		45.20		36.500f	0.000	0.500	6.750
1100	.367	48.510		49.72		36.500f	0.000	0.550	6.136
1200	.400	52.920		54.24		36.500f	0.000	0.600	5.625
1300	.433	57.330		58.76		36.500f	0.000	0.650	5.192
1400	.467	61.740		63.28		36.500f	0.000	0.700	4.821
1500	.500	66.150		67.80		36.500f	0.000	0.750	4.500
1600	.533	70.560		72.32		36.500f	0.000	0.800	4.219
1700	.567	74.970		76.84		36.500f	0.000	0.850	3.971
1800	.600	79.380		81.36		36.500f	0.000	0.900	3.750
1900	.633	83.790		85.88		36.500f	0.000	0.950	3.553
2000	.667	88.200		90.40		36.500f	0.000	1.000	3.375
2100	.700	92.610		94.92		36.500f	0.000	1.050	3.214
2200	.733	97.020		99.45		36.500f	0.000	1.100	3.068
2300	.767	101.430		103.97		36.500f	0.000	1.150	2.935
2400	.800	105.840		108.49		36.500f	0.000	1.200	2.813
2500	.833	110.250		113.01		36.500f	0.000	1.250	2.700
2600	.867	114.660		117.53		36.500f	0.000	1.300	2.596
2700	.900	119.070		122.05		36.500f	0.000	1.350	2.500
2800	.933	123.480		126.57		36.500f	0.000	1.400	2.411
2900	.967	127.890		131.09		36.500f	0.000	1.450	2.328
3000	1.000	132.293		135.60		36.500f	0.000	1.500	

Soundings in mm.-----Other distances in METERS.-----

N04_WBTK.C Reference Point: Long.= 36.500f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.

GHS 16.68C

PONTOON

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N05_WBTK.P, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	45.500f	5.025p	0.050	68.170	99.934
200	.062	8.996	9.22	45.500f	5.050p	0.101	34.413	105.933
300	.094	13.627	13.97	45.500f	5.076p	0.151	23.153	112.166
400	.127	18.346	18.80	45.500f	5.101p	0.203	17.526	118.640
500	.160	23.153	23.73	45.500f	5.127p	0.254	14.144	125.349
600	.194	28.004	28.70	45.500f	5.148p	0.305	11.694	125.349
700	.227	32.855	33.68	45.500f	5.163p	0.356	9.968	125.349
800	.261	37.706	38.65	45.500f	5.174p	0.407	8.685	125.349
900	.295	42.557	43.62	45.500f	5.183p	0.457	7.695	125.349
1000	.328	47.408	48.59	45.500f	5.190p	0.508	6.908	125.349
1100	.362	52.259	53.57	45.500f	5.196p	0.558	6.267	125.349
1200	.395	57.110	58.54	45.500f	5.200p	0.608	5.734	125.349
1300	.429	61.962	63.51	45.500f	5.204p	0.659	5.285	125.349
1400	.463	66.813	68.48	45.500f	5.207p	0.709	4.902	125.349
1500	.496	71.664	73.46	45.500f	5.210p	0.759	4.570	125.349
1600	.530	76.515	78.43	45.500f	5.213p	0.809	4.280	125.349
1700	.563	81.366	83.40	45.500f	5.215p	0.859	4.025	125.349
1800	.597	86.217	88.37	45.500f	5.217p	0.909	3.798	125.349
1900	.631	91.068	93.34	45.500f	5.219p	0.959	3.596	125.349
2000	.664	95.919	98.32	45.500f	5.220p	1.010	3.414	125.349
2100	.698	100.770	103.29	45.500f	5.222p	1.060	3.250	125.349
2200	.731	105.621	108.26	45.500f	5.223p	1.110	3.101	125.349
2300	.765	110.472	113.23	45.500f	5.224p	1.160	2.964	125.349
2400	.799	115.323	118.21	45.500f	5.225p	1.210	2.840	125.349
2500	.832	120.175	123.18	45.500f	5.226p	1.260	2.725	125.349
2600	.866	125.026	128.15	45.500f	5.227p	1.310	2.619	125.349
2700	.899	129.877	133.12	45.500f	5.228p	1.360	2.521	125.349
2800	.933	134.728	138.10	45.500f	5.229p	1.410	2.431	125.349
2900	.966	139.579	143.07	45.500f	5.230p	1.460	2.346	125.349
3000	1.000	144.422	148.03	45.500f	5.230p	1.510		

Soundings in mm.-----Other distances in METERS.-----

N05_WBTK.P Reference Point: Long.= 45.500f Trans.= 5.250p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N05_WBTK.S, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.454	4.57	45.500f	5.025s	0.050	68.170	99.934
200	.062	8.996	9.22	45.500f	5.050s	0.101	34.413	105.933
300	.094	13.627	13.97	45.500f	5.076s	0.151	23.153	112.166
400	.127	18.346	18.80	45.500f	5.101s	0.203	17.526	118.640
500	.160	23.153	23.73	45.500f	5.127s	0.254	14.144	125.349
600	.194	28.004	28.70	45.500f	5.148s	0.305	11.694	125.349
700	.227	32.855	33.68	45.500f	5.163s	0.356	9.968	125.349
800	.261	37.706	38.65	45.500f	5.174s	0.407	8.685	125.349
900	.295	42.557	43.62	45.500f	5.183s	0.457	7.695	125.349
1000	.328	47.408	48.59	45.500f	5.190s	0.508	6.908	125.349
1100	.362	52.259	53.57	45.500f	5.196s	0.558	6.267	125.349
1200	.395	57.110	58.54	45.500f	5.200s	0.608	5.734	125.349
1300	.429	61.962	63.51	45.500f	5.204s	0.659	5.285	125.349
1400	.463	66.813	68.48	45.500f	5.207s	0.709	4.902	125.349
1500	.496	71.664	73.46	45.500f	5.210s	0.759	4.570	125.349
1600	.530	76.515	78.43	45.500f	5.213s	0.809	4.280	125.349
1700	.563	81.366	83.40	45.500f	5.215s	0.859	4.025	125.349
1800	.597	86.217	88.37	45.500f	5.217s	0.909	3.798	125.349
1900	.631	91.068	93.34	45.500f	5.219s	0.959	3.596	125.349
2000	.664	95.919	98.32	45.500f	5.220s	1.010	3.414	125.349
2100	.698	100.770	103.29	45.500f	5.222s	1.060	3.250	125.349
2200	.731	105.621	108.26	45.500f	5.223s	1.110	3.101	125.349
2300	.765	110.472	113.23	45.500f	5.224s	1.160	2.964	125.349
2400	.799	115.323	118.21	45.500f	5.225s	1.210	2.840	125.349
2500	.832	120.175	123.18	45.500f	5.226s	1.260	2.725	125.349
2600	.866	125.026	128.15	45.500f	5.227s	1.310	2.619	125.349
2700	.899	129.877	133.12	45.500f	5.228s	1.360	2.521	125.349
2800	.933	134.728	138.10	45.500f	5.229s	1.410	2.431	125.349
2900	.966	139.579	143.07	45.500f	5.230s	1.460	2.346	125.349
3000	1.000	144.422	148.03	45.500f	5.230s	1.510		

Soundings in mm.-----Other distances in METERS.-----

N05_WBTK.S Reference Point: Long.= 45.500f Trans.= 5.250s Vert.= 0.000
(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N05_WBTK.C, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU.	METERS	METRIC	TON	LCG	TCG	VCG	GML
0	.000	0.000		0.00					
100	.033	4.410		4.52	45.500f	0.000	0.050	67.500	94.172
200	.067	8.820		9.04	45.500f	0.000	0.100	33.750	94.172
300	.100	13.230		13.56	45.500f	0.000	0.150	22.500	94.172
400	.133	17.640		18.08	45.500f	0.000	0.200	16.875	94.172
500	.167	22.050		22.60	45.500f	0.000	0.250	13.500	94.172
600	.200	26.460		27.12	45.500f	0.000	0.300	11.250	94.172
700	.233	30.870		31.64	45.500f	0.000	0.350	9.643	94.172
800	.267	35.280		36.16	45.500f	0.000	0.400	8.437	94.172
900	.300	39.690		40.68	45.500f	0.000	0.450	7.500	94.172
1000	.333	44.100		45.20	45.500f	0.000	0.500	6.750	94.172
1100	.367	48.510		49.72	45.500f	0.000	0.550	6.136	94.172
1200	.400	52.920		54.24	45.500f	0.000	0.600	5.625	94.172
1300	.433	57.330		58.76	45.500f	0.000	0.650	5.192	94.172
1400	.467	61.740		63.28	45.500f	0.000	0.700	4.821	94.172
1500	.500	66.150		67.80	45.500f	0.000	0.750	4.500	94.172
1600	.533	70.560		72.32	45.500f	0.000	0.800	4.219	94.172
1700	.567	74.970		76.84	45.500f	0.000	0.850	3.971	94.172
1800	.600	79.380		81.36	45.500f	0.000	0.900	3.750	94.172
1900	.633	83.790		85.88	45.500f	0.000	0.950	3.553	94.172
2000	.667	88.200		90.41	45.500f	0.000	1.000	3.375	94.172
2100	.700	92.610		94.93	45.500f	0.000	1.050	3.214	94.172
2200	.733	97.020		99.45	45.500f	0.000	1.100	3.068	94.172
2300	.767	101.430		103.97	45.500f	0.000	1.150	2.935	94.172
2400	.800	105.840		108.49	45.500f	0.000	1.200	2.812	94.172
2500	.833	110.250		113.01	45.500f	0.000	1.250	2.700	94.172
2600	.867	114.660		117.53	45.500f	0.000	1.300	2.596	94.172
2700	.900	119.070		122.05	45.500f	0.000	1.350	2.500	94.172
2800	.933	123.480		126.57	45.500f	0.000	1.400	2.411	94.172
2900	.967	127.890		131.09	45.500f	0.000	1.450	2.328	94.172
3000	1.000	132.293		135.60	45.500f	0.000	1.500		

Soundings in mm.-----Other distances in METERS.-----

N05_WBTK.C Reference Point: Long.= 45.500f Trans.= 0.000 Vert.= 0.000
(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N06_WBTK.P, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m. -MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.010	0.600	0.62	50.608f	5.025p	0.051	1.613	14.679
200	.022	1.324	1.36	50.669f	5.048p	0.105	1.170	17.893
300	.036	2.164	2.22	50.727f	5.070p	0.162	1.037	21.061
400	.052	3.118	3.20	50.784f	5.093p	0.218	1.007	24.496
500	.070	4.186	4.29	50.840f	5.114p	0.278	0.996	27.893
600	.090	5.358	5.49	50.896f	5.132p	0.338	0.999	30.415
700	.112	6.631	6.80	50.952f	5.146p	0.397	1.017	33.080
800	.135	8.000	8.20	51.007f	5.156p	0.458	1.040	35.512
900	.159	9.467	9.70	51.063f	5.164p	0.519	1.072	38.146
1000	.186	11.035	11.31	51.119f	5.171p	0.581	1.122	40.781
1100	.214	12.707	13.03	51.177f	5.177p	0.643	1.174	43.391
1200	.244	14.486	14.85	51.236f	5.182p	0.705	1.228	46.212
1300	.276	16.370	16.78	51.295f	5.186p	0.768	1.284	48.884
1400	.309	18.357	18.82	51.355f	5.189p	0.831	1.341	51.672
1500	.344	20.451	20.96	51.415f	5.193p	0.894	1.397	54.352
1600	.381	22.648	23.21	51.476f	5.195p	0.958	1.454	56.944
1700	.420	24.952	25.58	51.538f	5.198p	1.022	1.513	59.724
1800	.461	27.361	28.05	51.599f	5.200p	1.086	1.573	62.502
1900	.503	29.875	30.62	51.661f	5.202p	1.150	1.633	65.229
2000	.547	32.494	33.31	51.723f	5.204p	1.215	1.692	67.953
2100	.592	35.168	36.05	51.781f	5.206p	1.278	1.572	68.433
2200	.637	37.850	38.80	51.831f	5.209p	1.340	1.470	68.944
2300	.683	40.539	41.55	51.875f	5.211p	1.401	1.379	69.334
2400	.728	43.231	44.31	51.914f	5.213p	1.460	1.297	69.522
2500	.773	45.925	47.07	51.948f	5.215p	1.518	1.222	69.637
2600	.819	48.620	49.84	51.979f	5.217p	1.575	1.155	69.638
2700	.864	51.316	52.60	52.006f	5.219p	1.631	1.094	69.638
2800	.909	54.011	55.36	52.031f	5.221p	1.687	1.039	69.637
2900	.955	56.706	58.12	52.053f	5.222p	1.743	0.990	69.638
3000	1.000	59.396	60.88	52.074f	5.223p	1.797		

Soundings in mm.-----Other distances in METERS.-----

N06_WBTK.P Reference Point: Long.= 52.500f Trans.= 5.250p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: N06_WBTK.S, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM
		CU. METERS	METRIC TON			LCG	TCG	VCG	
0	.000	0.000	0.00						
100	.010	0.600	0.62	50.608f	5.025s	0.051	1.613	14.679	
200	.022	1.324	1.36	50.669f	5.048s	0.105	1.170	17.893	
300	.036	2.164	2.22	50.727f	5.070s	0.162	1.037	21.061	
400	.052	3.118	3.20	50.784f	5.093s	0.218	1.007	24.496	
500	.070	4.186	4.29	50.840f	5.114s	0.278	0.996	27.893	
600	.090	5.358	5.49	50.896f	5.132s	0.338	0.999	30.415	
700	.112	6.631	6.80	50.952f	5.146s	0.397	1.017	33.080	
800	.135	8.000	8.20	51.007f	5.156s	0.458	1.040	35.512	
900	.159	9.467	9.70	51.063f	5.164s	0.519	1.072	38.146	
1000	.186	11.035	11.31	51.119f	5.171s	0.581	1.122	40.781	
1100	.214	12.707	13.03	51.177f	5.177s	0.643	1.174	43.391	
1200	.244	14.486	14.85	51.236f	5.182s	0.705	1.228	46.212	
1300	.276	16.370	16.78	51.295f	5.186s	0.768	1.284	48.884	
1400	.309	18.357	18.82	51.355f	5.189s	0.831	1.341	51.672	
1500	.344	20.451	20.96	51.415f	5.193s	0.894	1.397	54.352	
1600	.381	22.648	23.21	51.476f	5.195s	0.958	1.454	56.944	
1700	.420	24.952	25.58	51.538f	5.198s	1.022	1.513	59.724	
1800	.461	27.361	28.05	51.599f	5.200s	1.086	1.573	62.502	
1900	.503	29.875	30.62	51.661f	5.202s	1.150	1.633	65.229	
2000	.547	32.494	33.31	51.723f	5.204s	1.215	1.692	67.953	
2100	.592	35.168	36.05	51.781f	5.206s	1.278	1.572	68.433	
2200	.637	37.850	38.80	51.831f	5.209s	1.340	1.470	68.944	
2300	.683	40.539	41.55	51.875f	5.211s	1.401	1.379	69.334	
2400	.728	43.231	44.31	51.914f	5.213s	1.460	1.297	69.522	
2500	.773	45.925	47.07	51.948f	5.215s	1.518	1.222	69.637	
2600	.819	48.620	49.84	51.979f	5.217s	1.575	1.155	69.638	
2700	.864	51.316	52.60	52.006f	5.219s	1.631	1.094	69.638	
2800	.909	54.011	55.36	52.031f	5.221s	1.687	1.039	69.637	
2900	.955	56.706	58.12	52.053f	5.222s	1.743	0.990	69.638	
3000	1.000	59.396	60.88	52.074f	5.223s	1.797			

Soundings in mm.-----Other distances in METERS.-----

N06_WBTK.S Reference Point: Long.= 52.500f Trans.= 5.250s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

11-12-19 15:57:44 Cybermarine Technologies PTE. Ltd.

GHS 16.68C

PONTOON

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: N06_WBTK.C, Contents: WB at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.011	0.594	0.61	50.608f	0.000	0.051	1.601	13.871
200	.024	1.299	1.33	50.669f	0.000	0.104	1.161	16.165
300	.039	2.104	2.16	50.728f	0.000	0.161	1.025	18.228
400	.055	3.006	3.08	50.786f	0.000	0.216	0.989	20.292
500	.073	4.003	4.10	50.844f	0.000	0.275	0.981	22.253
600	.093	5.090	5.22	50.900f	0.000	0.334	0.990	24.168
700	.115	6.267	6.42	50.957f	0.000	0.393	1.010	26.084
800	.138	7.532	7.72	51.013f	0.000	0.454	1.038	27.976
900	.163	8.887	9.11	51.070f	0.000	0.514	1.068	29.871
1000	.189	10.333	10.59	51.127f	0.000	0.575	1.122	31.912
1100	.218	11.876	12.17	51.185f	0.000	0.637	1.175	33.953
1200	.248	13.513	13.85	51.244f	0.000	0.699	1.230	35.995
1300	.279	15.247	15.63	51.303f	0.000	0.762	1.286	38.036
1400	.313	17.076	17.50	51.364f	0.000	0.825	1.344	40.077
1500	.348	19.000	19.48	51.425f	0.000	0.888	1.402	42.118
1600	.385	21.021	21.55	51.486f	0.000	0.952	1.461	44.159
1700	.424	23.136	23.71	51.547f	0.000	1.016	1.520	46.200
1800	.465	25.348	25.98	51.609f	0.000	1.080	1.579	48.241
1900	.507	27.654	28.35	51.671f	0.000	1.144	1.639	50.281
2000	.551	30.057	30.81	51.734f	0.000	1.209	1.698	52.317
2100	.596	32.507	33.32	51.792f	0.000	1.272	1.570	52.317
2200	.641	34.957	35.83	51.841f	0.000	1.334	1.460	52.317
2300	.686	37.407	38.34	51.884f	0.000	1.394	1.365	52.317
2400	.731	39.857	40.85	51.922f	0.000	1.452	1.281	52.317
2500	.776	42.307	43.36	51.956f	0.000	1.510	1.207	52.317
2600	.820	44.757	45.88	51.985f	0.000	1.567	1.141	52.317
2700	.865	47.207	48.39	52.012f	0.000	1.623	1.081	52.317
2800	.910	49.657	50.90	52.036f	0.000	1.679	1.028	52.317
2900	.955	52.107	53.41	52.058f	0.000	1.734	0.980	52.317
3000	1.000	54.553	55.92	52.078f	0.000	1.788		

Soundings in mm.-----Other distances in METERS.-----

N06_WBTK.C Reference Point: Long.= 52.500f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

HYDROSTATIC PARTICULARS

10-12-19 16:45:55 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/55.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
27.500f---	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	KML-----	KMT	
0.112	93.21	39.386f	0.099	5.75	33.596f	11.65	687.62	121.186	
0.212	159.74	35.816f	0.133	7.56	28.287f	26.14	900.09	94.307	
0.312	236.55	33.326f	0.173	7.76	28.153f	27.59	641.46	67.127	
0.412	314.90	32.033f	0.218	7.91	28.112f	28.50	497.75	52.501	
0.512	394.58	31.233f	0.265	8.03	28.043f	29.38	409.56	43.251	
0.612	475.36	30.685f	0.313	8.13	27.977f	30.28	350.30	36.761	
0.712	556.99	30.283f	0.363	8.20	27.917f	31.02	306.33	31.778	
0.812	639.29	29.980f	0.414	8.26	27.921f	31.78	273.36	28.006	
0.912	722.23	29.744f	0.464	8.33	27.928f	32.54	247.83	25.087	
1.012	805.81	29.558f	0.515	8.39	27.960f	33.25	226.94	22.712	
1.112	890.01	29.407f	0.567	8.46	27.973f	34.07	210.53	20.839	
1.212	974.86	29.281f	0.618	8.52	27.964f	34.84	196.54	19.252	
1.312	1,060.35	29.175f	0.669	8.58	27.961f	35.66	184.97	17.938	
1.412	1,146.50	29.084f	0.721	8.65	27.957f	36.53	175.21	16.828	
1.512	1,233.30	29.005f	0.773	8.71	27.975f	37.34	166.52	15.850	
1.612	1,320.74	28.937f	0.824	8.77	27.975f	38.17	158.94	14.999	
1.712	1,408.81	28.877f	0.876	8.84	27.977f	39.03	152.38	14.262	
1.812	1,497.46	28.823f	0.928	8.88	27.917f	39.67	145.69	13.592	
1.912	1,586.46	28.770f	0.980	8.92	27.830f	40.12	139.09	12.978	
2.012	1,675.79	28.717f	1.032	8.95	27.738f	40.60	133.24	12.436	
2.112	1,765.44	28.665f	1.084	8.98	27.646f	41.08	127.97	11.952	
2.212	1,855.43	28.613f	1.136	9.02	27.548f	41.56	123.20	11.518	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 55.00m.

Draft is from USK.

10-12-19 16:45:55 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

HYDROSTATIC PROPERTIES

Trim: Fwd 0.250/55.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
27.500f---	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	KML-----	KMT	
0.112	82.87	35.217f	0.062	7.44	27.824f	25.58	1697.99	173.768	
0.212	158.06	31.699f	0.104	7.60	27.826f	26.56	924.36	95.591	
0.312	234.84	30.432f	0.152	7.76	27.828f	27.57	645.76	67.494	
0.412	313.22	29.780f	0.202	7.92	27.828f	28.60	502.23	53.058	
0.512	393.11	29.377f	0.252	8.05	27.769f	29.51	412.87	43.813	
0.612	474.08	29.097f	0.302	8.14	27.720f	30.37	352.30	37.037	
0.712	555.78	28.895f	0.353	8.21	27.721f	31.10	307.76	31.935	
0.812	638.11	28.743f	0.404	8.26	27.696f	31.74	273.60	28.037	
0.912	720.99	28.624f	0.456	8.32	27.722f	32.42	247.29	25.039	
1.012	804.50	28.531f	0.508	8.38	27.726f	33.20	226.96	22.718	
1.112	888.67	28.455f	0.559	8.45	27.731f	34.01	210.51	20.840	
1.212	973.51	28.392f	0.611	8.52	27.735f	34.86	196.97	19.295	
1.312	1,059.01	28.338f	0.663	8.58	27.721f	35.68	185.29	17.971	
1.412	1,145.14	28.293f	0.715	8.64	27.732f	36.45	175.06	16.819	
1.512	1,231.90	28.253f	0.767	8.71	27.736f	37.28	166.46	15.847	
1.612	1,319.31	28.219f	0.819	8.77	27.740f	38.16	159.07	15.012	
1.712	1,407.37	28.189f	0.871	8.84	27.745f	39.07	152.70	14.291	
1.812	1,496.06	28.162f	0.923	8.90	27.740f	39.88	146.61	13.618	
1.912	1,585.37	28.138f	0.975	8.95	27.706f	40.61	140.89	13.025	
2.012	1,675.07	28.113f	1.028	8.99	27.613f	41.10	134.94	12.480	
2.112	1,765.11	28.085f	1.080	9.02	27.517f	41.61	129.64	11.997	
2.212	1,855.36	28.057f	1.132	9.03	27.509f	41.72	123.67	11.528	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 55.00m.

Draft is from USK.

10-12-19 16:45:55 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
27.500f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	82.32	27.500f	0.044	7.44	27.500f	25.58	1708.69	174.783	
0.212	157.50	27.500f	0.095	7.60	27.500f	26.56	927.34	95.855	
0.312	234.27	27.500f	0.145	7.76	27.500f	27.56	647.13	67.606	
0.412	312.64	27.500f	0.196	7.92	27.500f	28.60	503.15	53.134	
0.512	392.63	27.500f	0.247	8.08	27.500f	29.64	415.23	44.252	
0.612	473.68	27.500f	0.299	8.14	27.500f	30.36	352.52	37.063	
0.712	555.37	27.500f	0.350	8.21	27.500f	31.09	307.90	31.952	
0.812	637.67	27.501f	0.401	8.26	27.523f	31.75	273.85	28.066	
0.912	720.58	27.503f	0.453	8.32	27.495f	32.41	247.36	25.048	
1.012	804.09	27.502f	0.505	8.38	27.493f	33.19	227.02	22.725	
1.112	888.25	27.501f	0.557	8.45	27.489f	34.00	210.56	20.845	
1.212	973.08	27.500f	0.609	8.52	27.498f	34.81	196.73	19.274	
1.312	1,058.55	27.500f	0.661	8.58	27.507f	35.65	185.21	17.964	
1.412	1,144.68	27.500f	0.713	8.64	27.502f	36.44	175.10	16.824	
1.512	1,231.44	27.500f	0.765	8.71	27.498f	37.28	166.50	15.852	
1.612	1,318.84	27.500f	0.817	8.77	27.501f	38.12	158.98	15.005	
1.712	1,406.88	27.500f	0.869	8.84	27.502f	39.01	152.49	14.275	
1.812	1,495.57	27.500f	0.921	8.90	27.510f	39.91	146.77	13.632	
1.912	1,584.91	27.500f	0.973	8.96	27.499f	40.77	141.48	13.044	
2.012	1,674.87	27.500f	1.026	9.02	27.499f	41.56	136.46	12.517	
2.112	1,765.09	27.500f	1.078	9.02	27.497f	41.64	129.75	11.996	
2.212	1,855.34	27.500f	1.130	9.03	27.499f	41.71	123.64	11.524	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 55.00m.

Draft is from USK.

10-12-19 16:45:55 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 0.250/55.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
27.500f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	82.87	19.783f	0.062	7.44	27.176f	25.58	1697.99	173.760	
0.212	158.06	23.301f	0.104	7.60	27.174f	26.56	924.36	95.586	
0.312	234.84	24.568f	0.152	7.76	27.172f	27.57	645.76	67.490	
0.412	313.22	25.220f	0.202	7.92	27.172f	28.60	502.24	53.056	
0.512	393.11	25.623f	0.252	8.05	27.231f	29.51	412.87	43.812	
0.612	474.09	25.903f	0.302	8.14	27.280f	30.37	352.30	37.037	
0.712	555.74	26.107f	0.354	8.20	27.301f	31.02	306.95	31.853	
0.812	638.03	26.260f	0.405	8.26	27.299f	31.76	273.80	28.059	
0.912	720.97	26.378f	0.456	8.33	27.293f	32.53	248.17	25.128	
1.012	804.56	26.473f	0.507	8.40	27.285f	33.33	227.82	22.802	
1.112	888.73	26.547f	0.559	8.45	27.261f	33.99	210.35	20.826	
1.212	973.55	26.609f	0.611	8.51	27.263f	34.79	196.54	19.256	
1.312	1,059.01	26.662f	0.663	8.58	27.266f	35.62	184.98	17.941	
1.412	1,145.13	26.707f	0.715	8.65	27.270f	36.48	175.21	16.831	
1.512	1,231.91	26.747f	0.766	8.71	27.267f	37.29	166.48	15.849	
1.612	1,319.31	26.781f	0.819	8.77	27.261f	38.14	158.99	15.006	
1.712	1,407.36	26.811f	0.871	8.84	27.260f	39.00	152.41	14.267	
1.812	1,496.05	26.838f	0.923	8.90	27.259f	39.90	146.68	13.623	
1.912	1,585.37	26.862f	0.975	8.96	27.301f	40.65	141.03	13.037	
2.012	1,675.09	26.887f	1.027	8.99	27.384f	41.11	134.98	12.483	
2.112	1,765.12	26.915f	1.080	9.02	27.477f	41.58	129.56	11.990	
2.212	1,855.36	26.943f	1.132	9.03	27.490f	41.70	123.60	11.522	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 55.00m.

Draft is from USK.

10-12-19 16:45:55 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/55.000, No Heel, Fixed VCG = 0.000

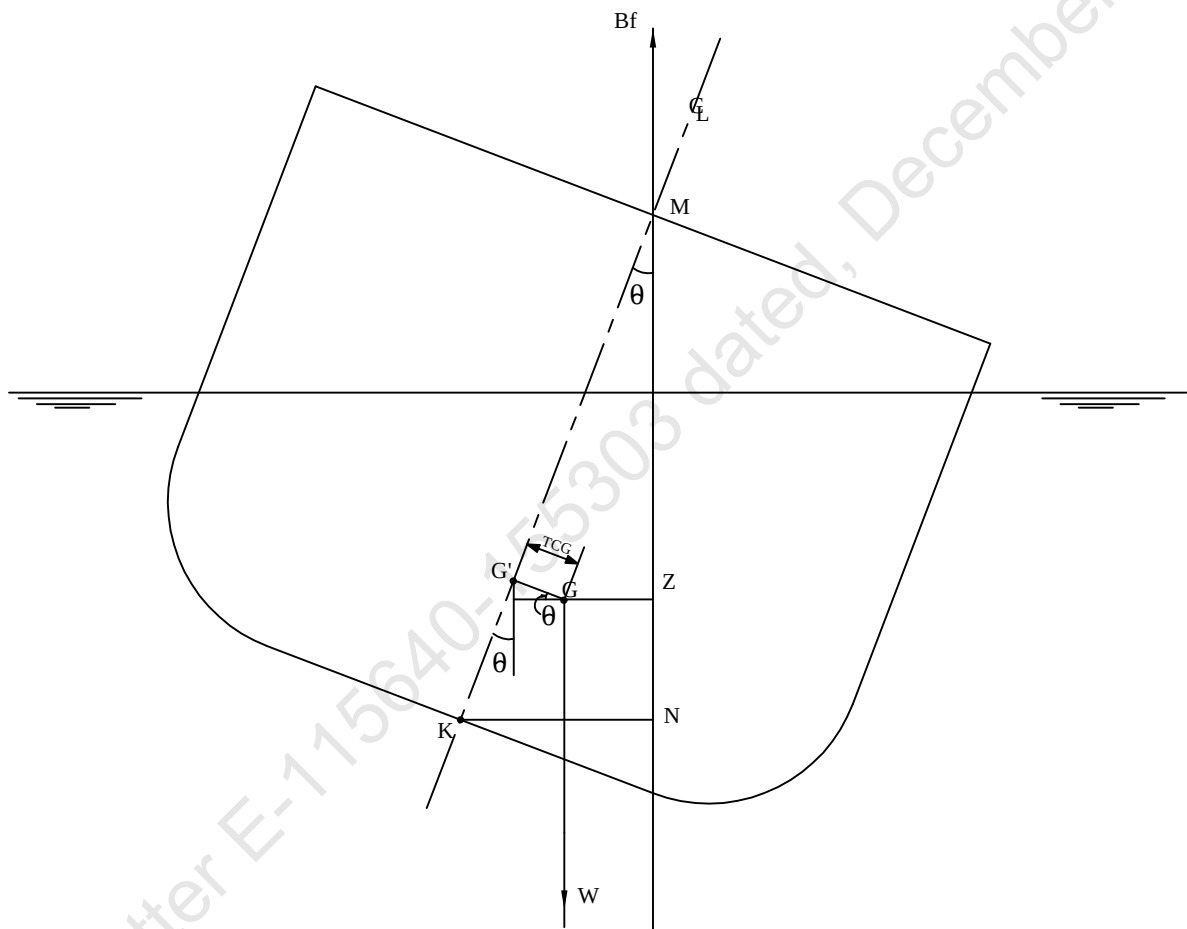
Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
27.500f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	93.21	15.614f	0.099	5.75	21.404f	11.65	687.62	121.175	
0.212	159.74	19.184f	0.133	7.56	26.713f	26.14	900.09	94.298	
0.312	236.55	21.674f	0.173	7.76	26.847f	27.59	641.45	67.120	
0.412	314.90	22.967f	0.218	7.91	26.888f	28.50	497.75	52.497	
0.512	394.58	23.767f	0.265	8.02	26.979f	29.30	408.46	43.134	
0.612	475.31	24.318f	0.315	8.12	27.046f	30.19	349.38	36.666	
0.712	556.92	24.720f	0.364	8.20	27.078f	31.04	306.56	31.804	
0.812	639.24	25.022f	0.415	8.27	27.071f	31.81	273.64	28.035	
0.912	722.21	25.256f	0.465	8.33	27.061f	32.59	248.17	25.122	
1.012	805.83	25.443f	0.515	8.40	27.062f	33.34	227.54	22.773	
1.112	890.08	25.595f	0.566	8.46	27.060f	34.12	210.80	20.867	
1.212	974.92	25.721f	0.617	8.51	27.030f	34.81	196.36	19.235	
1.312	1,060.39	25.826f	0.669	8.58	27.026f	35.64	184.84	17.924	
1.412	1,146.51	25.916f	0.721	8.65	27.030f	36.47	174.93	16.802	
1.512	1,233.29	25.994f	0.772	8.71	27.033f	37.34	166.50	15.848	
1.612	1,320.73	26.063f	0.824	8.78	27.026f	38.17	158.96	15.000	
1.712	1,408.81	26.123f	0.876	8.84	27.019f	39.05	152.43	14.267	
1.812	1,497.46	26.177f	0.928	8.88	27.076f	39.65	145.64	13.588	
1.912	1,586.46	26.230f	0.980	8.92	27.169f	40.13	139.13	12.981	
2.012	1,675.80	26.283f	1.032	8.95	27.268f	40.62	133.32	12.443	
2.112	1,765.47	26.335f	1.084	8.98	27.356f	41.07	127.94	11.950	
2.212	1,855.44	26.387f	1.136	9.01	27.448f	41.54	123.12	11.512	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 55.00m.

Draft is from USK.

KN CONCEPT



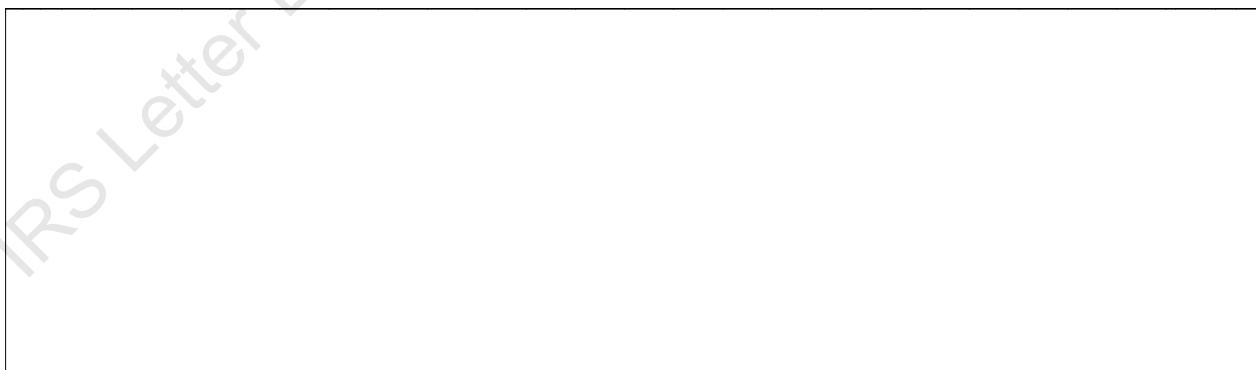
$$\underline{GZ = KN - KG' \sin \theta - TCG \cos \theta}$$

VOLUME USED FOR KN COMPUTATION

Profile View



Plan View



CROSS CURVE TABLES

11-12-19 09:54:26 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.500/55.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
93.21	5.929s	6.387s	6.318s	5.963s	5.394s	4.640s
159.74	5.658s	6.228s	6.226s	5.925s	5.385s	4.644s
236.55	5.370s	6.055s	6.129s	5.876s	5.361s	4.631s
314.90	5.078s	5.884s	6.029s	5.808s	5.305s	4.583s
394.58	4.802s	5.723s	5.924s	5.714s	5.217s	4.507s
475.36	4.543s	5.572s	5.804s	5.595s	5.105s	4.414s
556.99	4.300s	5.428s	5.668s	5.455s	4.978s	4.311s
639.29	4.071s	5.282s	5.512s	5.300s	4.841s	4.201s
722.23	3.854s	5.129s	5.341s	5.135s	4.697s	4.087s
805.81	3.646s	4.965s	5.158s	4.962s	4.548s	3.968s
890.01	3.447s	4.787s	4.966s	4.784s	4.394s	3.847s
974.86	3.255s	4.597s	4.766s	4.600s	4.237s	3.724s
1,060.35	3.074s	4.396s	4.561s	4.412s	4.077s	3.598s
1,146.50	2.905s	4.186s	4.351s	4.221s	3.914s	3.470s
1,233.30	2.748s	3.969s	4.136s	4.026s	3.749s	3.341s
1,320.74	2.600s	3.746s	3.918s	3.829s	3.582s	3.211s
1,408.81	2.458s	3.518s	3.696s	3.629s	3.413s	3.079s
1,497.46	2.317s	3.285s	3.471s	3.428s	3.243s	2.945s
1,586.46	2.174s	3.048s	3.244s	3.224s	3.071s	2.811s
1,675.79	2.028s	2.809s	3.016s	3.020s	2.899s	2.677s
1,765.44	1.877s	2.569s	2.786s	2.814s	2.725s	2.542s
1,855.43	1.723s	2.330s	2.554s	2.607s	2.551s	2.406s
Distances in METERS.---Specific Gravity = 1.025.-----						

11-12-19 09:54:26 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.250/55.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
82.87	6.217s	6.545s	6.394s	5.972s	5.350s	4.576s
158.06	5.828s	6.316s	6.268s	5.934s	5.391s	4.662s
234.84	5.467s	6.107s	6.154s	5.900s	5.400s	4.673s
313.22	5.144s	5.920s	6.053s	5.849s	5.351s	4.622s
393.11	4.851s	5.750s	5.956s	5.761s	5.259s	4.541s
474.08	4.582s	5.595s	5.846s	5.638s	5.143s	4.445s
555.78	4.332s	5.449s	5.709s	5.494s	5.013s	4.339s
638.11	4.098s	5.310s	5.551s	5.336s	4.873s	4.228s
720.99	3.877s	5.163s	5.377s	5.169s	4.728s	4.112s
804.50	3.667s	5.000s	5.192s	4.995s	4.577s	3.993s
888.67	3.466s	4.821s	4.999s	4.815s	4.422s	3.871s
973.51	3.273s	4.630s	4.799s	4.630s	4.264s	3.747s
1,059.01	3.087s	4.428s	4.592s	4.441s	4.104s	3.620s
1,145.14	2.912s	4.217s	4.381s	4.249s	3.940s	3.492s
1,231.90	2.754s	4.000s	4.166s	4.054s	3.775s	3.363s
1,319.31	2.612s	3.776s	3.947s	3.857s	3.607s	3.232s
1,407.37	2.476s	3.547s	3.725s	3.656s	3.438s	3.099s
1,496.06	2.338s	3.314s	3.499s	3.454s	3.266s	2.965s
1,585.37	2.194s	3.076s	3.271s	3.249s	3.094s	2.830s
1,675.07	2.046s	2.835s	3.041s	3.043s	2.920s	2.695s
1,765.11	1.895s	2.591s	2.809s	2.835s	2.745s	2.558s
1,855.36	1.740s	2.348s	2.576s	2.627s	2.569s	2.421s
Distances in METERS.---Specific Gravity = 1.025.-----						

11-12-19 09:54:26 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONTON

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
82.32	6.360s	6.621s	6.429s	5.973s	5.319s	4.520s
157.50	5.884s	6.346s	6.281s	5.934s	5.384s	4.676s
234.27	5.500s	6.123s	6.162s	5.903s	5.417s	4.688s
312.64	5.167s	5.932s	6.059s	5.866s	5.367s	4.635s
392.63	4.867s	5.759s	5.965s	5.777s	5.273s	4.552s
473.68	4.594s	5.602s	5.861s	5.652s	5.155s	4.455s
555.37	4.342s	5.455s	5.723s	5.507s	5.024s	4.349s
637.67	4.107s	5.318s	5.564s	5.348s	4.884s	4.237s
720.58	3.885s	5.176s	5.389s	5.180s	4.738s	4.120s
804.09	3.674s	5.012s	5.203s	5.005s	4.587s	4.001s
888.25	3.473s	4.833s	5.010s	4.825s	4.432s	3.879s
973.08	3.279s	4.641s	4.809s	4.640s	4.273s	3.754s
1,058.55	3.092s	4.438s	4.602s	4.451s	4.112s	3.628s
1,144.68	2.914s	4.228s	4.391s	4.259s	3.949s	3.500s
1,231.44	2.756s	4.010s	4.176s	4.064s	3.783s	3.370s
1,318.84	2.615s	3.786s	3.957s	3.866s	3.616s	3.239s
1,406.88	2.483s	3.557s	3.734s	3.665s	3.446s	3.106s
1,495.57	2.345s	3.323s	3.508s	3.462s	3.275s	2.972s
1,584.91	2.202s	3.085s	3.280s	3.258s	3.101s	2.837s
1,674.87	2.053s	2.843s	3.049s	3.050s	2.927s	2.701s
1,765.09	1.901s	2.599s	2.816s	2.842s	2.751s	2.564s
1,855.34	1.746s	2.353s	2.583s	2.634s	2.575s	2.427s
Distances in METERS.---Specific Gravity = 1.025.-----						

11-12-19 09:54:26 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.250/55.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
82.87	6.216s	6.545s	6.394s	5.972s	5.350s	4.576s
158.06	5.829s	6.316s	6.268s	5.934s	5.391s	4.662s
234.84	5.468s	6.107s	6.154s	5.900s	5.400s	4.673s
313.22	5.144s	5.920s	6.053s	5.849s	5.351s	4.622s
393.11	4.851s	5.750s	5.956s	5.761s	5.259s	4.541s
474.09	4.581s	5.595s	5.846s	5.638s	5.143s	4.445s
555.74	4.332s	5.449s	5.709s	5.494s	5.013s	4.339s
638.03	4.099s	5.310s	5.551s	5.337s	4.874s	4.228s
720.97	3.877s	5.163s	5.377s	5.169s	4.728s	4.112s
804.56	3.667s	5.000s	5.192s	4.995s	4.577s	3.993s
888.73	3.466s	4.821s	4.999s	4.815s	4.422s	3.871s
973.55	3.273s	4.629s	4.798s	4.630s	4.264s	3.747s
1,059.01	3.087s	4.428s	4.592s	4.441s	4.104s	3.620s
1,145.13	2.912s	4.217s	4.381s	4.249s	3.940s	3.492s
1,231.91	2.754s	4.000s	4.166s	4.054s	3.775s	3.363s
1,319.31	2.612s	3.776s	3.947s	3.857s	3.607s	3.232s
1,407.36	2.476s	3.547s	3.725s	3.656s	3.438s	3.099s
1,496.05	2.338s	3.314s	3.499s	3.454s	3.267s	2.966s
1,585.37	2.194s	3.076s	3.271s	3.249s	3.094s	2.830s
1,675.09	2.046s	2.835s	3.041s	3.043s	2.920s	2.695s
1,765.12	1.895s	2.591s	2.809s	2.835s	2.745s	2.558s
1,855.36	1.741s	2.348s	2.575s	2.627s	2.569s	2.421s
Distances in METERS.---Specific Gravity = 1.025.-----						

11-12-19 09:54:26 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.500/55.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
93.21	5.929s	6.386s	6.317s	5.963s	5.394s	4.640s
159.74	5.659s	6.227s	6.226s	5.925s	5.385s	4.644s
236.55	5.370s	6.055s	6.128s	5.876s	5.360s	4.631s
314.90	5.077s	5.884s	6.029s	5.808s	5.304s	4.583s
394.58	4.801s	5.723s	5.923s	5.714s	5.217s	4.507s
475.31	4.543s	5.572s	5.805s	5.596s	5.105s	4.414s
556.92	4.300s	5.428s	5.668s	5.455s	4.978s	4.311s
639.24	4.071s	5.282s	5.512s	5.301s	4.841s	4.201s
722.21	3.854s	5.129s	5.341s	5.135s	4.697s	4.087s
805.83	3.646s	4.965s	5.158s	4.962s	4.548s	3.968s
890.08	3.447s	4.787s	4.966s	4.783s	4.394s	3.847s
974.92	3.255s	4.596s	4.766s	4.600s	4.237s	3.724s
1,060.39	3.074s	4.395s	4.561s	4.412s	4.077s	3.598s
1,146.51	2.904s	4.186s	4.351s	4.221s	3.914s	3.470s
1,233.29	2.748s	3.969s	4.136s	4.026s	3.749s	3.341s
1,320.73	2.600s	3.746s	3.918s	3.829s	3.582s	3.211s
1,408.81	2.458s	3.518s	3.696s	3.629s	3.413s	3.079s
1,497.46	2.317s	3.285s	3.471s	3.427s	3.243s	2.945s
1,586.46	2.174s	3.048s	3.244s	3.224s	3.071s	2.811s
1,675.80	2.028s	2.809s	3.016s	3.020s	2.899s	2.677s
1,765.47	1.877s	2.569s	2.786s	2.814s	2.725s	2.542s
1,855.44	1.724s	2.330s	2.554s	2.607s	2.551s	2.406s
Distances in METERS.---Specific Gravity = 1.025.-----						

DEADWEIGHT TABLE

Draft	Disp	Dwt
M	T	T
0.612	473.68	50.33
0.712	555.37	132.02
0.812	637.67	214.32
0.912	720.58	297.23
1.012	804.09	380.74
1.112	888.25	464.90
1.212	973.08	549.73
1.312	1058.55	635.20
1.412	1144.68	721.33
1.512	1231.44	808.09
1.612	1318.84	895.49
1.712	1406.88	983.53
1.812	1495.57	1072.22
1.912	1584.91	1161.56
2.012	1674.87	1251.52
2.112	1765.09	1341.74
2.212	1855.34	1431.99

Full Load Displacement

11-12-19 09:56:35 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

USK draft refers to the line:

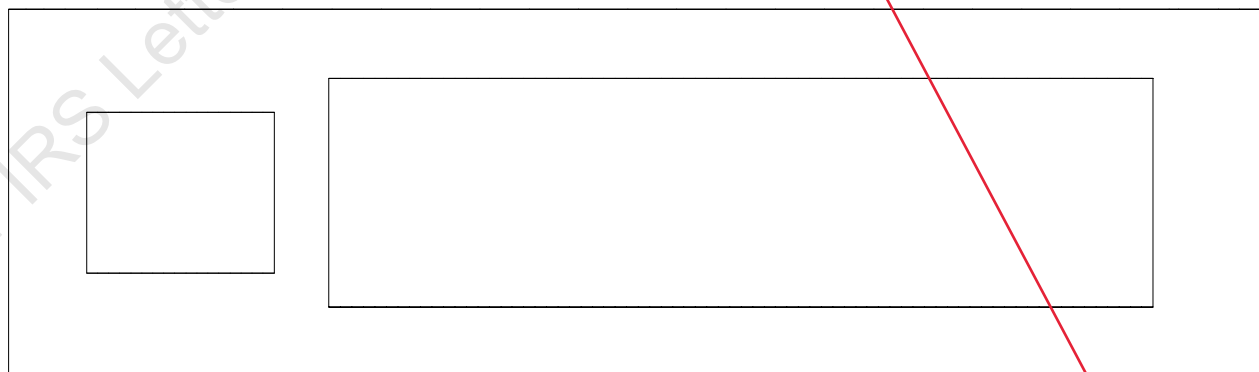
0.012 below baseline @ 55.000f and 0.012 below baseline @ 0.000

MAX. PERMISSIBLE VCG

Profile View



Plan View



11-12-19 09:56:35 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 0.500/55.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	16.374	107%	0d	11d
600.00	15.320	66%	0d	8d
800.00	14.323	40%	0d	6d
1,000.00	13.055	30%	0d	5d
1,200.00	11.691	15%	0d	4d
1,400.00	10.168	0%	0d	4d
1,600.00	8.123	0%	2d	3d
1,800.00	6.175	0%	5d	2d
2,000.00	4.336	0%	10d	1d

Heeling moment is present from: wind

Trim = Fwd 0.250/55.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	16.441	108%	0d	13d
600.00	15.426	71%	0d	9d
800.00	14.414	39%	0d	7d
1,000.00	13.096	31%	0d	6d
1,200.00	11.770	17%	0d	5d
1,400.00	10.244	0%	0d	4d
1,600.00	8.162	0%	2d	3d
1,800.00	6.156	0%	5d	3d
2,000.00	4.346	0%	10d	2d

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	16.466	108%	0d	15d
600.00	15.408	71%	0d	11d
800.00	14.449	37%	0d	8d
1,000.00	13.121	30%	0d	6d
1,200.00	11.660	22%	0d	5d
1,400.00	10.263	0%	0d	4d
1,600.00	8.177	0%	2d	4d
1,800.00	6.165	0%	5d	3d
2,000.00	4.352	0%	10d	2d

Heeling moment is present from: wind

Trim = Aft 0.250/55.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---

METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	16.444	107%	0d	13d
600.00	15.375	75%	0d	9d
800.00	14.448	38%	0d	7d
1,000.00	13.078	31%	0d	6d
1,200.00	11.782	16%	0d	5d

11-12-19 09:56:35 Cybermarine Technologies PTE. Ltd.
GHS 16.68C PONT00N

1,400.00	10.247	0%	0d	4d
1,600.00	8.162	0%	2d	3d
1,800.00	6.152	0%	5d	3d
2,000.00	4.345	0%	10d	2d

Heeling moment is present from: wind

Trim = Aft 0.500/55.000 at zero heel (trim righting arm held at zero)

Displacement --- Margins ---
METRIC TONS Max VCG LIM1 LIM2 LIM3

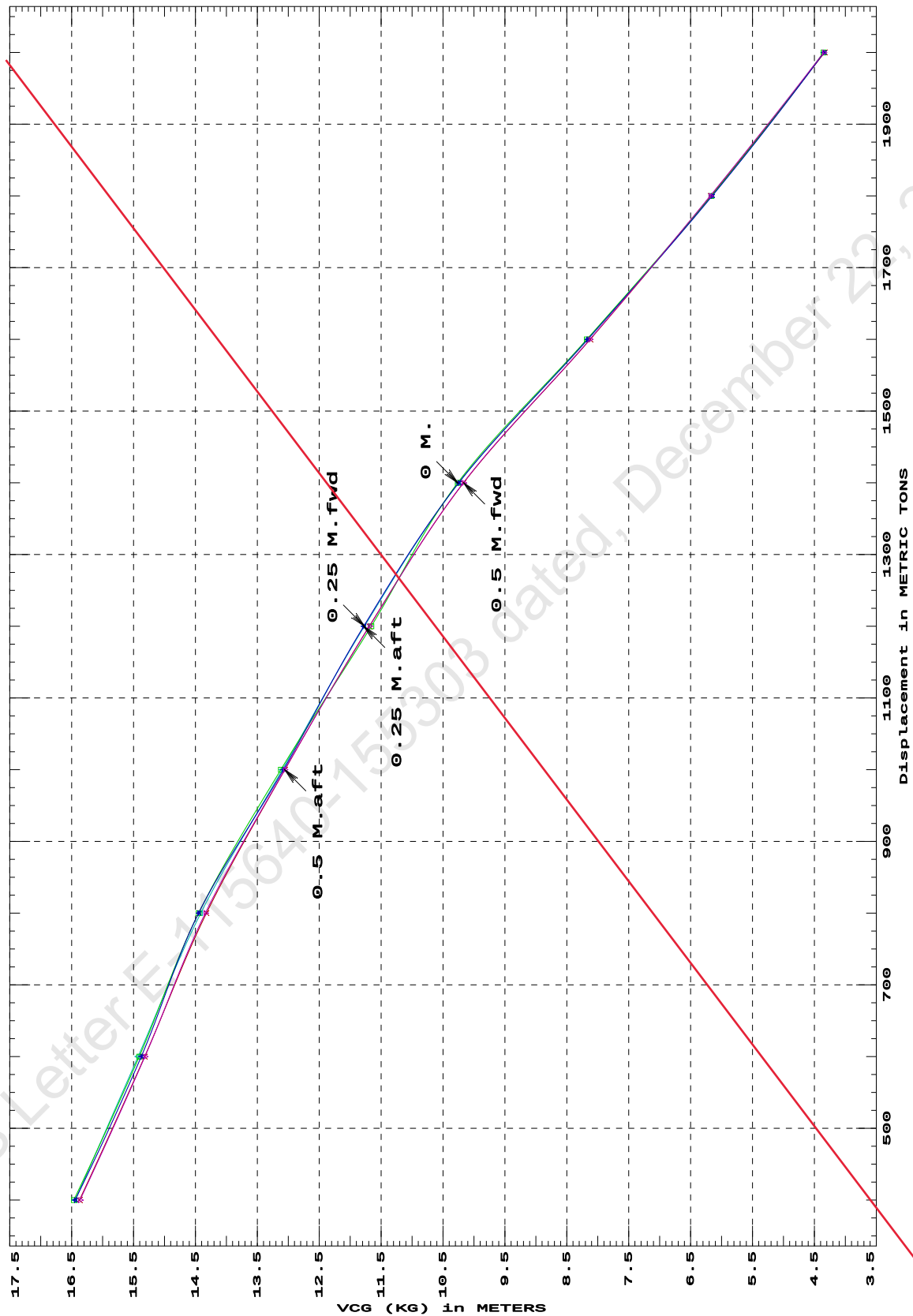
400.00	16.366	106%	0d	11d
600.00	15.317	66%	0d	8d
800.00	14.341	39%	0d	6d
1,000.00	13.058	30%	0d	5d
1,200.00	11.689	15%	0d	4d
1,400.00	10.171	0%	0d	4d
1,600.00	8.126	0%	2d	3d
1,800.00	6.184	0%	5d	2d
2,000.00	4.349	1%	10d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max

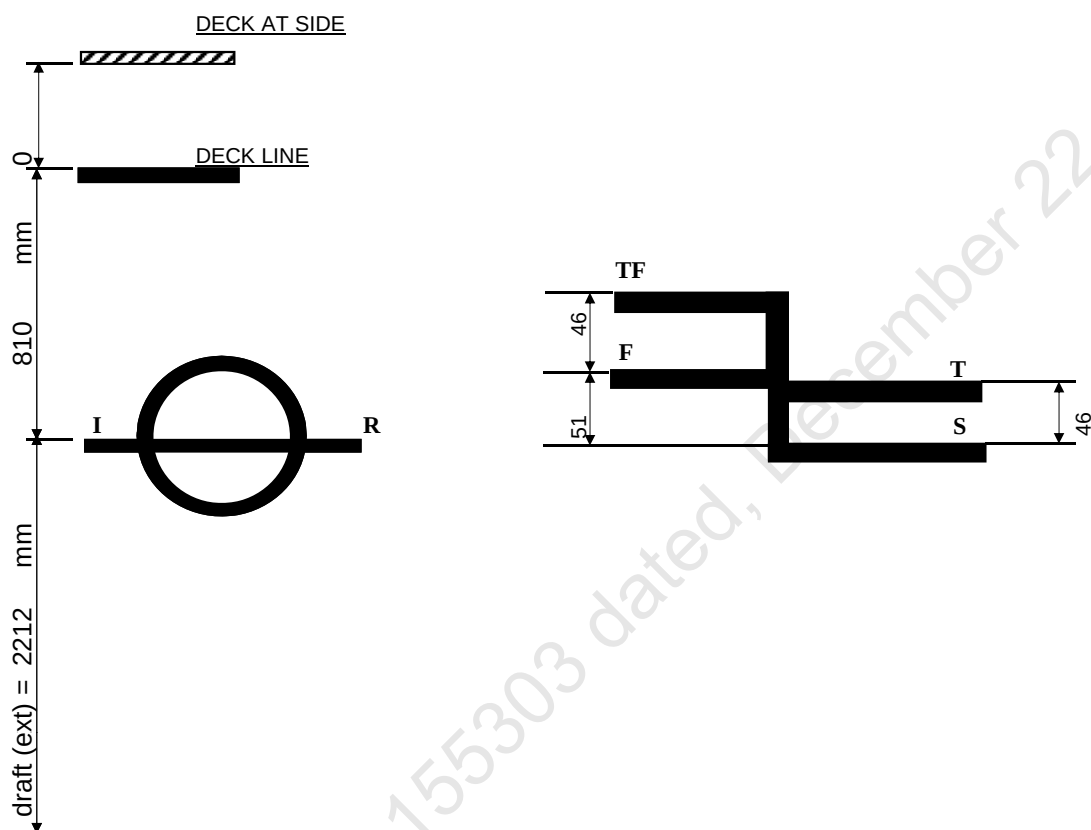
- (1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

IS CODE 2008 PART B CHAPTER-2 SEC.2 MAXIMUM VCG (KG)
at various trims (initial)



Specific Gravity = 1.025 "K" = BPL
Trim is per 55 M.

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1855.24	1431.89
FRESH WATER	759	2.263	1855.00	1431.65
TROPICAL	764	2.258	1896.86	1473.51
TROPICAL FRESH WATER	713	2.309	1895.52	1472.17

Deck Plate Thickness = 10 mm

Keel Plate Thickness = 12 mm

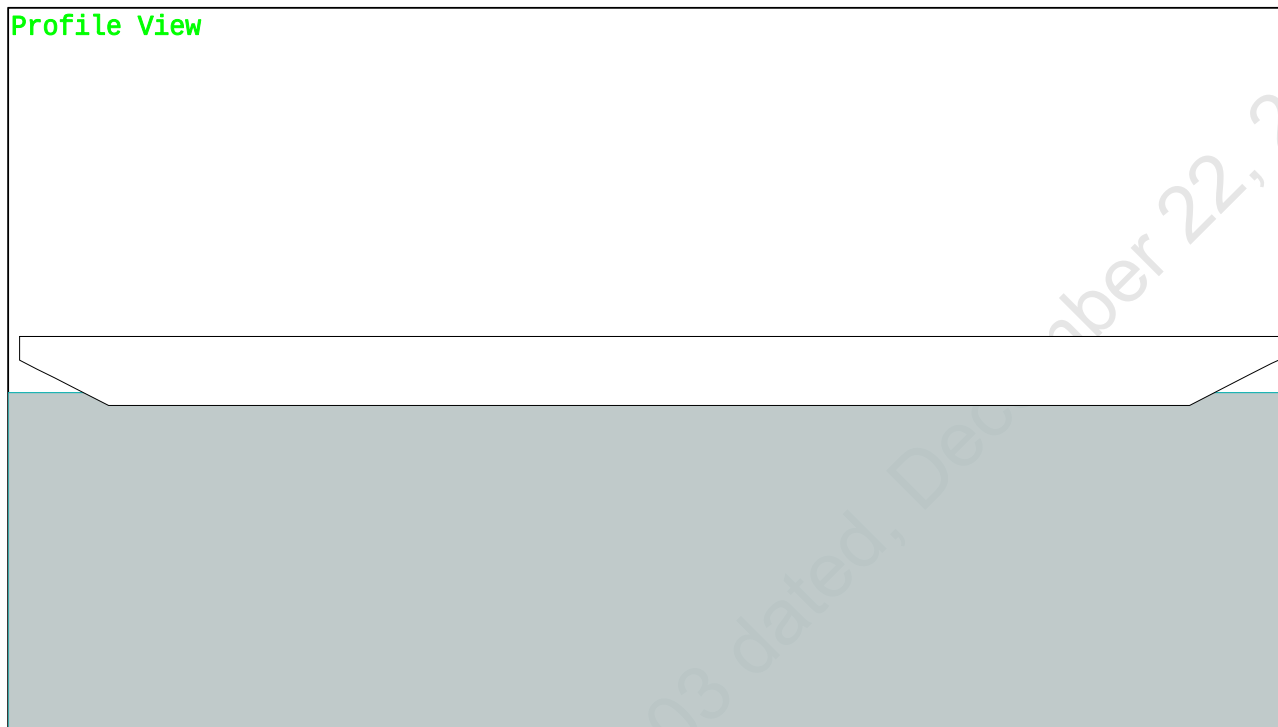
LOADING CONDITIONS

SUMMARY OF CONDITIONS					
CONDITION NO.	1	2	3	4	5
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Aft	Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 51.30T @ 12m Radius Fwd
TOTAL TANK LOAD (T)	0.000	121.760	435.000	435.000	435.000
DWT CONST.(T)	0.000	0.000	0.000	0.000	0.000
DEADWEIGHT (T)	0.000	0.000	0.000	0.000	0.000
LIGHT WEIGHT (T)	423.350	423.350	423.350	423.350	423.350
DISPLACEMENT (T)	423.350	695.010	1855.310	1855.310	1855.340
FWD DRAFT (M)	0.550	0.880	2.110	2.186	2.262
AFT DRAFT (M)	0.550	0.883	2.314	2.238	2.162
MEAN DRAFT (M)	0.550	0.881	2.212	2.212	2.212
TRIM (M)	0.000	0.000	0.205a	0.052a	0.101f
L.C.G. (FROM AP) (M)	27.500f	27.489f	27.055f	27.387f	27.719f
L.C.B. (FROM AP) (M)	27.500f	27.490f	27.044f	27.384f	27.724f
M.C.T. (T-CM)	29.680	31.860	40.310	40.310	40.320
KG (M)	3.000	3.264	4.124	4.124	4.124
GM _T (M)	38.201	22.492	7.170	7.171	7.173
MAX. GZ (M)	4.648	4.013	1.001	1.004	1.004

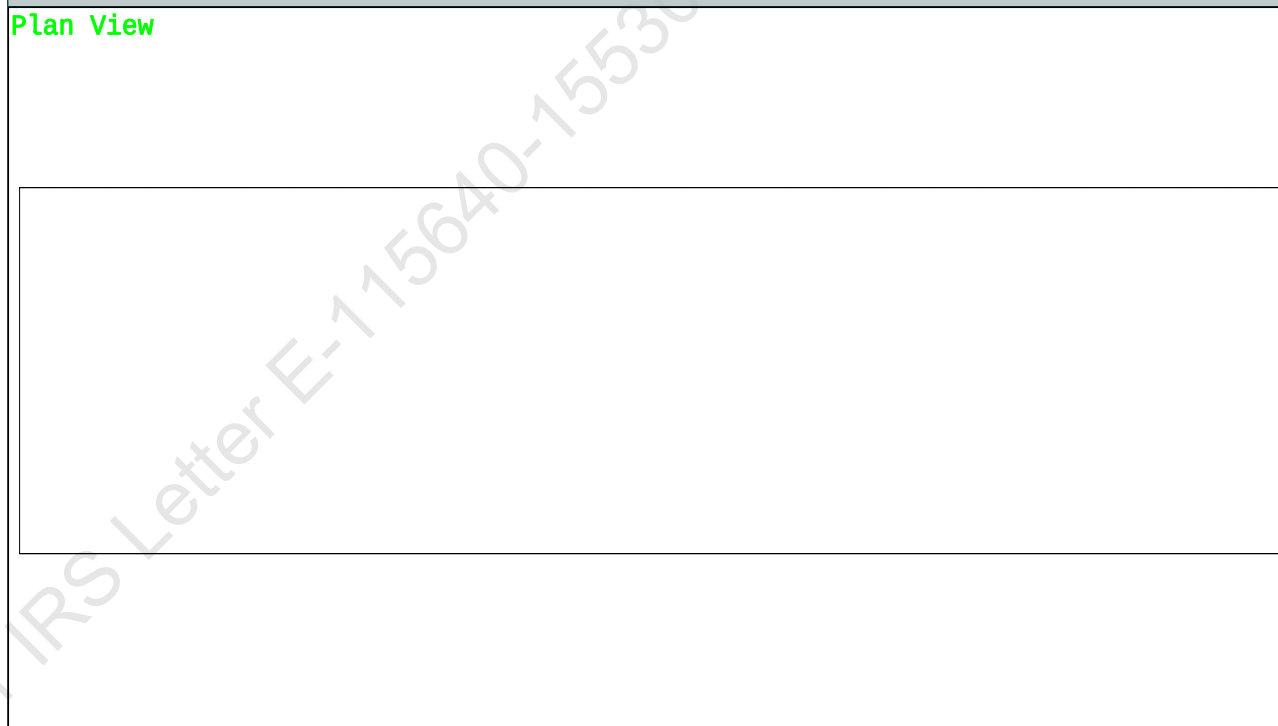
SUMMARY OF CONDITIONS					
CONDITION NO.	6	7	8	9	10
CONDITION	Condition 5 : Lightship + Crane + Ballast + Deck Cargo Towing Condition with Max. draft 2.212m	Condition 6 : Lightship + Max. Deck Cargo Max. deck cargo loading condition	Condition 7 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 8 : Lightship + Load at Stbd + Ballast Crane Lifting 36.6T @ 18m Radius Stbd	Condition 9 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd
TOTAL TANK LOAD (T)	379.080	0.000	369.180	369.180	369.180
DWT CONST.(T)	0.000	0.000	0.000	0.000	0.000
DEADWEIGHT (T)	0.000	0.000	0.000	0.000	0.000
LIGHT WEIGHT (T)	423.350	423.350	423.350	423.350	423.350
DISPLACEMENT (T)	1855.240	1855.220	1855.200	1855.200	1855.200
FWD DRAFT (M)	2.143	2.212	2.092	2.175	2.257
AFT DRAFT (M)	2.281	2.212	2.331	2.249	2.167
MEAN DRAFT (M)	2.212	2.212	2.212	2.212	2.212
TRIM (M)	0.138a	0.000	0.239a	0.074a	0.090f
L.C.G. (FROM AP) (M)	27.199f	27.500f	26.984f	27.339f	27.694f
L.C.B. (FROM AP) (M)	27.193f	27.500f	26.968f	27.334f	27.700f
M.C.T. (T-CM)	40.560	40.440	40.040	40.050	40.060
KG (M)	3.397	3.772	4.913	4.913	4.913
GM _T (M)	7.926	7.753	6.428	6.429	6.432
MAX. GZ (M)	1.178	1.141	0.850	0.854	0.854

Condition Graphic - Draft: 0.550 Trim: zero Heel: zero

Profile View



Plan View



17-12-20 15:13:55 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 00 : Lightship****Lightship condition****WEIGHT and DISPLACEMENT and WATERPLANE STATUS**

USK draft: 0.550 @ 55.00f, 0.550 @ 27.50f, 0.550 @ 0.00

Trim: zero, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WEIGHT	423.35	27.500f	0.000	3.000	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	423.35	27.500f	0.000	0.267

	Righting Arms:		0.000	0.000	
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	790.5	27.500f	0.000	388.36 40.934
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		8.10		29.68	385.63 38.201
Distances in METERS.-----			Moments in m.-MT.		

SUMMARY OF LOADING

0.0 Cu.M. (0%) WB

0.0 Cu.M. (0%) VOID

DRAFT AT FP (M) = 0.550
 DRAFT AT MS (M) = 0.550
 DRAFT AT AP (M) = 0.550
 DRAFT AT FWD MARK (M) = 0.550
 DRAFT AT AFT MARK (M) = 0.550

17-12-20 15:13:55 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 00 : Lightship****Lightship condition****HYDROSTATIC PROPERTIES**

No Trim, No Heel, VCG = 3.000

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
 Draft---Weight(MT)---LCB---VCB---cm---LCF---cm trim---GML---GMT
 0.550 423.35 27.500f 0.267 8.10 27.500f 29.68 385.63 38.201
 Distances in METERS.---Specific Gravity = 1.025.---Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.550 @ 55.00f, 0.550 @ 27.50f, 0.550 @ 0.00

Trim: zero, Heel: zero

Part-----LPA*Sf-----HCP-----Arm-----Pressure-----Moment
 HULL 131.6 1.262 1.537 0.05140 10.40
 Distances in METERS.---Pressure in MT/Sqm.---Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 27.500f TCG = 0.000 VCG = 3.000

Origin Depth---	Degrees of Trim---	Heel---	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. 50% DeckImm Area---Angle
0.538	0.00	0.00	423.24	0.000 -0.025	0.0000 14.493
0.538	0.00	0.02s	423.35	0.000 0.000	-0.0000 14.477
0.510	0.00a	5.02s	423.35	0.000 3.055	0.1333 9.477
0.299	0.00a	10.02s	423.39	0.000 4.217	0.4611 4.477
0.002	0.00a	14.49s	423.22	0.000 4.552	0.8081 0.000
-0.037	0.00	15.02s	423.41	0.000 4.572	0.8497 -0.523
-0.408	0.00	19.66s	423.35	0.000 4.648	1.2257 -5.162
-0.438	0.00	20.02s	423.35	0.000 4.647	1.2549 -5.523
-0.880	0.00	25.02s	423.35	0.000 4.574	1.6584 -10.523
-1.346	0.00	30.02s	423.59	0.000 4.405	2.0508 -15.523
-1.813	0.00	35.02s	423.56	0.000 4.133	2.4241 -20.523
-2.265	0.00	40.02s	423.54	0.000 3.778	2.7699 -25.523
-2.701	0.00	45.02s	423.53	0.000 3.364	3.0820 -30.523
-3.115	0.00	50.02s	423.53	0.000 2.906	3.3559 -35.523
-3.506	0.00	55.02s	423.53	0.000 2.413	3.5882 -40.523
-3.871	0.00	60.02s	423.53	0.000 1.892	3.7763 -45.523

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: No tank loads are present.

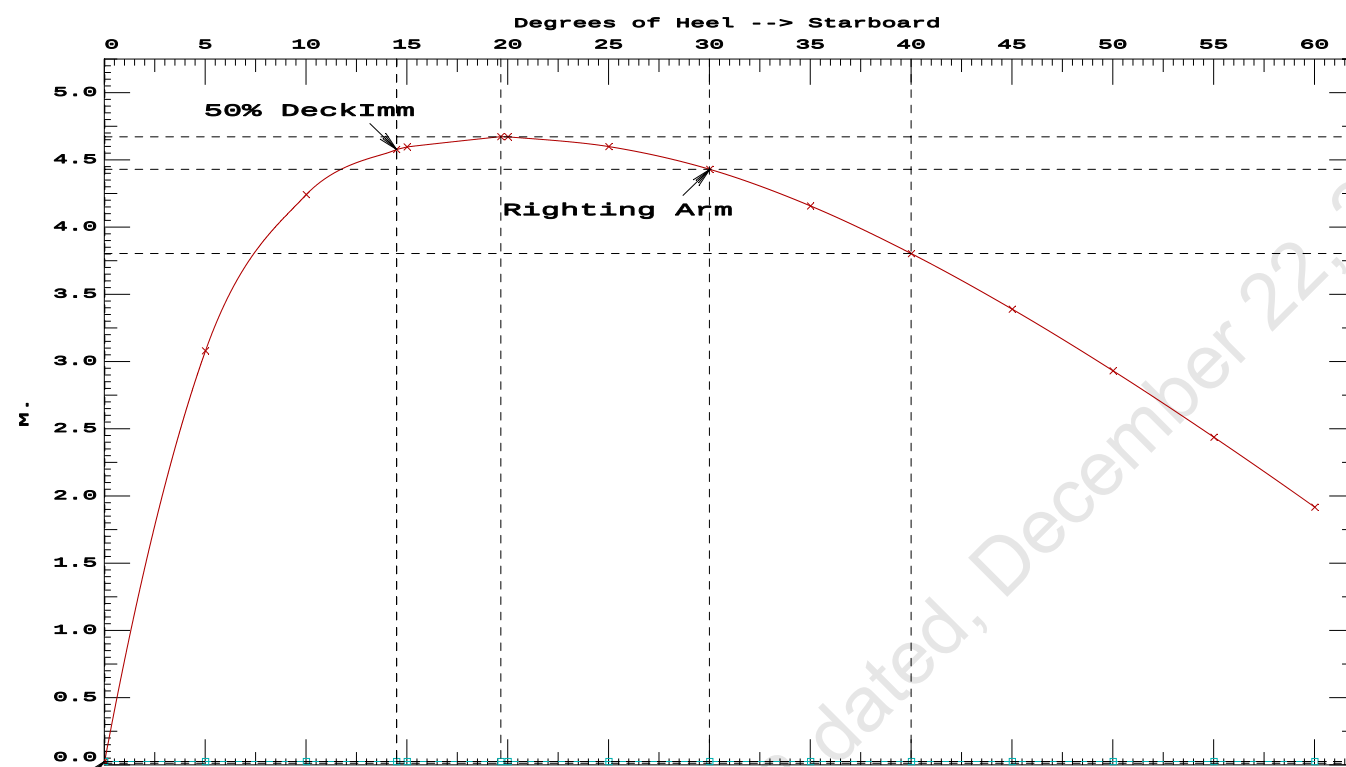
Note: The Residual Righting Arms shown above are in excess of the
 wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 10.40 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained
 (1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.2257 P
 (2) Angle from Equilibrium to RAZero or Flood > 20.00 deg LARGE
 (3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 14.48 P

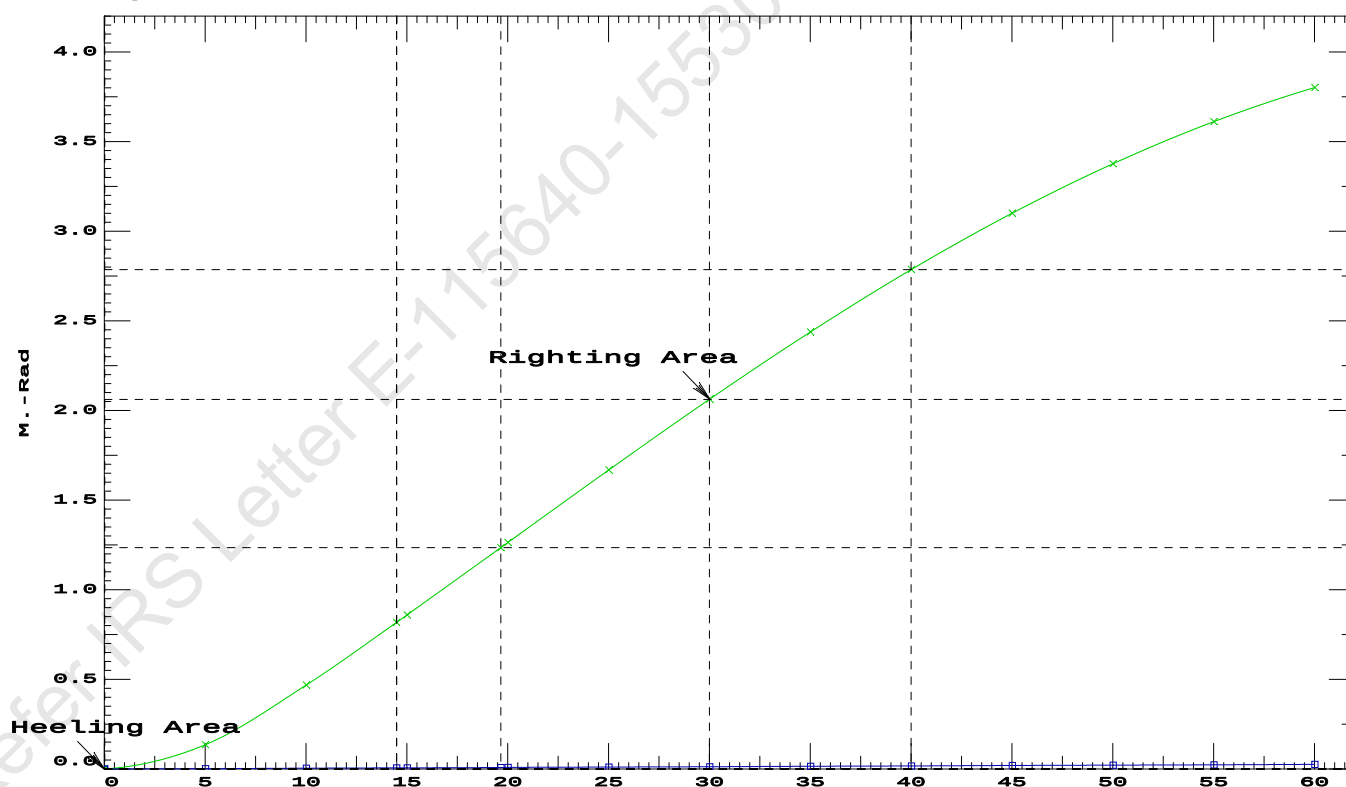
17-12-20 15:13:55 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 00 : LightshipLightship condition

Heeling Arm



17-12-20 15:14:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 1 : Lightship + Crane + Ballast

Condition Graphic - Draft: 0.880 @ 55.000f, 0.883 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

19 N06_WBTK.P

20 N06_WBTK.S

17-12-20 15:14:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 1 : Lightship + Crane + Ballast****WEIGHT and DISPLACEMENT and WATERPLANE STATUS**

USK draft: 0.880 @ 55.00f, 0.881 @ 27.50f, 0.883 @ 0.00

Trim: 0.000/55.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	423.35	27.500f	0.000	3.000	
Crane	150.00	7.500f	0.000	5.200	
Total Fixed----->	573.35	22.268f	0.000	3.576	
Load-----	SpGr-----	Weight(MT)	LCG	TCG	VCG
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s
Total Tanks----->			121.76	52.074f	0.000
Total Weight----->			695.11	27.489f	0.000
			Displ(MT)	LCB	TCB
HULL	1.025		695.01	27.490f	0.000

Righting Arms:			0.001f	0.000	
Part-----	SpGr-----	WPA	LCF	TCF	BML
Total Waterplane---->	1.025	810.5	27.520f	0.000	254.96
		MT/cm	m.	MT/cm	GML
		8.31		31.86	252.13
Distances in METERS.-----		Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

118.8 Cu.M. (6%) WB

0.0 Cu.M. (0%) VOID

150.00 MT Crane

DRAFT AT FP (M) = 0.880
 DRAFT AT MS (M) = 0.881
 DRAFT AT AP (M) = 0.883
 DRAFT AT FWD MARK (M) = 0.880
 DRAFT AT AFT MARK (M) = 0.882

17-12-20 15:14:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 1 : Lightship + Crane + Ballast****HYDROSTATIC PROPERTIES**

Trim: 0.000/55.000, No Heel, VCG = 3.264

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
 Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
 0.881 695.01 27.490f 0.437 8.31 27.520f 31.86 252.13 22.492
 Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.
 Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 139.3 m.-MT

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.880 @ 55.00f, 0.881 @ 27.50f, 0.883 @ 0.00

Trim: 0.000/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	115.2	1.086	1.523	0.05140	9.02
CRANE.C	36.2	4.380	4.817	0.05140	8.95
Total wind heeling moment to starboard----->					17.97
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.489f TCG = 0.000 VCG = 3.264

Free Surface Adjustment: 0.200

Adjusted CG: LCG = 27.489f TCG = 0.000 VCG = 3.464

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle	
0.871 0.00 0.00	695.06	0.000 -0.026	0.0000	9.371
0.871 0.00a 0.02s	695.11	0.000 0.000	-0.0000	9.347
0.857 0.01a 5.02s	695.13	0.000 1.929	0.0842	4.347
0.783 0.01a 9.37s	695.11	0.000 3.215	0.2816	0.000
0.761 0.01a 10.02s	695.12	0.000 3.328	0.3189	-0.653
0.516 0.01a 15.02s	695.11	0.000 3.843	0.6374	-5.653
0.187 0.01a 20.02s	695.11	0.000 4.011	0.9827	-10.653
0.149 0.01a 20.57s	695.11	0.000 4.013	1.0210	-11.200
-0.161 0.01a 25.02s	695.11	0.000 3.924	1.3307	-15.653
-0.508 0.01a 30.02s	694.87	0.000 3.685	1.6640	-20.653
-0.851 0.01a 35.02s	694.91	0.000 3.360	1.9721	-25.653
-1.187 0.01a 40.02s	694.93	0.000 2.979	2.2491	-30.653
-1.514 0.01a 45.02s	694.94	0.000 2.556	2.4909	-35.653
-1.830 0.01a 50.02s	694.95	0.000 2.102	2.6943	-40.653
-2.131 0.01a 55.02s	695.11	0.000 1.623	2.8570	-45.653
-2.417 0.01a 60.02s	695.11	0.000 1.128	2.9772	-50.653
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 139.3 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 17.97 (constant)

17-12-20 15:14:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

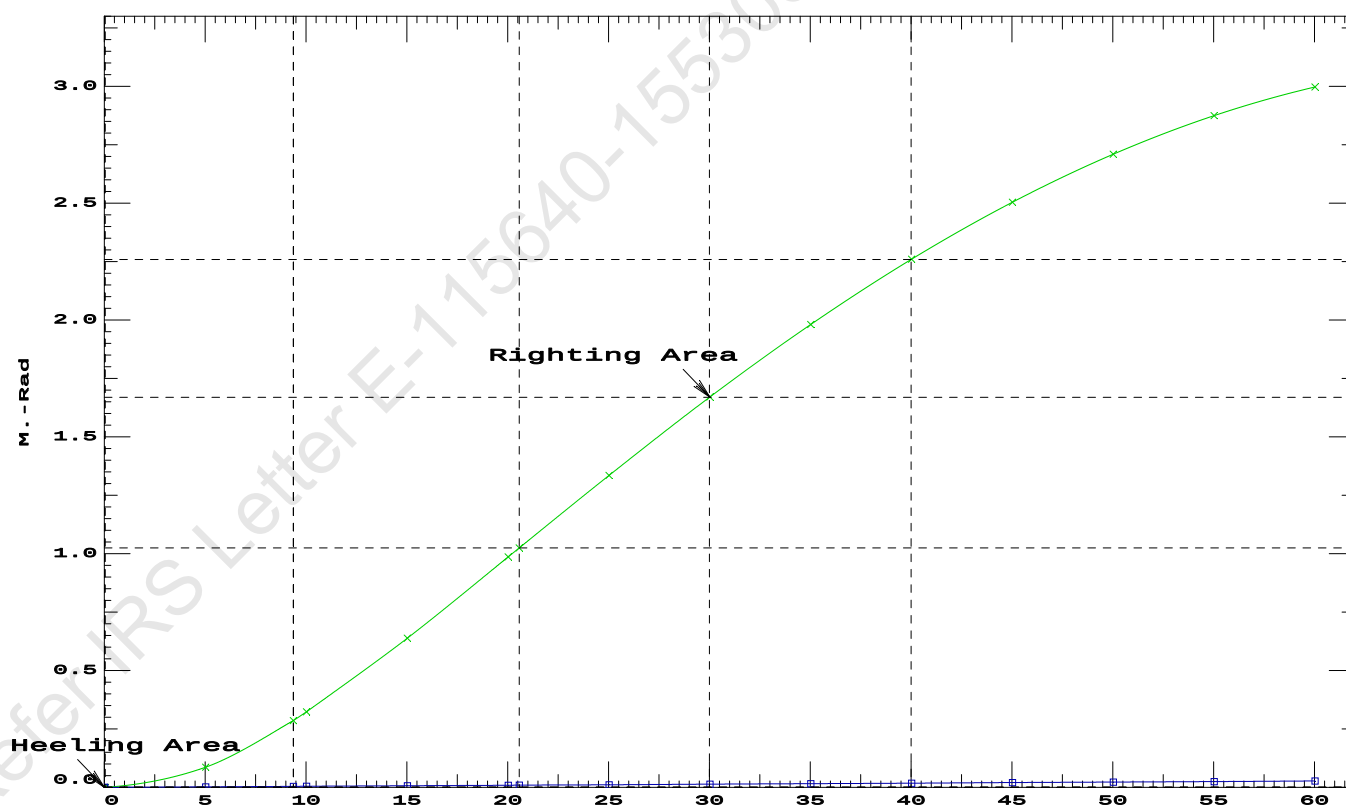
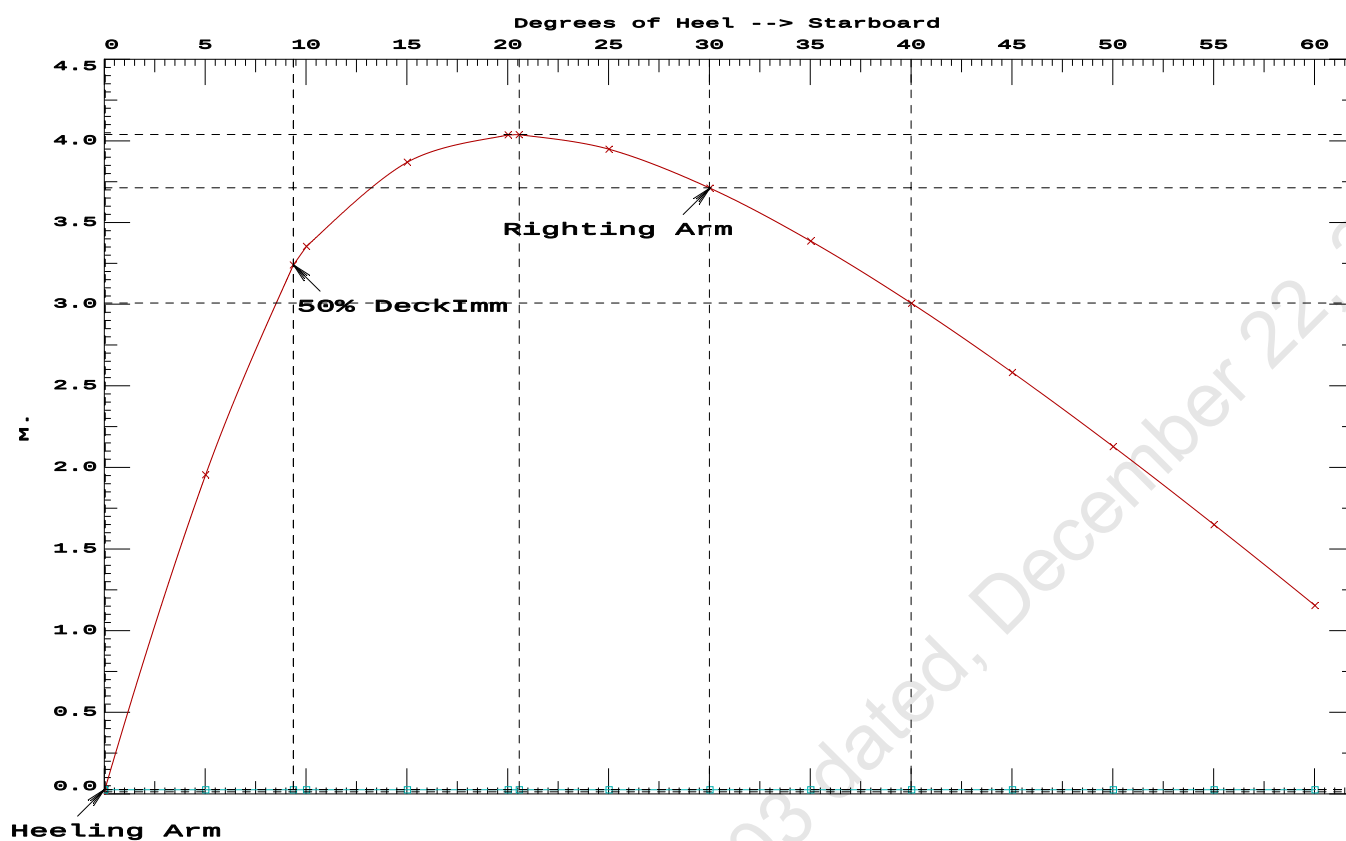
PONTOON**Condition 1 : Lightship + Crane + Ballast**

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.0210 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 9.35 P

17-12-20 15:14:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 1 : Lightship + Crane + Ballast

17-12-20 15:25:51 Cybermarine Technologies PTE. Ltd.

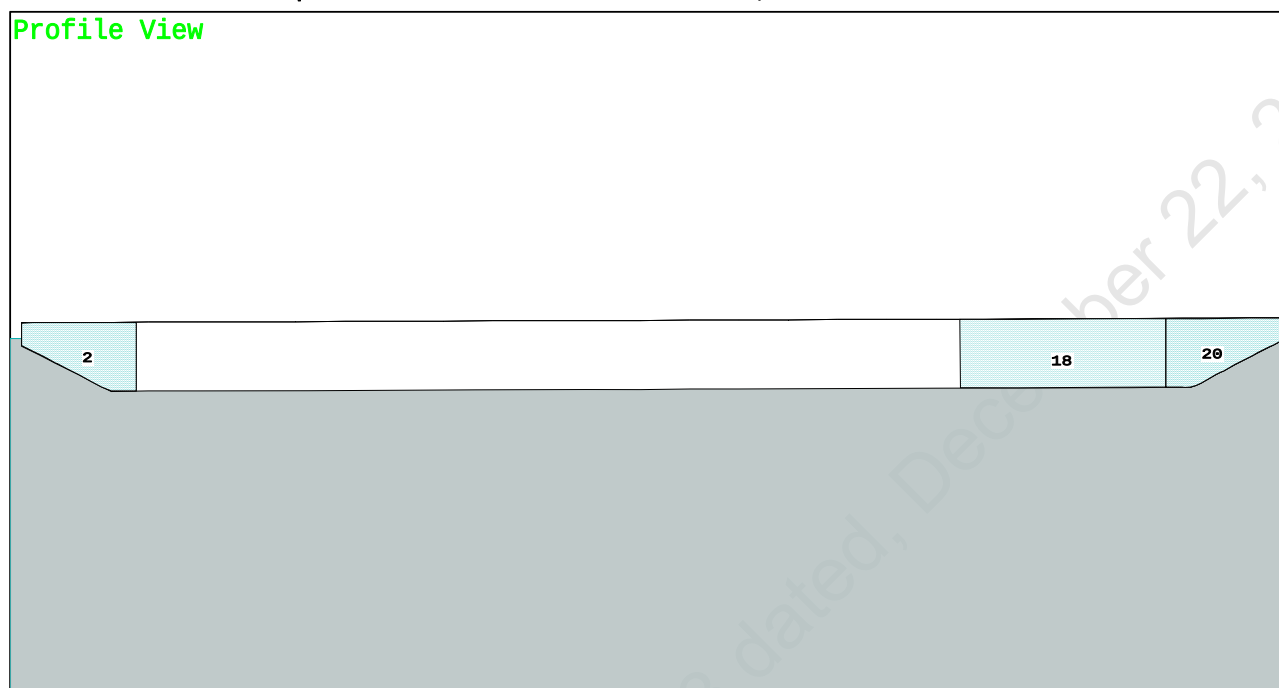
GHS 17.22

PONT00N

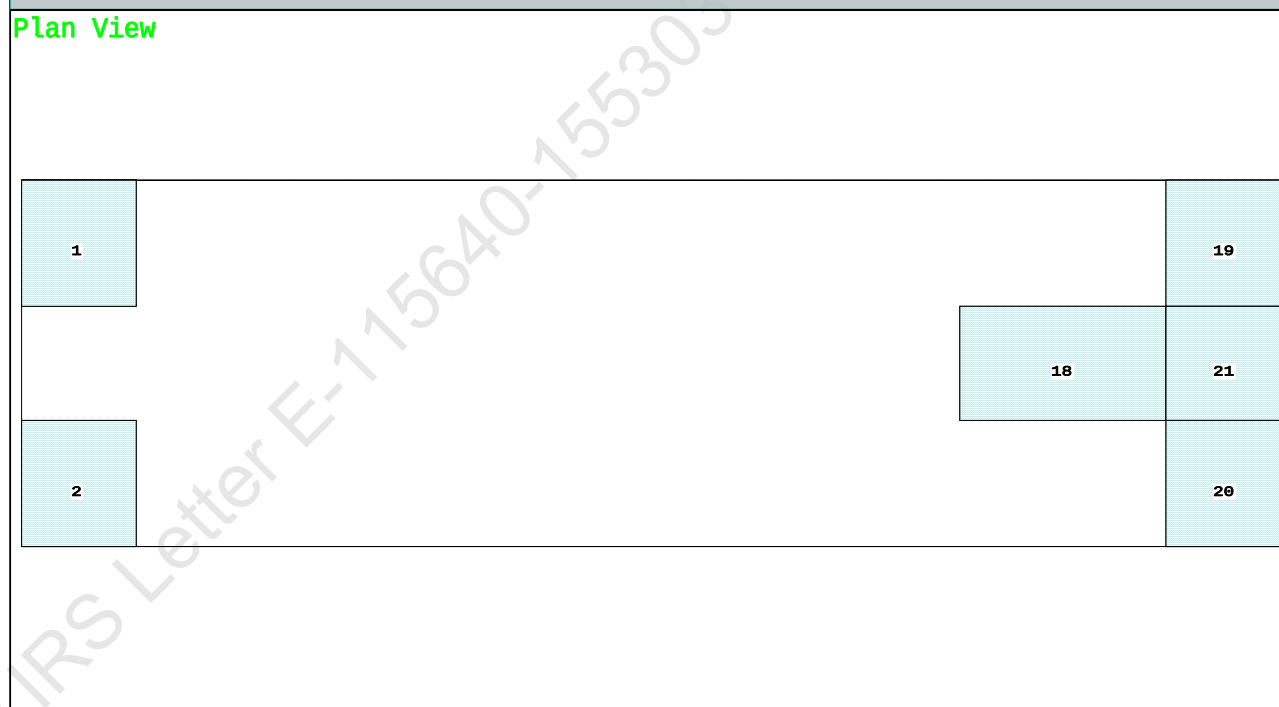
Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Aft

Condition Graphic - Draft: 2.110 @ 55.000f, 2.314 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

1 N01_WBTK.P

2 N01_WBTK.S

18 N05_WBTK.C

19 N06_WBTK.P

20 N06_WBTK.S

21 N06_WBTK.C

17-12-20 15:25:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.110 @ 55.00f, 2.212 @ 27.50f, 2.314 @ 0.00

Trim: Aft 0.205/55.000, Heel: zero

Part-----			Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	32.695	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,420.34	24.232f	0.000	4.865	
	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,855.34	27.055f	0.000	4.124	
			Displ(MT)	LCB	TCB	VCB	
HULL	1.025		1,855.31	27.044f	0.000	1.131	

Righting Arms:				0.000	0.000		
Part-----	SpGr		WPA	LCF	TCF	BML	BMT
Total Waterplane----->	1.025		880.6	27.492f	0.000	122.48	10.162
			MT/cm	m.	MT/cm	GML	GMT
			9.03		40.31	119.49	7.170
Distances in METERS.-----				Moments in m.-MT			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

424.4 Cu.M. (21%) WB

0.0 Cu.M. (0%) VOID

150.00 MT Crane

51.30 MT Crane Load

795.69 MT Deck Cargo

DRAFT AT FP (M) = 2.110
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.314
 DRAFT AT FWD MARK (M) = 2.124
 DRAFT AT AFT MARK (M) = 2.299

HYDROSTATIC PROPERTIES

Trim: Aft 0.205/55.000, No Heel, VCG = 4.124

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,855.31	27.044f	1.131	9.03
				27.492f
				40.31
				119.49
				7.170

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 425.0 m.-MT

17-12-20 15:25:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.110 @ 55.00f, 2.212 @ 27.50f, 2.314 @ 0.00

Trim: Aft 0.205/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	44.5	0.407	1.483	0.05140	3.39
CRANE.C	36.2	2.976	4.051	0.05140	7.53
DECKCARGO.C	70.6	1.817	2.892	0.05140	10.49
Total wind heeling moment to starboard----->					21.41
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.055f TCG = 0.000 VCG = 4.124

Free Surface Adjustment: 0.229

Adjusted CG: LCG = 27.056f TCG = 0.000 VCG = 4.353

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle
2.302	0.21a	0.00	1,855.3	0.000 -0.011 0.0000 2.521
2.302	0.21a	0.09s	1,855.3	0.000 0.000 -0.0000 2.433
2.300	0.21a	2.52s	1,855.3	0.000 0.308 0.0065 0.000
2.294	0.22a	5.09s	1,855.3	0.000 0.629 0.0275 -2.567
2.387	0.30a	10.09s	1,855.3	0.000 0.977 0.0996 -7.567
2.486	0.34a	12.61s	1,855.3	0.000 1.001 0.1436 -10.089
2.601	0.39a	15.09s	1,855.3	0.000 0.978 0.1863 -12.567
2.862	0.47a	20.09s	1,855.3	0.000 0.847 0.2675 -17.567
3.140	0.58a	25.09s	1,855.3	0.000 0.640 0.3329 -22.567
3.401	0.70a	30.09s	1,855.3	0.000 0.385 0.3779 -27.567
3.638	0.81a	35.09s	1,855.3	0.000 0.106 0.3995 -32.567
3.718	0.85a	36.92s	1,855.3	0.000 0.000 0.4012 -34.400
3.847	0.92a	40.09s	1,855.3	0.000 -0.186 0.3961 -37.567
4.029	1.02a	45.09s	1,855.3	0.000 -0.482 0.3670 -42.567
4.179	1.12a	50.09s	1,855.3	0.000 -0.780 0.3119 -47.567
4.297	1.20a	55.09s	1,855.3	0.000 -1.074 0.2310 -52.567
4.380	1.28a	60.09s	1,855.3	0.000 -1.363 0.1246 -57.567
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 425.0 m.-MT was applied to artificially modify the CG.

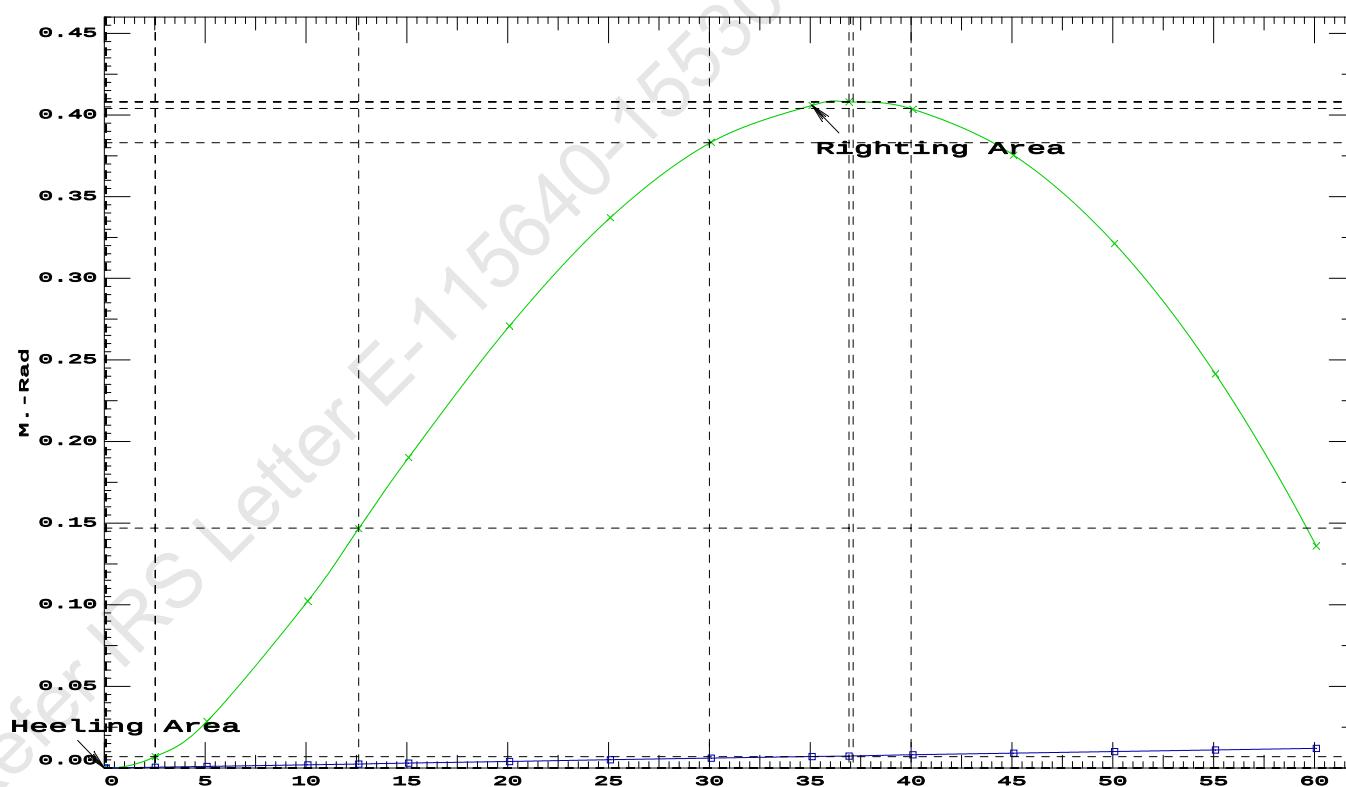
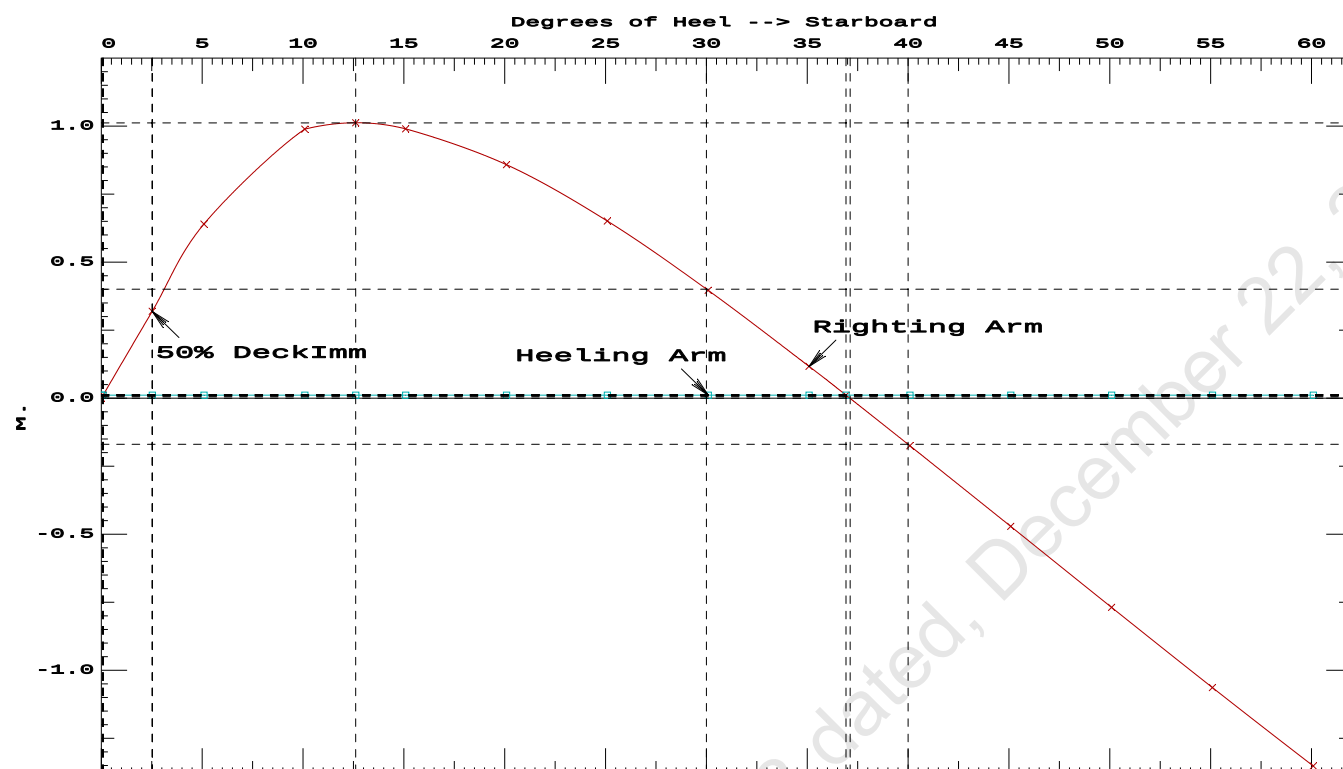
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.41 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1436 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	36.83 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.43 P

17-12-20 15:25:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Aft

17-12-20 15:25:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo****Crane Lifting 51.30T @ 12m Radius Aft****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 27.055f TCG = 0.000 VCG = 4.124

Free Surface Adjustment: 0.229

Adjusted CG: LCG = 27.056f TCG = 0.000 VCG = 4.353

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area
2.302	0.21a	0.00s	1,855.3	0.000	0.000	0.0000
2.294	0.22a	5.00s	1,855.3	0.000	0.629	0.0274
2.384	0.30a	10.00s	1,855.3	0.000	0.986	0.0999
2.486	0.34a	12.61s	1,855.3	0.000	1.011	0.1458
2.597	0.38a	15.00s	1,855.3	0.000	0.990	0.1876
2.857	0.47a	20.00s	1,855.3	0.000	0.860	0.2698
3.135	0.58a	25.00s	1,855.3	0.000	0.654	0.3365
3.397	0.69a	30.00s	1,855.3	0.000	0.400	0.3828
3.634	0.81a	35.00s	1,855.3	0.000	0.121	0.4058
3.726	0.86a	37.11s	1,855.3	0.000	0.000	0.4080
3.844	0.92a	40.00s	1,855.3	0.000	-0.170	0.4037
4.026	1.02a	45.00s	1,855.3	0.000	-0.467	0.3760
4.177	1.12a	50.00s	1,855.3	0.000	-0.764	0.3223
4.295	1.20a	55.00s	1,855.3	0.000	-1.059	0.2427
4.379	1.28a	60.00s	1,855.3	0.000	-1.347	0.1377

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 425.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

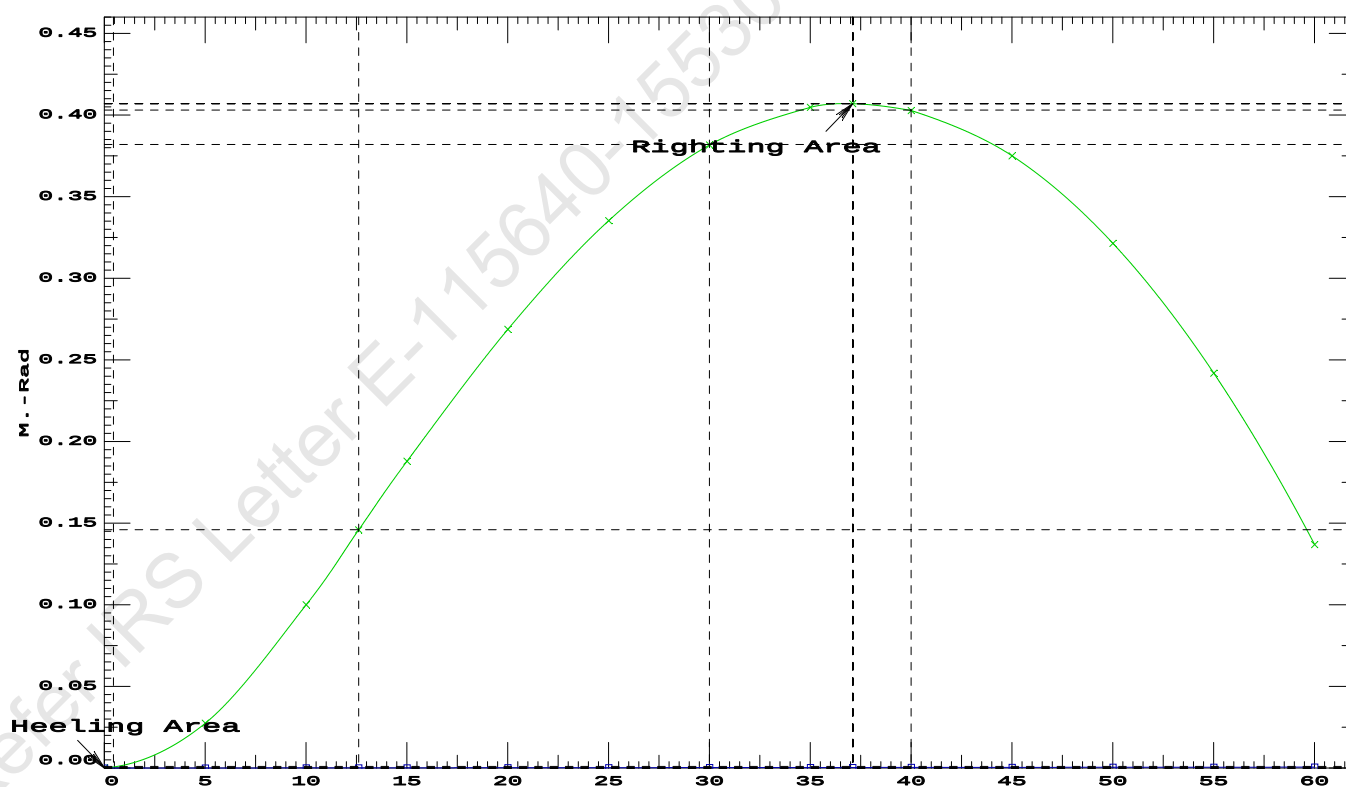
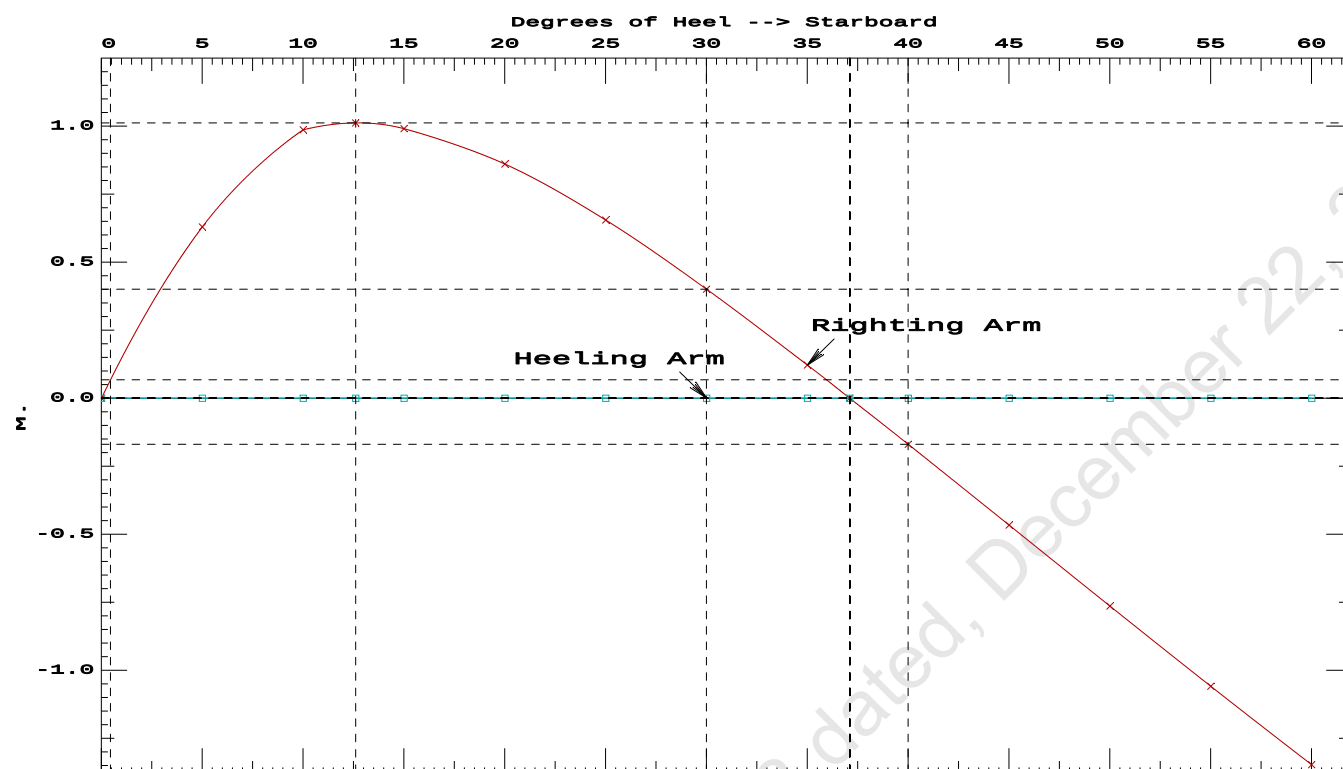
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	187644 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1169.98 P

17-12-20 15:25:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Aft

17-12-20 15:25:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo**Crane Lifting 51.30T @ 12m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.110 @ 55.00f, 2.212 @ 27.50f, 2.314 @ 0.00

Trim: Aft 0.205/55.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	32.695	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,420.34	24.232f	0.000	4.865	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,855.34	27.055f	0.000	4.124	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.34	27.044f	0.001	1.131	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	880.6	27.492f	0.000	122.48	10.162
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.03		40.31	119.49	7.170
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

17-12-20 15:25:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.256 @ 55.00f, 2.155 @ 27.50f, 2.055 @ 0.00

Trim: Fwd 0.201/55.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,369.04	25.309f	0.000	3.822	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,804.04	27.952f	0.000	3.311	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,804.04	27.960f	0.001	1.102	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		880.5	27.507f	0.000	125.94	10.449
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.03		40.58	123.73	8.240
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 25.309f TCG = 0.000 VCG = 3.822

Origin	Degrees of	Displacement	Residual Arms	Res.	
Depth---	Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area
2.042	0.21f	0.00s	1,804.0	0.000	0.0000
2.043	0.21f	0.07s	1,804.0	0.000	0.0000
2.030	0.21f	5.00s	1,804.0	0.000	0.0324
2.031	0.28f	10.00s	1,804.2	0.000	0.1212
2.146	0.36f	15.00s	1,804.0	0.000	0.2391
2.182	0.38f	16.19s	1,804.0	0.000	0.2678
2.308	0.45f	20.00s	1,804.1	0.000	0.3596

17-12-20 15:25:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

2.457	0.56f	25.00s	1,804.0	0.000	1.232	0.4733
2.585	0.67f	30.00s	1,804.0	0.000	1.054	0.5735

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1369.04 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

17-12-20 15:25:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 25.309f TCG = 0.000 VCG = 3.822

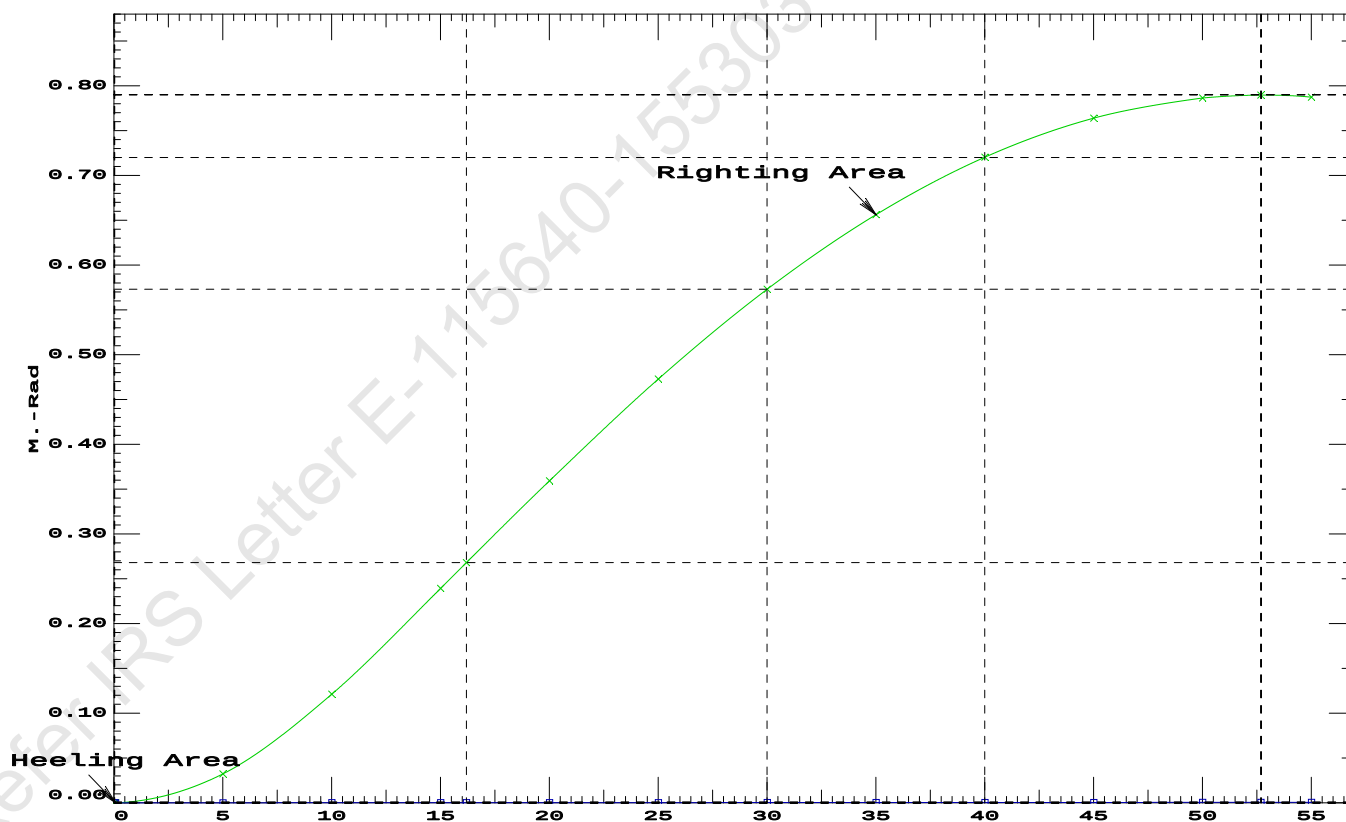
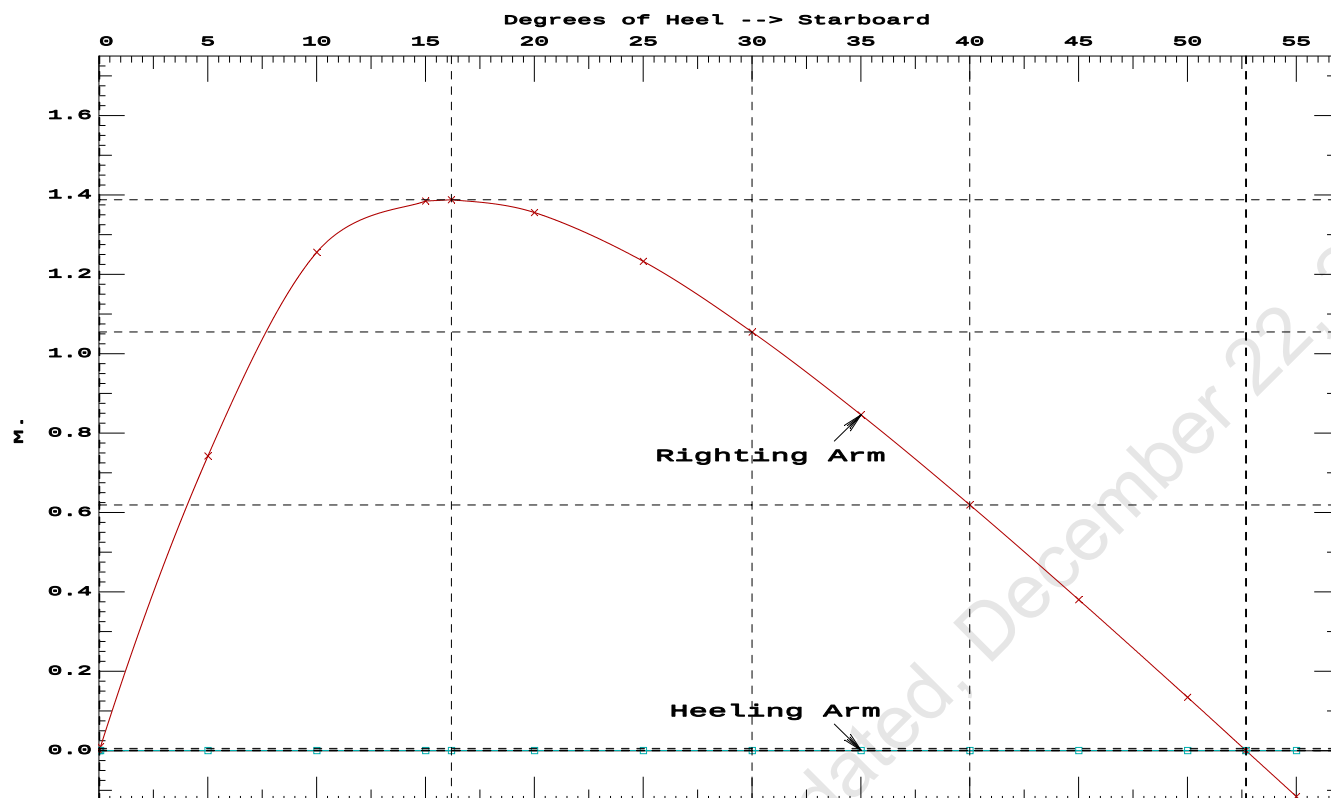
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.042	0.21f	0.00s	1,804.0	0.000	0.000	0.0000
2.043	0.21f	0.07s	1,804.0	0.000	0.010	0.0000
2.030	0.21f	5.00s	1,804.0	0.000	0.742	0.0324
2.031	0.28f	10.00s	1,804.2	0.000	1.255	0.1212
2.146	0.36f	15.00s	1,804.0	0.000	1.384	0.2391
2.182	0.38f	16.19s	1,804.0	0.000	1.387	0.2678
2.308	0.45f	20.00s	1,804.1	0.000	1.355	0.3596
2.457	0.56f	25.00s	1,804.0	0.000	1.232	0.4733
2.585	0.67f	30.00s	1,804.0	0.000	1.054	0.5735
2.692	0.78f	35.00s	1,804.0	0.000	0.845	0.6566
2.777	0.88f	40.00s	1,804.0	0.000	0.618	0.7206
2.841	0.98f	45.00s	1,804.0	0.000	0.380	0.7642
2.882	1.07f	50.00s	1,804.0	0.000	0.134	0.7867
2.896	1.12f	52.69s	1,804.0	0.000	0.000	0.7898
2.902	1.16f	55.00s	1,804.0	0.000	-0.116	0.7875

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1369.04 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 52.62 P



17-12-20 15:26:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

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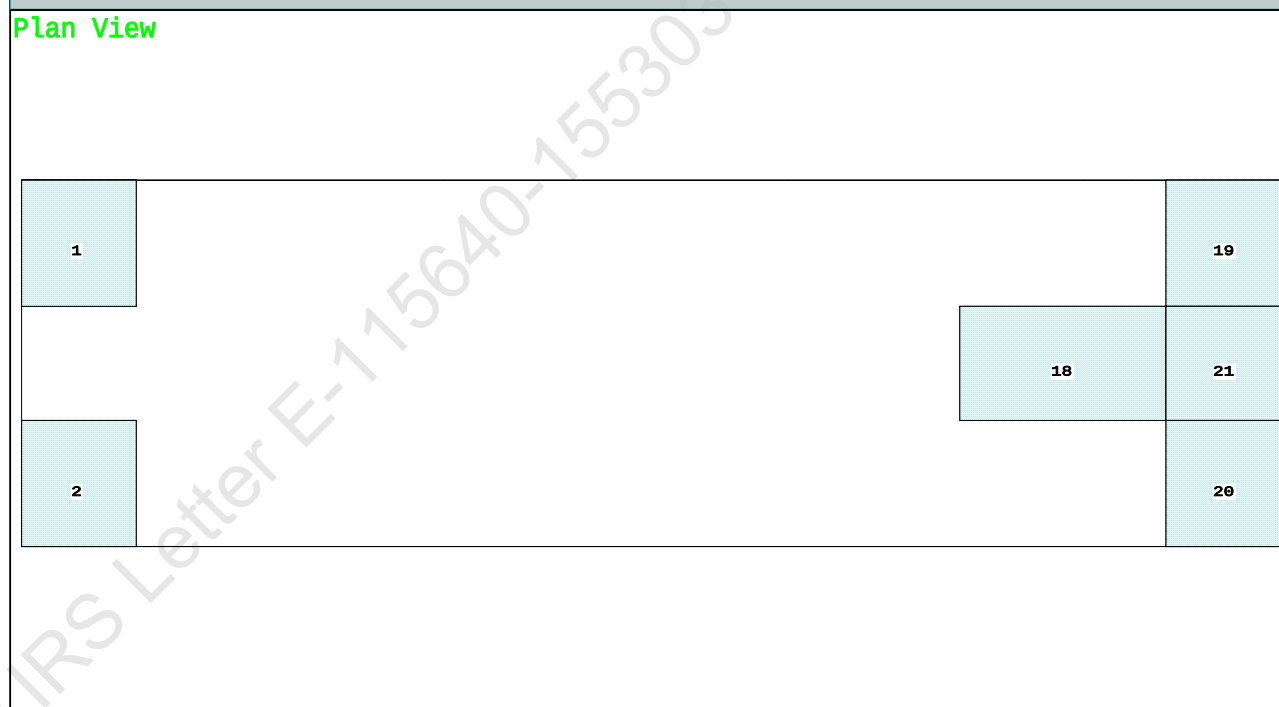
Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Stbd

Condition Graphic - Draft: 2.186 @ 55.000f, 2.238 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

1 N01_WBTK.P

2 N01_WBTK.S

18 N05_WBTK.C

19 N06_WBTK.P

20 N06_WBTK.S

21 N06_WBTK.C

17-12-20 15:26:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.186 @ 55.00f, 2.212 @ 27.50f, 2.238 @ 0.00

Trim: Aft 0.052/55.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	32.695	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,420.34	24.665f	0.000	4.865	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,855.34	27.387f	0.000	4.124	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.31	27.384f	0.000	1.131	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	880.6	27.497f	0.000	122.50	10.164
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.03		40.31	119.51	7.171
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

424.4 Cu.M. (21%) WB

0.0 Cu.M. (0%) VOID

150.00 MT Crane

51.30 MT Crane Load

795.69 MT Deck Cargo

DRAFT AT FP (M) = 2.186
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.238
 DRAFT AT FWD MARK (M) = 2.190
 DRAFT AT AFT MARK (M) = 2.234

HYDROSTATIC PROPERTIES

Trim: Aft 0.052/55.000, No Heel, VCG = 4.124

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,855.31	27.384f	1.131	9.03
				27.497f
				40.31
				119.51
				7.171

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 425.0 m.-MT

17-12-20 15:26:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo****Crane Lifting 51.30T @ 12m Radius Stbd****HEELING MOMENT specification****Lateral Plane Method****Wind pressure toward starboard**

USK draft: 2.186 @ 55.00f, 2.212 @ 27.50f, 2.238 @ 0.00

Trim: Aft 0.052/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	44.5	0.405	1.480	0.05140	3.39
CRANE.C	36.2	3.031	4.106	0.05140	7.63
DECKCARGO.C	70.6	1.804	2.879	0.05140	10.44
Total wind heeling moment to starboard----->					21.46
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.387f TCG = 0.000 VCG = 4.124

Free Surface Adjustment: 0.229

Adjusted CG: LCG = 27.387f TCG = 0.000 VCG = 4.353

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle
2.226	0.05a	0.00	1,855.3	0.000 -0.011 0.0000 2.799
2.226	0.05a	0.09s	1,855.3	0.000 0.000 -0.0000 2.710
2.223	0.05a	2.80s	1,855.3	0.000 0.343 0.0081 -0.000
2.216	0.06a	5.09s	1,855.3	0.000 0.629 0.0275 -2.290
2.278	0.08a	10.09s	1,855.4	0.000 0.981 0.0998 -7.290
2.362	0.09a	12.62s	1,855.3	0.000 1.004 0.1440 -9.820
2.463	0.10a	15.09s	1,855.3	0.000 0.982 0.1868 -12.290
2.693	0.12a	20.09s	1,855.3	0.000 0.850 0.2682 -17.290
2.933	0.15a	25.09s	1,855.3	0.000 0.644 0.3340 -22.290
3.152	0.18a	30.09s	1,855.3	0.000 0.390 0.3795 -27.290
3.348	0.21a	35.09s	1,855.3	0.000 0.110 0.4015 -32.290
3.417	0.22a	37.00s	1,855.3	0.000 0.000 0.4033 -34.201
3.519	0.24a	40.09s	1,855.3	0.000 -0.181 0.3985 -37.290
3.663	0.26a	45.09s	1,855.3	0.000 -0.478 0.3697 -42.290
3.779	0.29a	50.09s	1,855.3	0.000 -0.776 0.3150 -47.290
3.867	0.31a	55.09s	1,855.3	0.000 -1.071 0.2344 -52.290
3.924	0.33a	60.09s	1,855.3	0.000 -1.360 0.1283 -57.290
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 425.0 m.-MT was applied to artificially modify the CG.

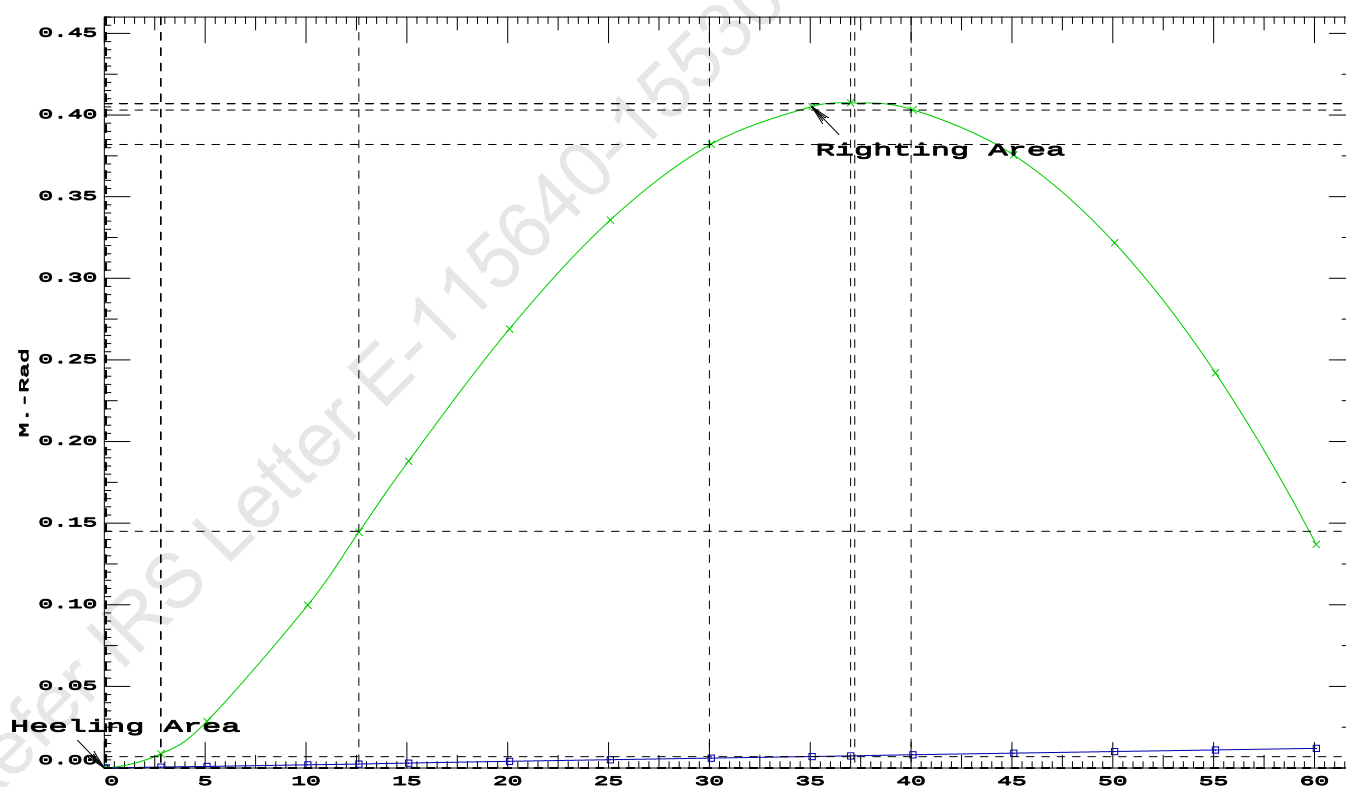
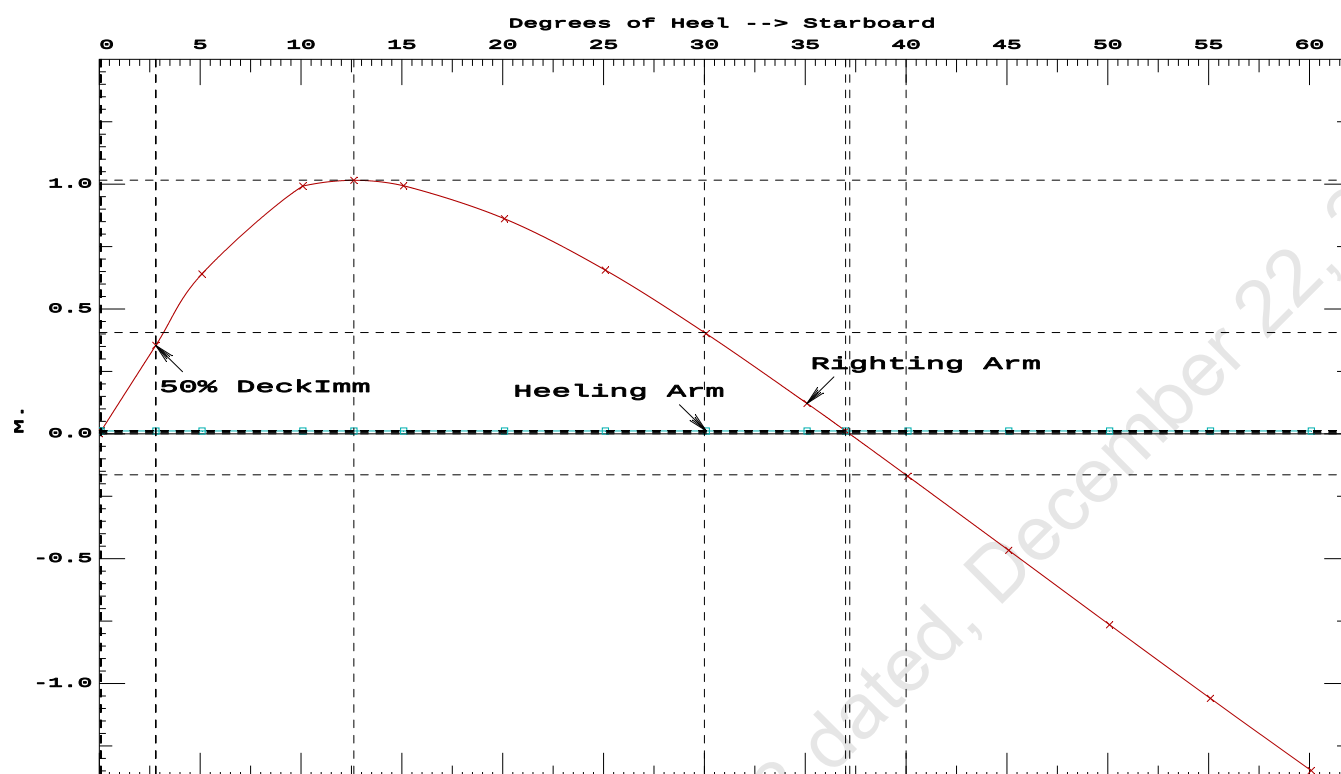
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.46 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1440 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	36.91 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.71 P

17-12-20 15:26:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Stbd

17-12-20 15:26:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo****Crane Lifting 51.30T @ 12m Radius Stbd****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 27.387f TCG = 0.000 VCG = 4.124

Free Surface Adjustment: 0.229

Adjusted CG: LCG = 27.387f TCG = 0.010p VCG = 4.353

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area
2.224	0.05a	2.54s	1,855.3	0.000	0.000	0.0000
2.224	0.06a	7.54s	1,855.3	0.000	0.567	0.0247
2.359	0.09a	12.54s	1,855.3	0.000	0.694	0.0820
2.572	0.11a	17.54s	1,855.3	0.000	0.618	0.1407
2.812	0.13a	22.54s	1,855.3	0.000	0.447	0.1878
3.043	0.16a	27.54s	1,855.3	0.000	0.212	0.2170
3.211	0.19a	31.52s	1,855.3	0.000	0.000	0.2245
3.252	0.19a	32.54s	1,855.3	0.000	-0.057	0.2240
3.436	0.22a	37.54s	1,855.3	0.000	-0.344	0.2067
3.593	0.25a	42.54s	1,855.3	0.000	-0.640	0.1638
3.724	0.27a	47.54s	1,855.3	0.000	-0.938	0.0950
3.826	0.30a	52.54s	1,855.3	0.000	-1.236	0.0002
3.899	0.32a	57.54s	1,855.3	0.000	-1.529	-0.1205
3.941	0.33a	62.54s	1,855.3	0.000	-1.814	-0.2664

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 425.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

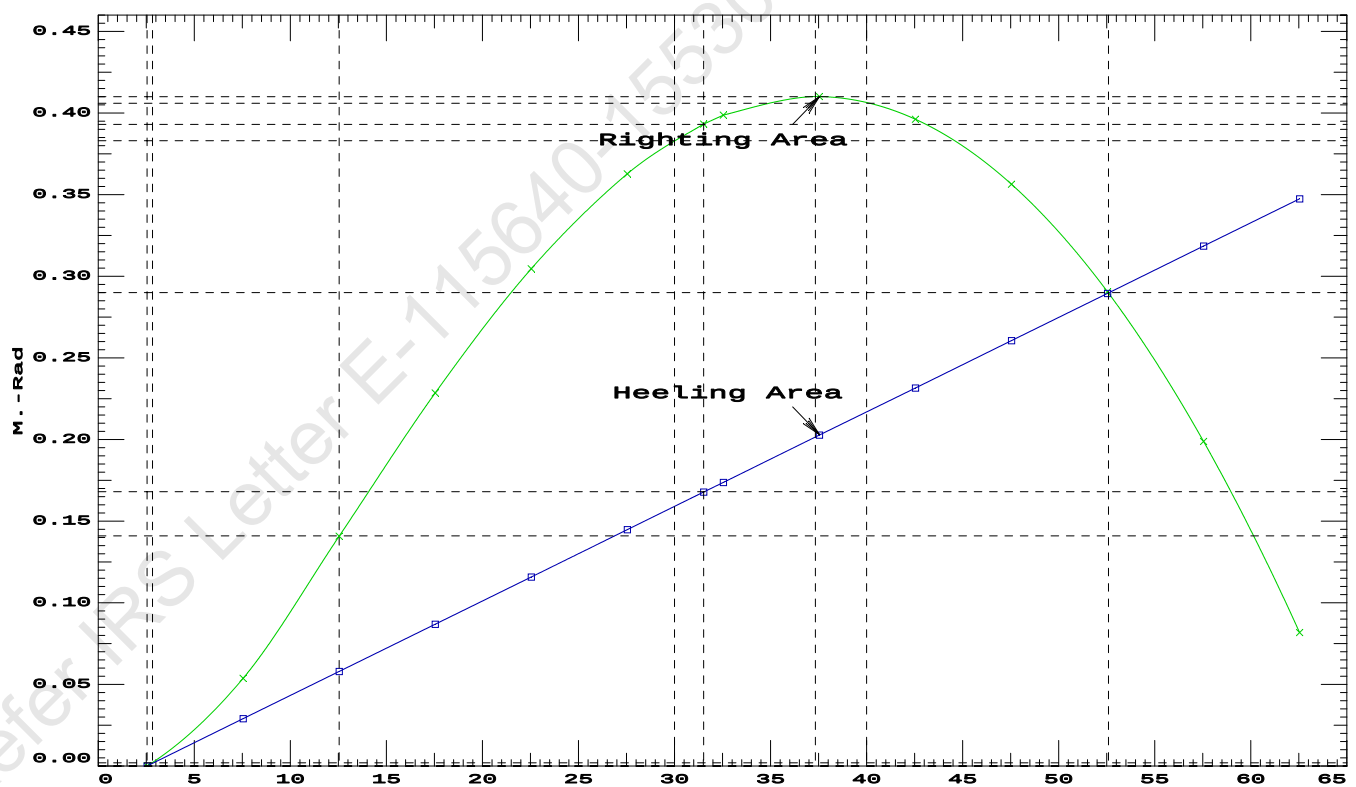
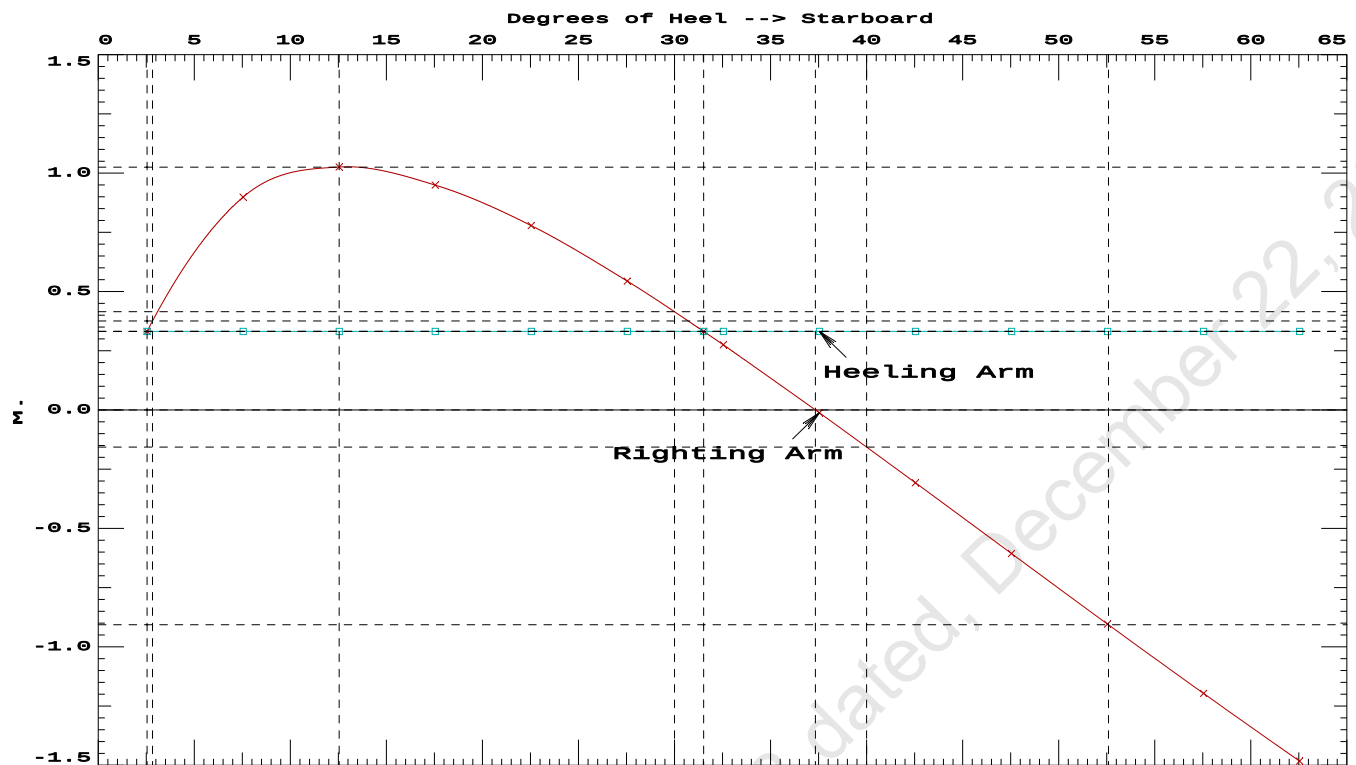
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	2.54 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	209.1 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	2.338 P

17-12-20 15:26:00 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 3 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Stbd

17-12-20 15:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

Condition 3 : Lightship + Load at Stbd + Ballast + Deck Cargo**Crane Lifting 51.30T @ 12m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.186 @ 55.00f, 2.212 @ 27.50f, 2.238 @ 0.00

Trim: Aft 0.052/55.000, Heel: Stbd 2.54 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	32.695	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,420.34	24.665f	0.000	4.865	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.073f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,855.34	27.387f	0.000	4.124	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.32	27.384f	0.465s	1.141	

	Righting Arms:			0.000	0.332s		
	External Arms:			0.000	0.332s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	879.9	27.508f	0.018s	121.96	10.151
			MT/cm				
			9.02				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.54

17-12-20 15:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.256 @ 55.00f, 2.155 @ 27.50f, 2.055 @ 0.00

Trim: Fwd 0.201/55.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,369.04	25.309f	0.000	3.822	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.073f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,804.04	27.952f	0.000	3.311	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,804.06	27.960f	0.001	1.102	

	Righting Arms:			0.001	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.001	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		880.5	27.507f	0.000	125.94	10.449
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.03		40.58	123.73	8.240
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 6749.97; t3: 30.00; t: 15.00
limmarg: 2755.63; t3: 15.00; t: 7.50
limmarg: 761.40; t3: 7.50; t: 3.75
limmarg: 112.61; t3: 3.70; t: 1.88
limmarg: -49.44; t3: 1.80; t: 0.94
limmarg: 13.43; t3: 2.70; t: 0.47
limmarg: -24.59; t3: 2.20; t: 0.23
limmarg: -10.31; t3: 2.40; t: 0.12
limmarg: -2.70; t3: 2.50; t: 0.06
limmarg: 5.21; t3: 2.60; t: 0.03
limmarg: 5.21; t3: 2.60; t: 0.01

Theta3 is 2.60 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 25.309f TCG = 0.000 VCG = 3.822

Origin	Degrees of	Displacement	Residual Arms	Res.		
Depth---	Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area	
2.039	0.21f	2.54s	1,804.0	0.000	-0.378	0.0000
2.043	0.21f	0.00s	1,804.0	0.000	0.000	-0.0084
2.039	0.21f	2.60p	1,804.0	0.000	0.388	0.0004
2.030	0.21f	5.00p	1,804.1	0.000	0.742	0.0241
2.031	0.28f	10.00p	1,804.2	0.000	1.256	0.1126
2.146	0.36f	15.00p	1,804.0	0.000	1.385	0.2306
2.182	0.38f	16.19p	1,804.0	0.000	1.389	0.2595

17-12-20 15:26:06 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

2.308	0.45f	20.00p	1,804.1	0.000	1.356	0.3511
2.457	0.56f	25.00p	1,804.0	0.000	1.234	0.4650
2.585	0.67f	30.00p	1,804.0	0.000	1.055	0.5653

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1369.04 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -2.5 deg to abs 2.6 > 1.000 1.052 P

17-12-20 15:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 25.309f TCG = 0.000 VCG = 3.822

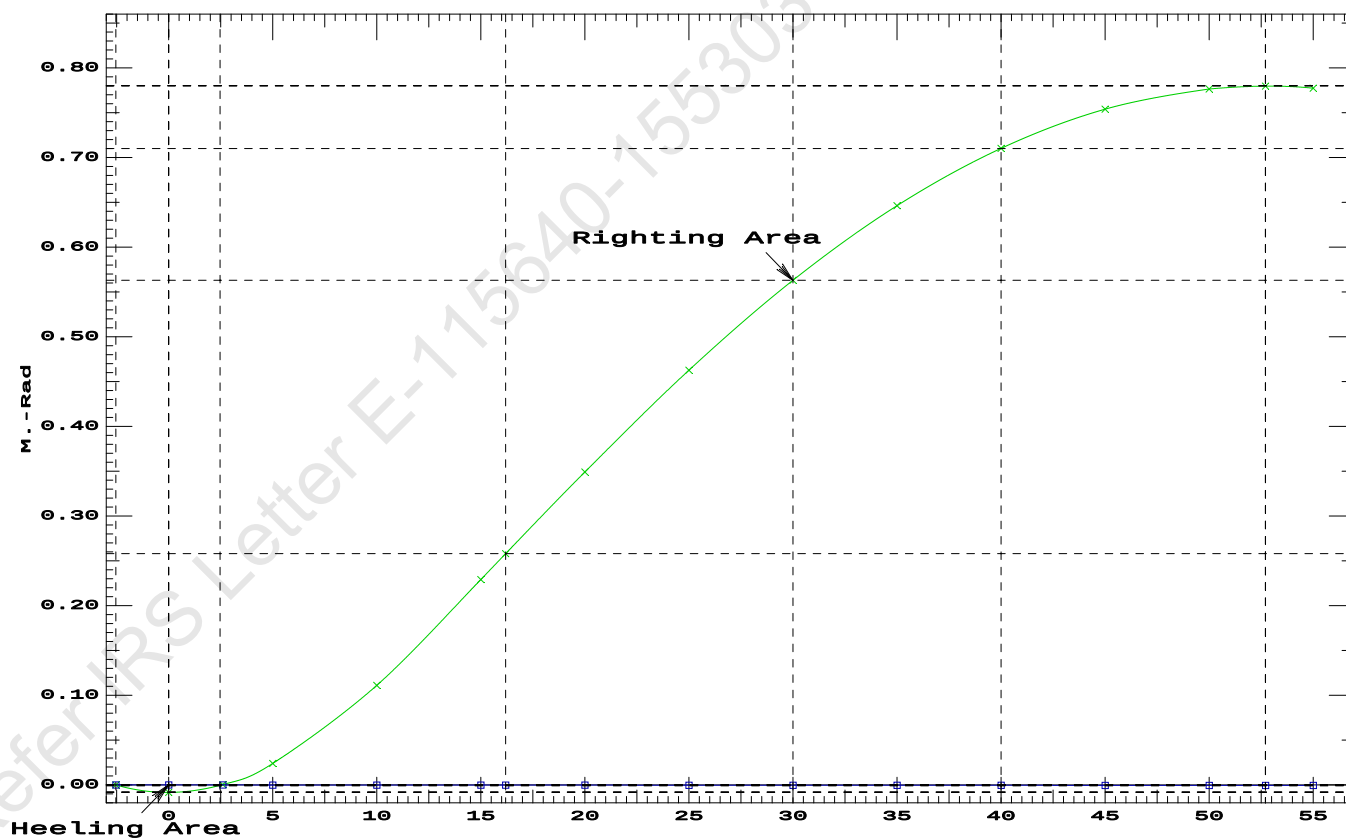
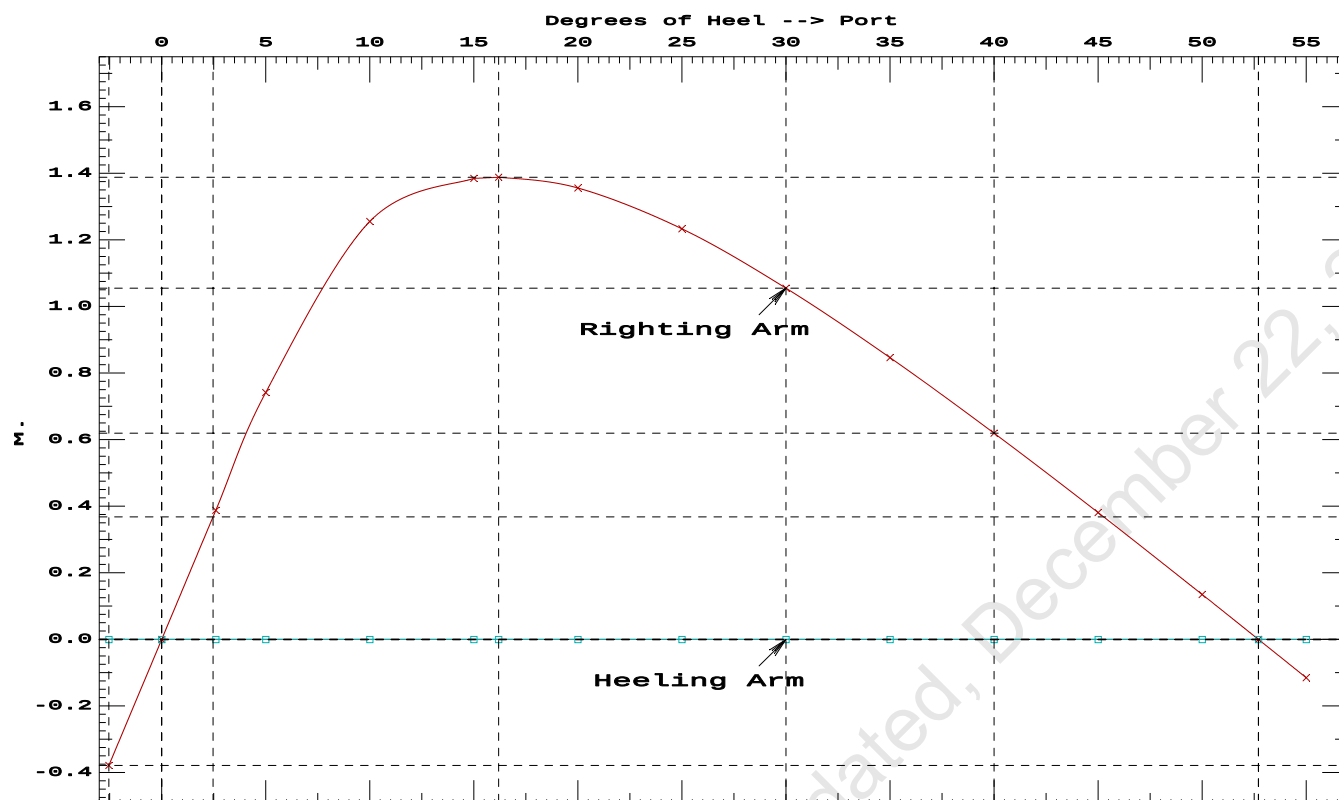
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.039	0.21f	2.54s	1,804.0	0.000	-0.378	0.0000
2.043	0.21f	0.00s	1,804.0	0.000	0.000	-0.0084
2.039	0.21f	2.60p	1,804.0	0.000	0.388	0.0004
2.030	0.21f	5.00p	1,804.1	0.000	0.742	0.0241
2.031	0.28f	10.00p	1,804.2	0.000	1.256	0.1126
2.146	0.36f	15.00p	1,804.0	0.000	1.385	0.2306
2.182	0.38f	16.19p	1,804.0	0.000	1.389	0.2595
2.308	0.45f	20.00p	1,804.1	0.000	1.356	0.3511
2.457	0.56f	25.00p	1,804.0	0.000	1.234	0.4650
2.585	0.67f	30.00p	1,804.0	0.000	1.055	0.5653
2.692	0.78f	35.00p	1,804.0	0.000	0.847	0.6485
2.777	0.88f	40.00p	1,804.0	0.000	0.620	0.7126
2.841	0.98f	45.00p	1,804.0	0.000	0.381	0.7564
2.882	1.07f	50.00p	1,804.0	0.000	0.135	0.7790
2.896	1.12f	52.71p	1,804.0	0.000	0.000	0.7822
2.902	1.16f	55.00p	1,804.0	0.000	-0.115	0.7799

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1369.04 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -2.5 deg to RAZero > 1.000 94.355 P
(2) Angle from abs 2.6 deg to RAZero > 20.00 deg 50.11 P



17-12-20 15:26:08 Cybermarine Technologies PTE. Ltd.

GHS 17.22

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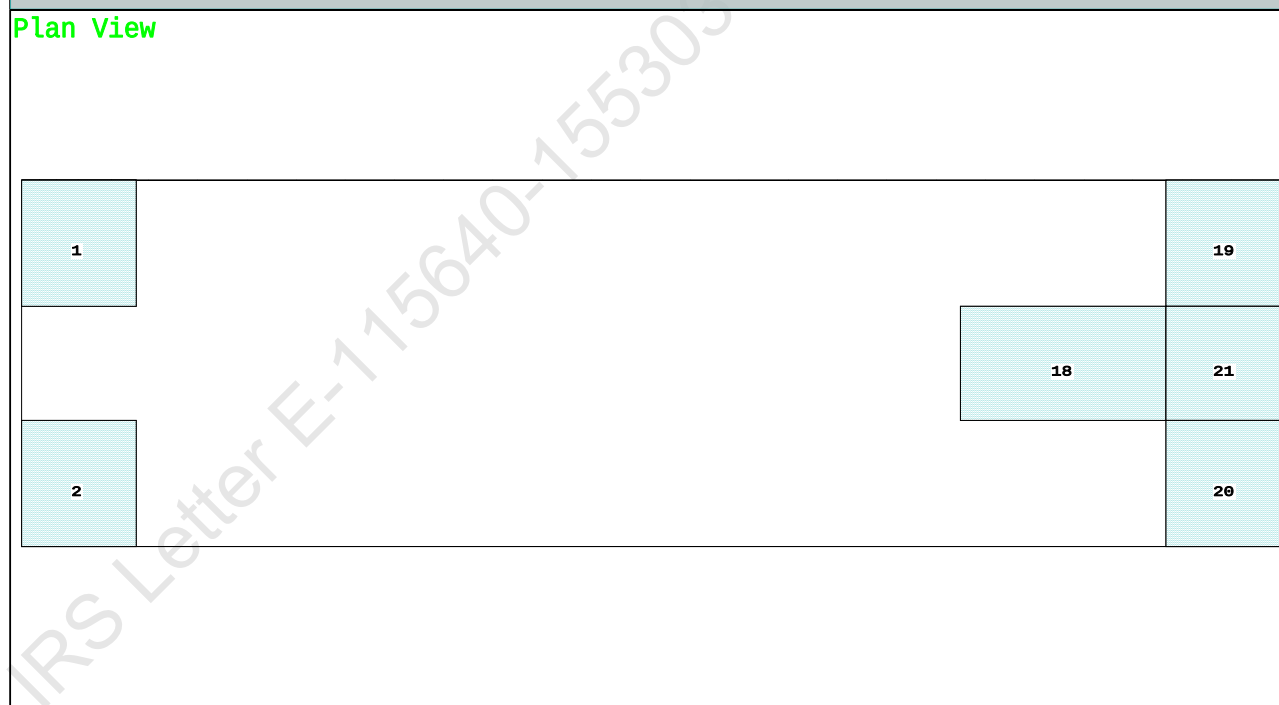
Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Fwd

Condition Graphic - Draft: 2.262 @ 55.000f, 2.162 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

1 N01_WBTK.P

2 N01_WBTK.S

18 N05_WBTK.C

19 N06_WBTK.P

20 N06_WBTK.S

21 N06_WBTK.C

17-12-20 15:26:08 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.262 @ 55.00f, 2.212 @ 27.50f, 2.162 @ 0.00

Trim: Fwd 0.101/55.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	32.695	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,420.34	25.099f	0.000	4.865	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,855.34	27.719f	0.000	4.124	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.34	27.724f	0.000	1.131	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	880.7	27.503f	0.000	122.53	10.166
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.03		40.32	119.53	7.173
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

424.4 Cu.M. (21%) WB

0.0 Cu.M. (0%) VOID

150.00 MT Crane

51.30 MT Crane Load

795.69 MT Deck Cargo

DRAFT AT FP (M) = 2.262
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.162
 DRAFT AT FWD MARK (M) = 2.255
 DRAFT AT AFT MARK (M) = 2.169

HYDROSTATIC PROPERTIES

Trim: Fwd 0.101/55.000, No Heel, VCG = 4.124

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----
2.212	1,855.34	27.724f	1.131	9.03
				27.503f
				40.32
				119.53
				7.173

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 425.0 m.-MT

17-12-20 15:26:08 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.262 @ 55.00f, 2.212 @ 27.50f, 2.162 @ 0.00

Trim: Fwd 0.101/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	44.5	0.405	1.480	0.05140	3.39
CRANE.C	36.2	3.087	4.162	0.05140	7.74
DECKCARGO.C	70.6	1.792	2.867	0.05140	10.40
Total wind heeling moment to starboard----->					21.52
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.719f TCG = 0.000 VCG = 4.124

Free Surface Adjustment: 0.229

Adjusted CG: LCG = 27.718f TCG = 0.000 VCG = 4.353

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---	Trim---	Heel---	Weight(MT)---	Area---Angle
2.149	0.11f	0.00	1,855.3	0.0000 2.709
2.150	0.10f	0.09s	1,855.3	0.0000 2.621
2.147	0.11f	2.71s	1,855.4	0.0000 0.000
2.138	0.11f	5.09s	1,855.4	0.0000 -2.379
2.171	0.15f	10.09s	1,855.4	0.0000 -7.379
2.240	0.17f	12.62s	1,855.3	0.0000 -9.915
2.325	0.19f	15.09s	1,855.3	0.0000 -12.379
2.526	0.23f	20.09s	1,855.3	0.0000 -17.379
2.726	0.28f	25.09s	1,855.3	0.0000 -22.379
2.904	0.34f	30.09s	1,855.3	0.0000 -27.379
3.059	0.40f	35.09s	1,855.3	0.0000 -32.379
3.112	0.42f	36.98s	1,855.3	0.0000 -34.275
3.190	0.45f	40.09s	1,855.3	0.0000 -37.379
3.297	0.50f	45.09s	1,855.3	0.0000 -42.379
3.379	0.55f	50.09s	1,855.3	0.0000 -47.379
3.436	0.59f	55.09s	1,855.3	0.0000 -52.379
3.467	0.63f	60.09s	1,855.3	0.0000 -57.379
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 425.0 m.-MT was applied to artificially modify the CG.

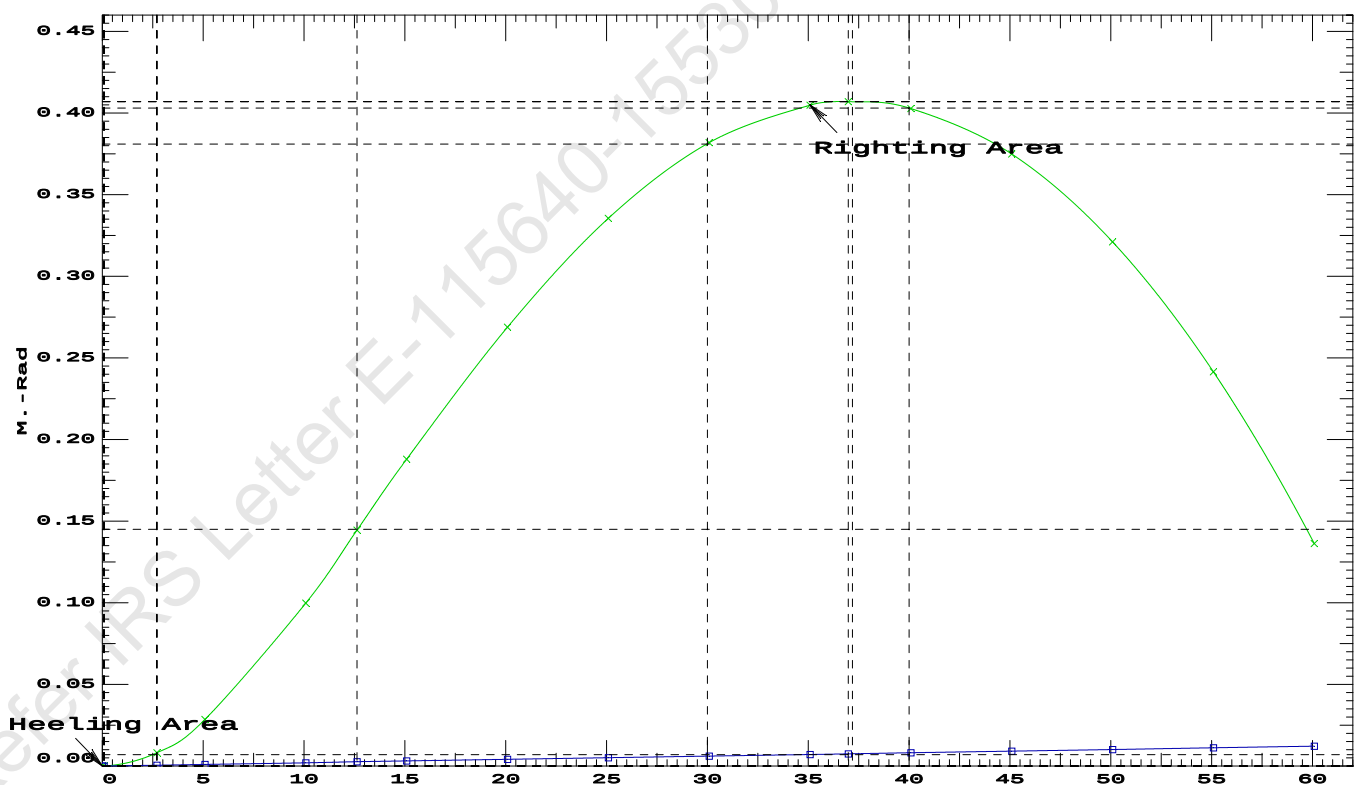
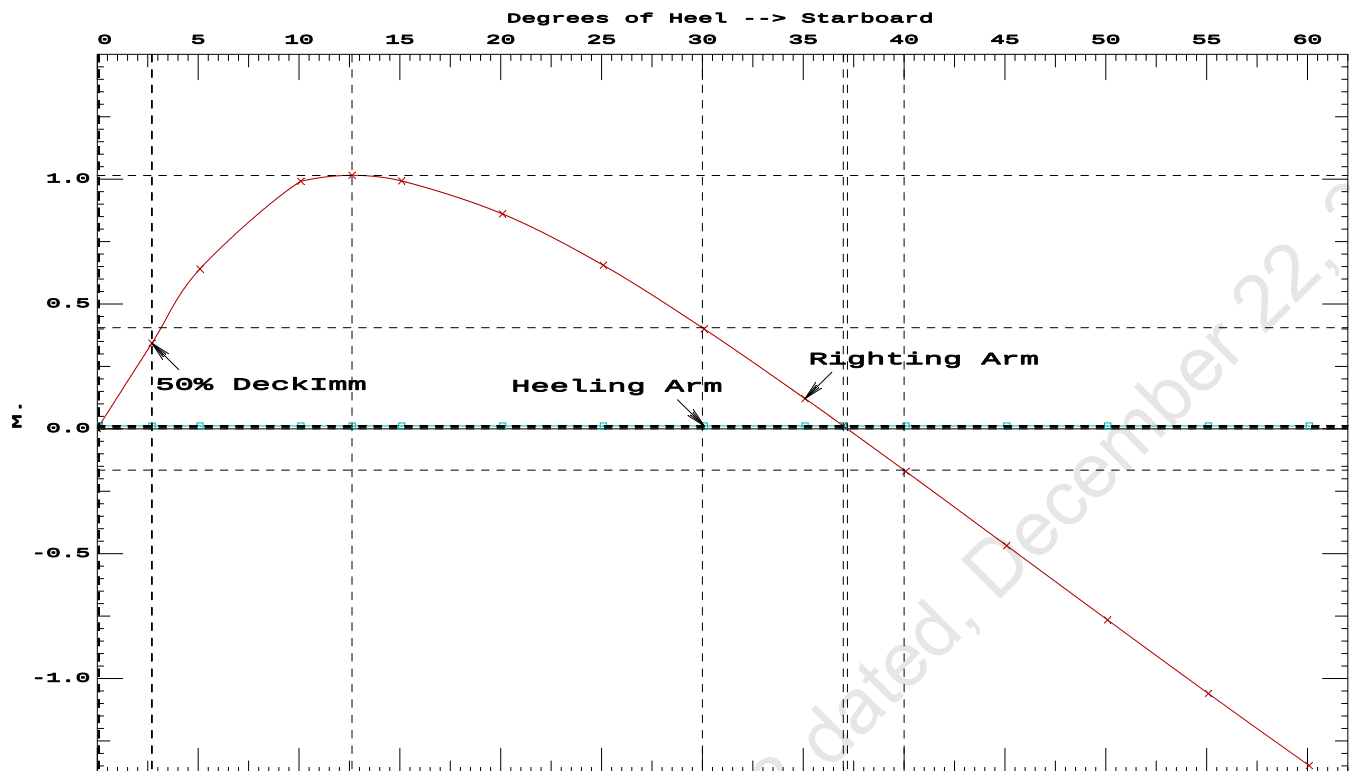
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.52 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1440 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	36.90 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.62 P

17-12-20 15:26:08 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Fwd

17-12-20 15:26:08 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo****Crane Lifting 51.30T @ 12m Radius Fwd****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 27.719f TCG = 0.000 VCG = 4.124

Free Surface Adjustment: 0.229

Adjusted CG: LCG = 27.718f TCG = 0.000 VCG = 4.353

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms		Res. Area
				in Trim--	in Heel---	
2.150	0.10f	0.00s	1,855.3	0.000	0.000	0.0000
2.139	0.11f	5.00s	1,855.4	0.000	0.629	0.0275
2.169	0.15f	10.00s	1,855.4	0.000	0.989	0.1000
2.240	0.17f	12.63s	1,855.3	0.000	1.014	0.1464
2.322	0.19f	15.00s	1,855.3	0.000	0.993	0.1879
2.522	0.23f	20.00s	1,855.3	0.000	0.863	0.2705
2.723	0.28f	25.00s	1,855.3	0.000	0.658	0.3374
2.901	0.34f	30.00s	1,855.3	0.000	0.404	0.3841
3.057	0.39f	35.00s	1,855.3	0.000	0.125	0.4074
3.117	0.42f	37.17s	1,855.3	0.000	0.000	0.4097
3.188	0.45f	40.00s	1,855.3	0.000	-0.166	0.4057
3.296	0.50f	45.00s	1,855.3	0.000	-0.463	0.3782
3.378	0.55f	50.00s	1,855.3	0.000	-0.761	0.3248
3.435	0.59f	55.00s	1,855.3	0.000	-1.056	0.2455
3.466	0.63f	60.00s	1,855.3	0.000	-1.345	0.1407

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 425.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

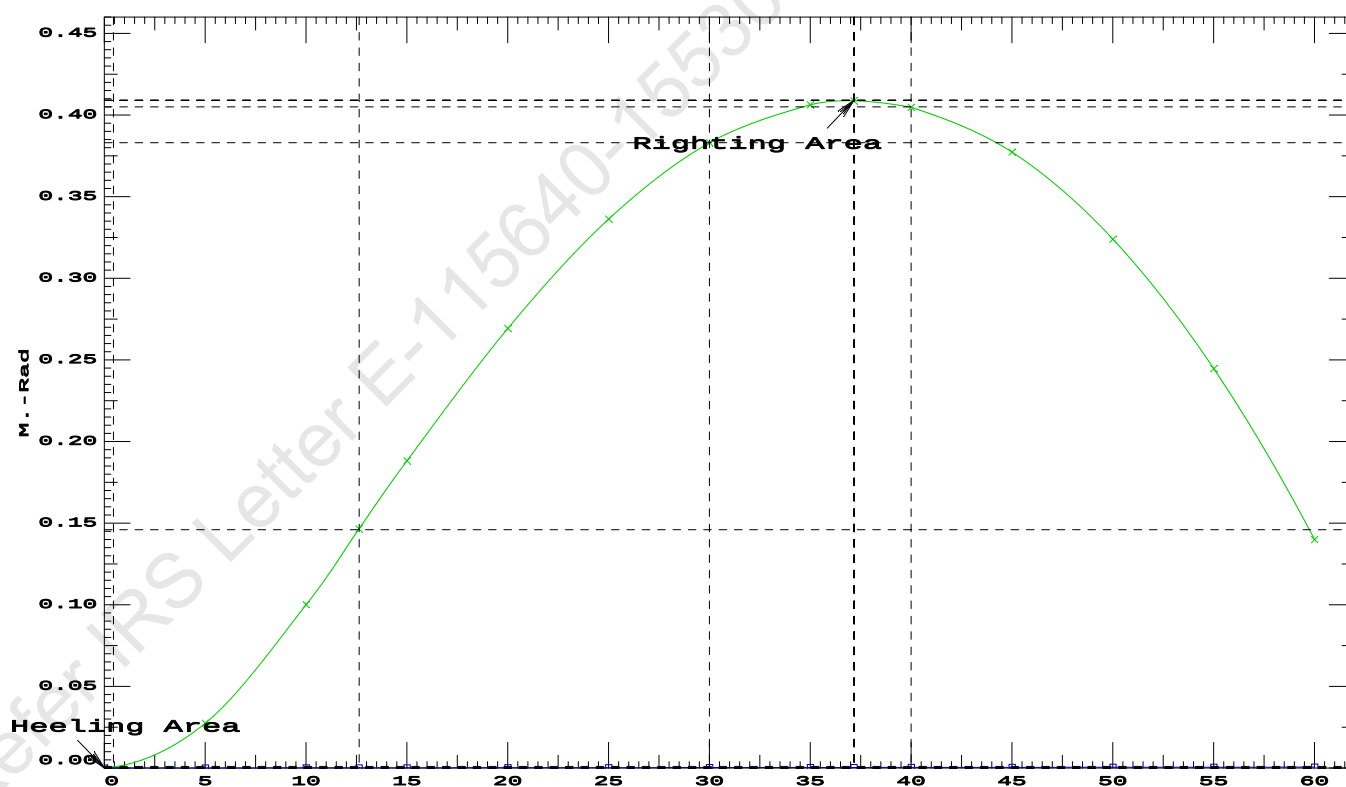
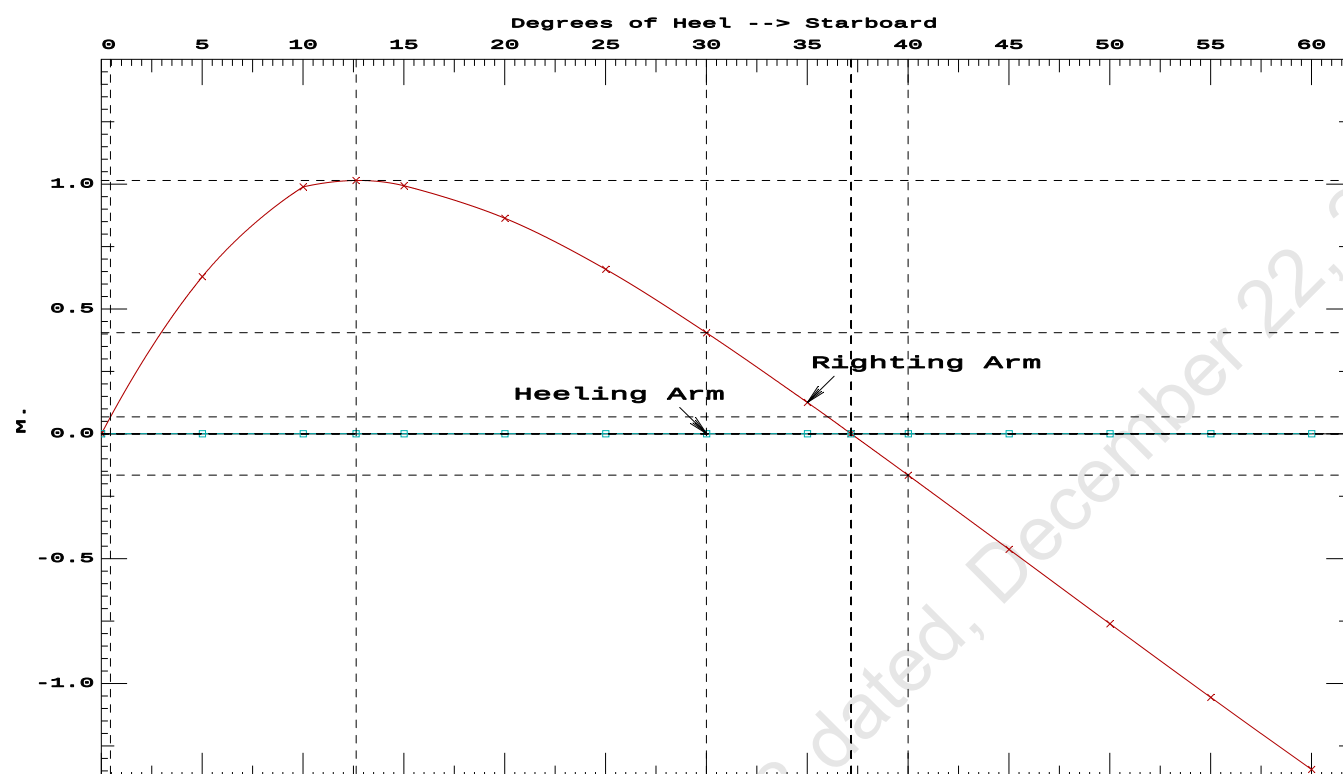
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	188177 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1172.99 P

17-12-20 15:26:08 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 51.30T @ 12m Radius Fwd

17-12-20 15:26:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo**Crane Lifting 51.30T @ 12m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.263 @ 55.00f, 2.212 @ 27.50f, 2.161 @ 0.00

Trim: Fwd 0.101/55.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	32.695	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,420.34	25.099f	0.000	4.865	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,855.34	27.719f	0.000	4.124	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.34	27.725f	0.001	1.131	

	Righting Arms:			0.001	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.001	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	880.7	27.503f	0.000	122.53	10.166
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.03		40.32	119.53	7.173
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

17-12-20 15:26:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.255 @ 55.00f, 2.155 @ 27.50f, 2.055 @ 0.00

Trim: Fwd 0.201/55.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Deck Cargo			795.69	27.500f	0.000	4.000	
Total Fixed----->			1,369.04	25.309f	0.000	3.822	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			435.00	36.273f	0.000	1.703	425.04
Total Weight----->			1,804.04	27.952f	0.000	3.311	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,803.86	27.960f	0.001	1.102	

	Righting Arms:			0.001	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.001	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		880.5	27.507f	0.000	125.95	10.450
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.03		40.58	123.74	8.241
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 25.309f TCG = 0.000 VCG = 3.822

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight(MT)---	in Trim--	in Heel---	Area
2.042	0.21f	0.00s	1,803.9	0.000	0.000	0.0000
2.042	0.21f	0.07s	1,803.9	0.000	0.010	0.0000
2.030	0.21f	5.00s	1,804.0	0.000	0.742	0.0324
2.031	0.28f	10.00s	1,804.2	0.000	1.255	0.1212
2.146	0.36f	15.00s	1,804.0	0.000	1.384	0.2391
2.182	0.38f	16.19s	1,804.0	0.000	1.387	0.2678
2.308	0.45f	20.00s	1,804.1	0.000	1.355	0.3596

17-12-20 15:26:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

2.457	0.56f	25.00s	1,804.0	0.000	1.232	0.4733
2.585	0.67f	30.00s	1,804.0	0.000	1.054	0.5735

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1369.04 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

17-12-20 15:26:13 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 25.309f TCG = 0.000 VCG = 3.822

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.042	0.21f	0.00s	1,803.9	0.000	0.000	0.0000
2.042	0.21f	0.07s	1,803.9	0.000	0.010	0.0000
2.030	0.21f	5.00s	1,804.0	0.000	0.742	0.0324
2.031	0.28f	10.00s	1,804.2	0.000	1.255	0.1212
2.146	0.36f	15.00s	1,804.0	0.000	1.384	0.2391
2.182	0.38f	16.19s	1,804.0	0.000	1.387	0.2678
2.308	0.45f	20.00s	1,804.1	0.000	1.355	0.3596
2.457	0.56f	25.00s	1,804.0	0.000	1.232	0.4733
2.585	0.67f	30.00s	1,804.0	0.000	1.054	0.5735
2.692	0.78f	35.00s	1,804.0	0.000	0.845	0.6566
2.777	0.88f	40.00s	1,804.0	0.000	0.618	0.7206
2.841	0.98f	45.00s	1,804.0	0.000	0.380	0.7642
2.882	1.07f	50.00s	1,804.0	0.000	0.134	0.7867
2.896	1.12f	52.69s	1,804.0	0.000	0.000	0.7898
2.902	1.16f	55.00s	1,804.0	0.000	-0.116	0.7875

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1369.04 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

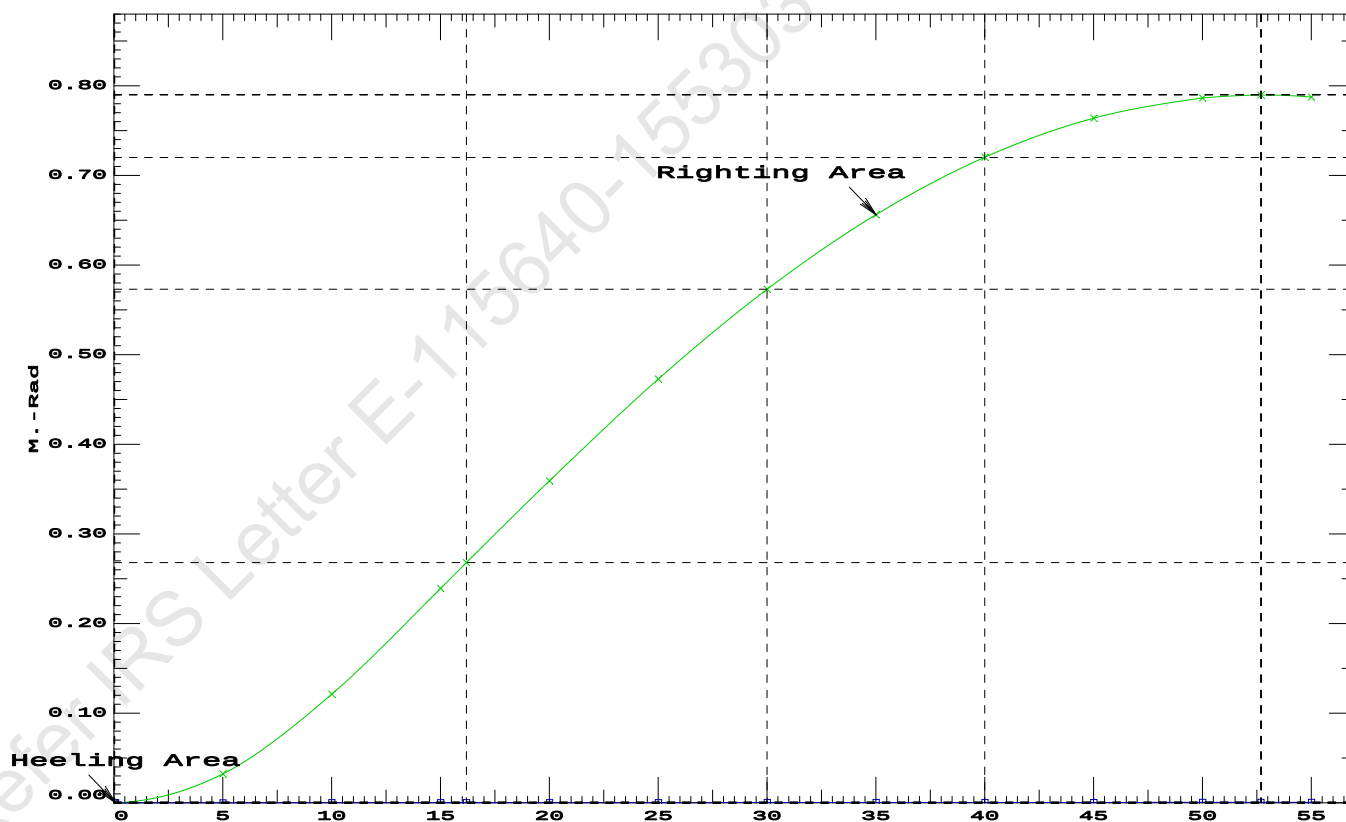
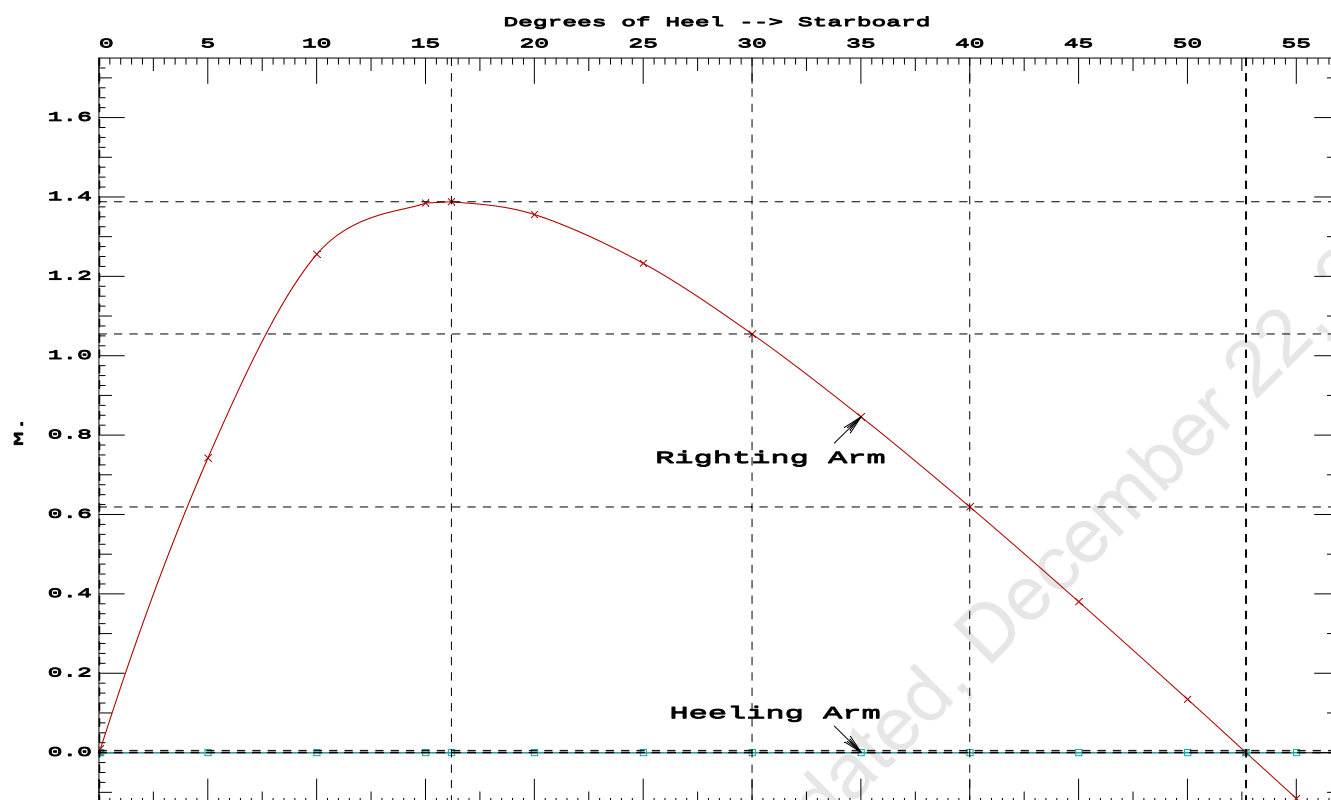
LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 52.62 P

17-12-20 15:26:13

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GHS 17.22

PONTON



17-12-20 15:30:57 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

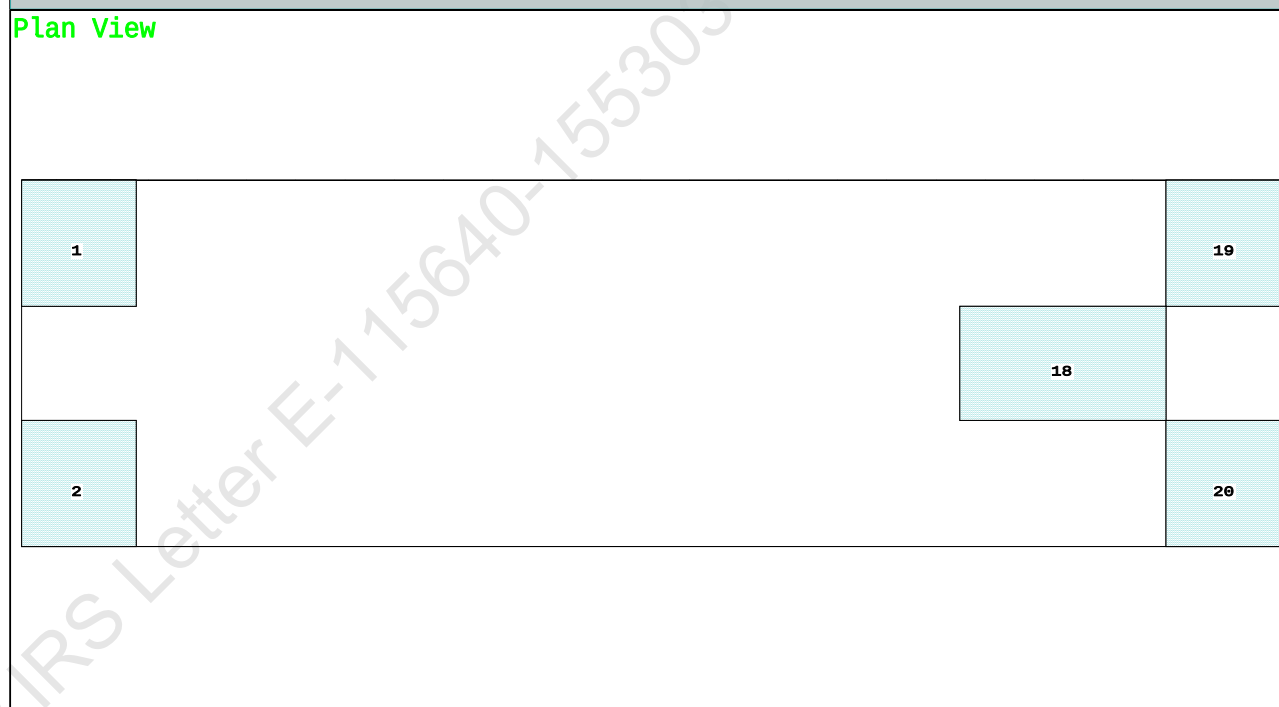
Condition 5 : Lightship + Crane + Ballast + Deck CargoTowing Condition with Max. draft 2.212m

Condition Graphic - Draft: 2.143 @ 55.000f, 2.281 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

1 N01_WBTK.P

2 N01_WBTK.S

18 N05_WBTK.C

19 N06_WBTK.P

20 N06_WBTK.S

17-12-20 15:30:57 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck CargoTowing Condition with Max. draft 2.212m

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.143 @ 55.00f, 2.212 @ 27.50f, 2.281 @ 0.00

Trim: Aft 0.138/55.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Deck Cargo			902.81	27.500f	0.000	4.000	
Total Fixed----->			1,476.16	25.468f	0.000	3.835	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.P	1.000	1.025	60.86	2.927f	5.223p	1.797	69.64*
N01_WBTK.S	1.000	1.025	60.86	2.927f	5.223s	1.797	69.64*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
Total Tanks----->			379.08	33.941f	0.000	1.691	372.72
Total Weight----->			1,855.24	27.199f	0.000	3.397	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.24	27.193f	0.000	1.131	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		880.6	27.494f	0.000	122.50	10.192
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			9.03		40.56	120.23	7.926
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

369.8 Cu.M. (18%) WB

0.0 Cu.M. (0%) VOID

150.00 MT Crane

902.81 MT Deck Cargo

DRAFT AT FP (M) = 2.143
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.281
 DRAFT AT FWD MARK (M) = 2.153
 DRAFT AT AFT MARK (M) = 2.271

17-12-20 15:30:57 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 5 : Lightship + Crane + Ballast + Deck Cargo****Towing Condition with Max. draft 2.212m****HYDROSTATIC PROPERTIES**

Trim: Aft 0.138/55.000, No Heel, VCG = 3.397

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
 Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
 2.212 1,855.24 27.193f 1.131 9.03 27.494f 40.56 120.23 7.926
 Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 372.7 m.-MT

HEELING MOMENT specification**Lateral Plane Method****Wind pressure toward starboard**

USK draft: 2.143 @ 55.00f, 2.212 @ 27.50f, 2.281 @ 0.00

Trim: Aft 0.138/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment
HULL	44.5	0.406	1.481	0.05140	3.39
CRANE.C	36.2	3.000	4.075	0.05140	7.57
DECKCARGO.C	70.6	1.811	2.886	0.05140	10.47
Total wind heeling moment to starboard----->					21.43
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.199f TCG = 0.000 VCG = 3.397

Free Surface Adjustment: 0.201

Adjusted CG: LCG = 27.200f TCG = 0.000 VCG = 3.598

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle	
2.269 0.14a 0.00	1,855.2	0.000 -0.011	0.0000	2.644
2.269 0.14a 0.08s	1,855.2	0.000 0.000	-0.0000	2.563
2.266 0.14a 2.64s	1,855.2	0.000 0.358	0.0080	0.000
2.260 0.15a 5.08s	1,855.2	0.000 0.695	0.0304	-2.437
2.338 0.20a 10.08s	1,855.2	0.000 1.112	0.1112	-7.437
2.513 0.25a 14.52s	1,855.2	0.000 1.178	0.2017	-11.873
2.539 0.26a 15.08s	1,855.2	0.000 1.177	0.2133	-12.437
2.786 0.31a 20.08s	1,855.3	0.000 1.108	0.3145	-17.437
3.046 0.39a 25.08s	1,855.2	0.000 0.963	0.4055	-22.437
3.288 0.46a 30.08s	1,855.2	0.000 0.767	0.4813	-27.437
3.506 0.54a 35.08s	1,855.2	0.000 0.543	0.5387	-32.437
3.698 0.61a 40.08s	1,855.2	0.000 0.304	0.5758	-37.437
3.863 0.68a 45.08s	1,855.2	0.000 0.055	0.5915	-42.437
3.895 0.69a 46.17s	1,855.2	0.000 0.000	0.5920	-43.531
3.997 0.74a 50.08s	1,855.2	0.000 -0.198	0.5853	-47.437
4.102 0.80a 55.08s	1,855.2	0.000 -0.453	0.5569	-52.437
4.174 0.85a 60.08s	1,855.2	0.000 -0.706	0.5063	-57.437
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 372.7 m.-MT was applied to artificially modify the CG.

17-12-20 15:30:57 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 5 : Lightship + Crane + Ballast + Deck Cargo****Towing Condition with Max. draft 2.212m**

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):

Stbd heeling moment = 21.43 (constant)

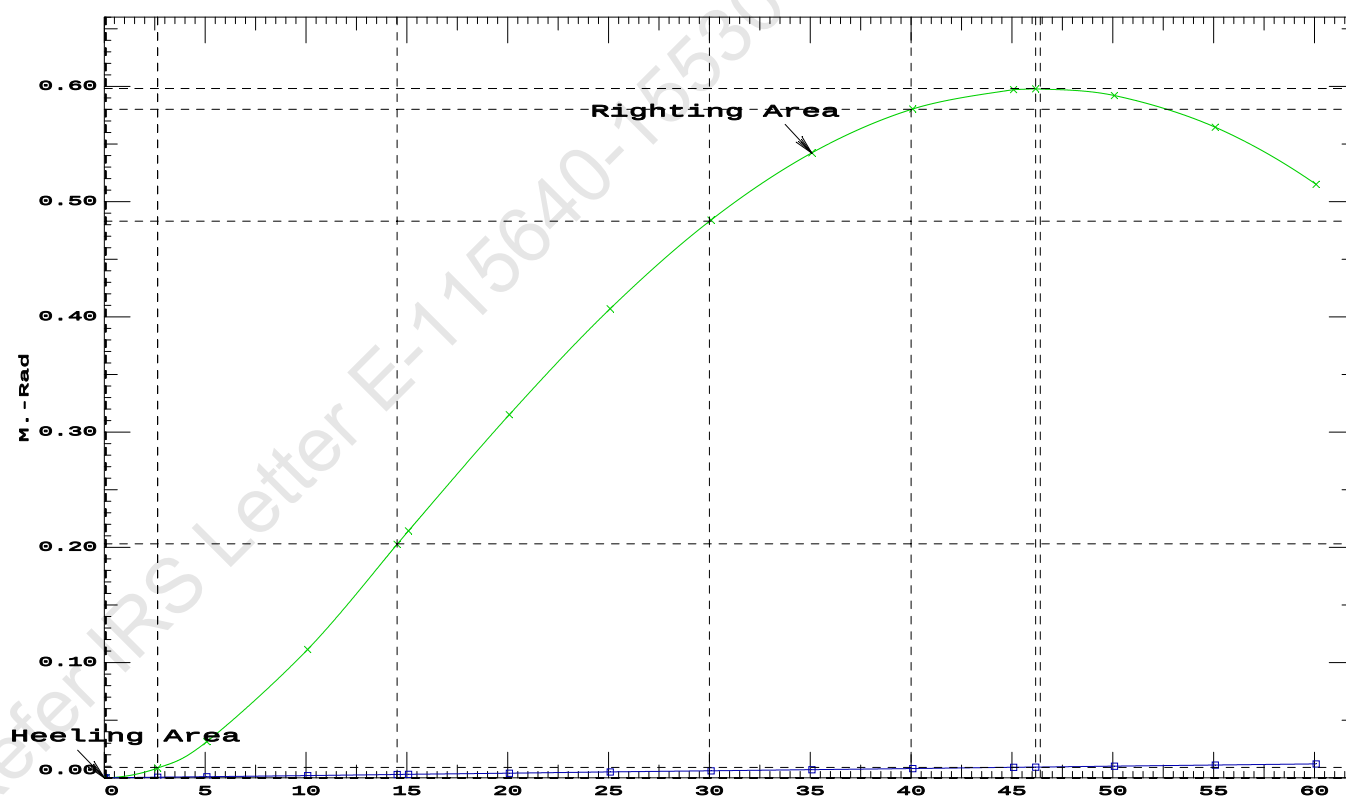
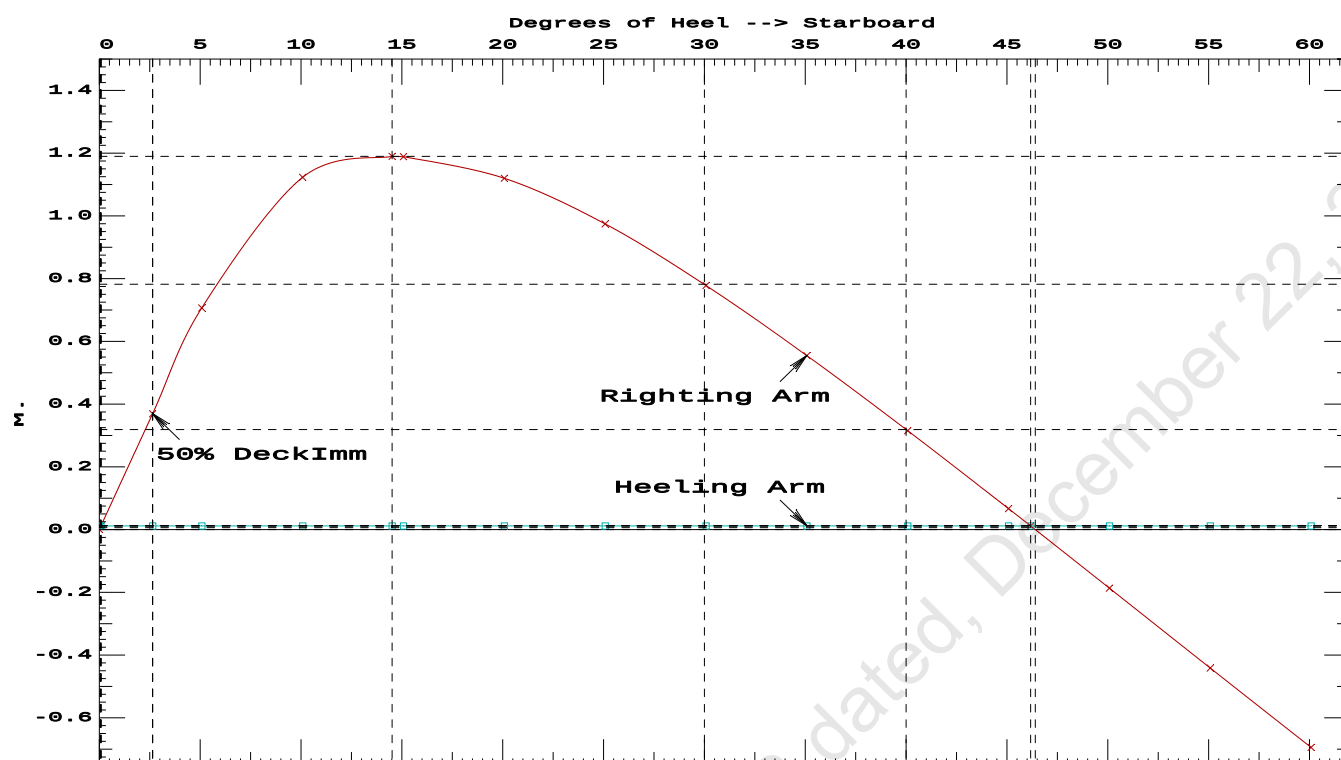
LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2017 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	46.09 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.56 P

17-12-20 15:30:57 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 5 : Lightship + Crane + Ballast + Deck CargoTowing Condition with Max. draft 2.212m

17-12-20 15:31:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

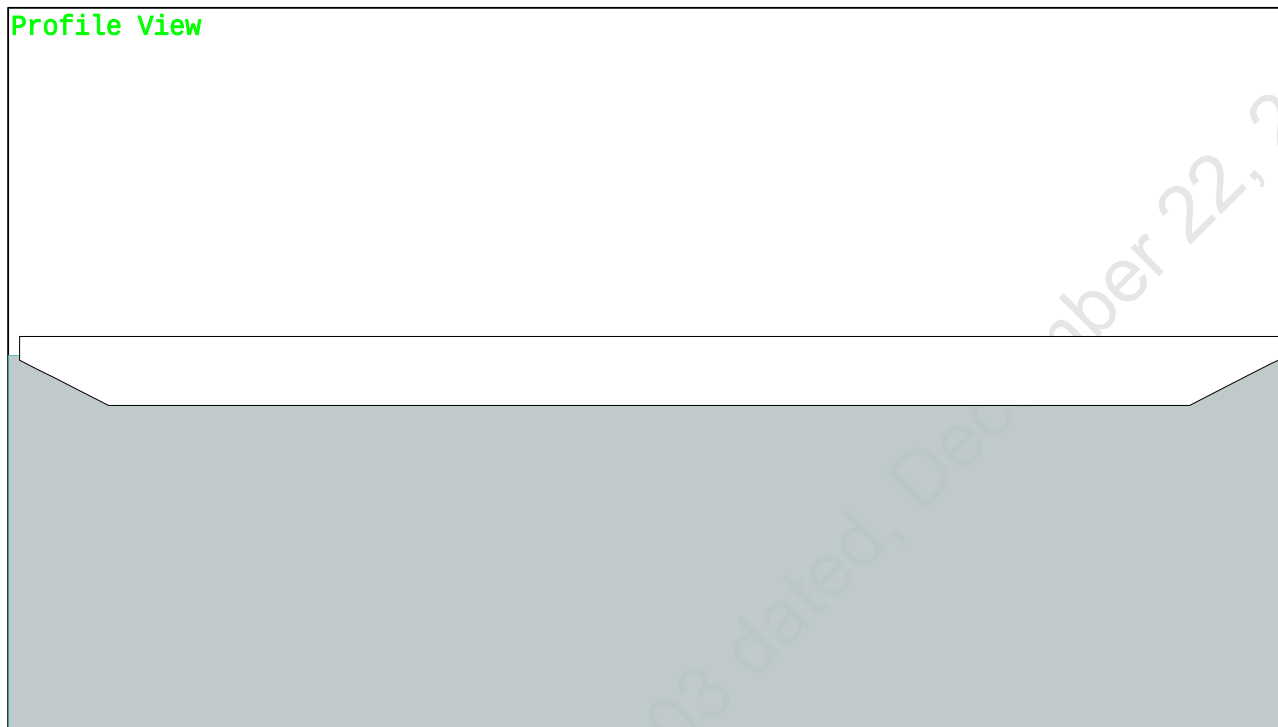
PONTOON

Condition 6 : Lightship + Max.Deck Cargo

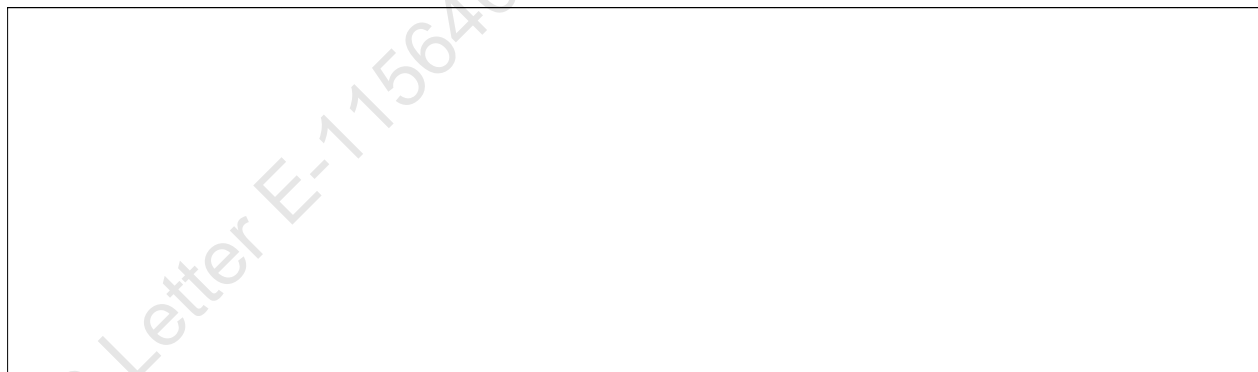
Max. deck cargo loading condition

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero

Profile View



Plan View



17-12-20 15:31:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 6 : Lightship + Max.Deck Cargo****Max. deck cargo loading condition****WEIGHT and DISPLACEMENT and WATERPLANE STATUS**

USK draft: 2.212 @ 55.00f, 2.212 @ 27.50f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	423.35	27.500f	0.000	3.000	
Deck Cargo	1,432.09	27.500f	0.000	4.000	
Total Weight----->	1,855.44	27.500f	0.000	3.772	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,855.22	27.500f	0.000	1.130

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	880.6	27.499f	0.000	122.52 10.395
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		9.03	40.44	119.88	7.753
Distances in METERS.-----				Moments in m.-MT.	

SUMMARY OF LOADING

0.0 Cu.M. (0%) WB

0.0 Cu.M. (0%) VOID

1,432.09 MT Deck Cargo

DRAFT AT FP (M) = 2.212
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.212
 DRAFT AT FWD MARK (M) = 2.212
 DRAFT AT AFT MARK (M) = 2.212

17-12-20 15:31:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22

PONTOON

Condition 6 : Lightship + Max.Deck Cargo
Max. deck cargo loading condition

HYDROSTATIC PROPERTIES

No Trim, No Heel, VCG = 3.772

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft---Weight(MT)---LCB---VCB---cm---LCF---cm trim---GML---GMT
2.212 1,855.22 27.500f 1.130 9.03 27.499f 40.44 119.88 7.753
Distances in METERS.---Specific Gravity = 1.025.---Moment in m.-MT.
Trim is per 55.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.212 @ 55.00f, 2.212 @ 27.50f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----LPA*SF-----HCP-----Arm---Pressure-----Moment
HULL 44.5 0.405 1.480 0.05140 3.39
DECKCARGO.C 70.6 1.800 2.875 0.05140 10.43
Total wind heeling moment to starboard-----> 13.81
Distances in METERS.---Pressure in MT/Sqm.---Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 27.500f TCG = 0.000 VCG = 3.772

Origin Depth---	Degrees of Trim---Heel---	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. 50% DeckImm Area---Angle
2.200	0.00 0.00	1,855.4	0.000 -0.007	0.0000 2.894
2.200	0.00 0.05s	1,855.4	0.000 0.000	-0.0000 2.839
2.197	0.00 2.89s	1,855.4	0.000 0.388	0.0096 0.000
2.190	0.00 5.05s	1,855.4	0.000 0.680	0.0297 -2.161
2.241	0.00 10.05s	1,855.4	0.000 1.085	0.1087 -7.161
2.374	0.00 14.04s	1,855.4	0.000 1.141	0.1875 -11.145
2.414	0.00 15.05s	1,855.4	0.000 1.137	0.2077 -12.161
2.634	0.00 20.05s	1,855.4	0.000 1.054	0.3049 -17.161
2.861	0.00 25.05s	1,855.4	0.000 0.896	0.3905 -22.161
3.067	0.00 30.05s	1,855.4	0.000 0.686	0.4599 -27.161
3.249	0.00 35.05s	1,855.4	0.000 0.450	0.5097 -32.161
3.407	0.00 40.05s	1,855.4	0.000 0.199	0.5381 -37.161
3.510	0.00 43.89s	1,855.4	0.000 0.000	0.5448 -40.994
3.538	0.00 45.05s	1,855.4	0.000 -0.061	0.5442 -42.161
3.643	0.00 50.05s	1,855.4	0.000 -0.325	0.5274 -47.161
3.720	0.00 55.05s	1,855.4	0.000 -0.589	0.4875 -52.161
3.768	0.00 60.05s	1,855.4	0.000 -0.850	0.4247 -57.161

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: No tank loads are present.

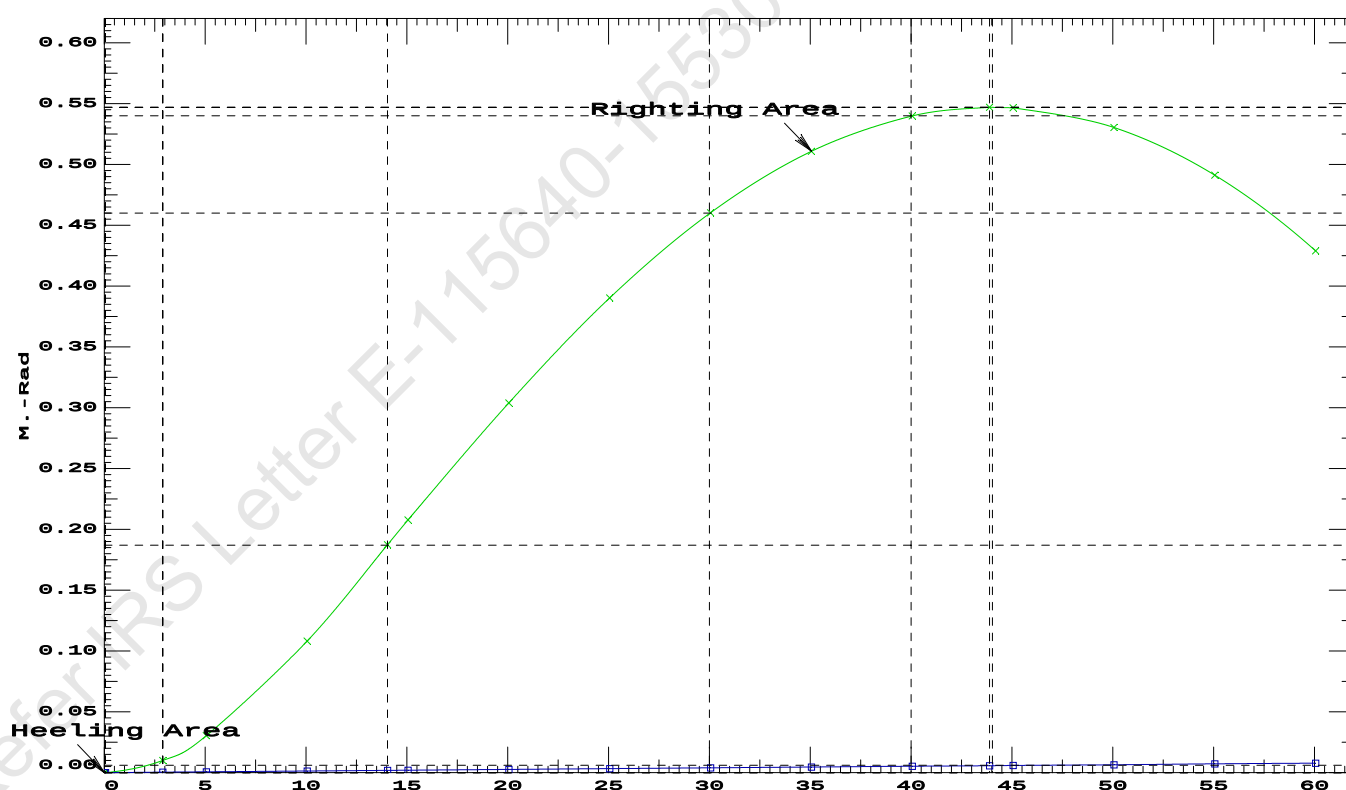
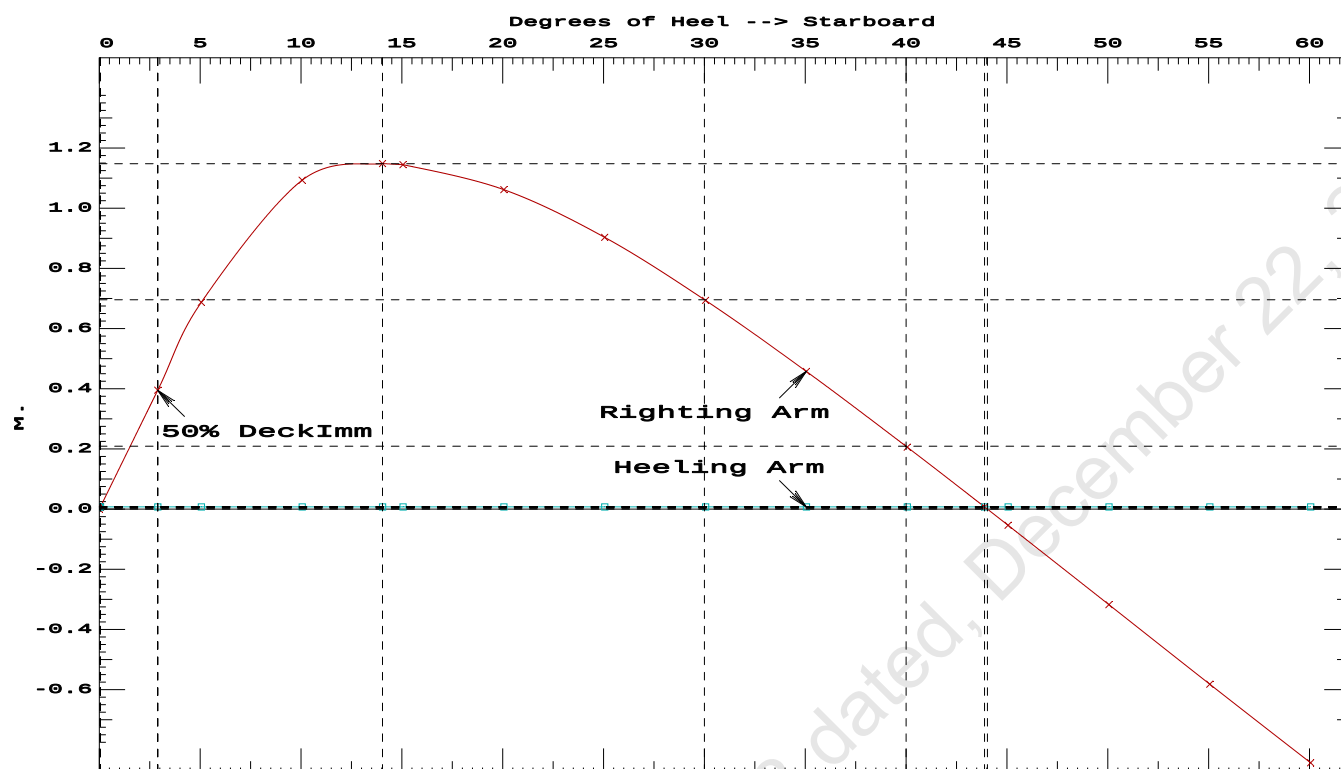
Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.81 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.1875 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg 43.83 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 2.84 P

17-12-20 15:31:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTON

Condition 6 : Lightship + Max.Deck CargoMax. deck cargo loading condition

17-12-20 15:26:31 Cybermarine Technologies PTE. Ltd.

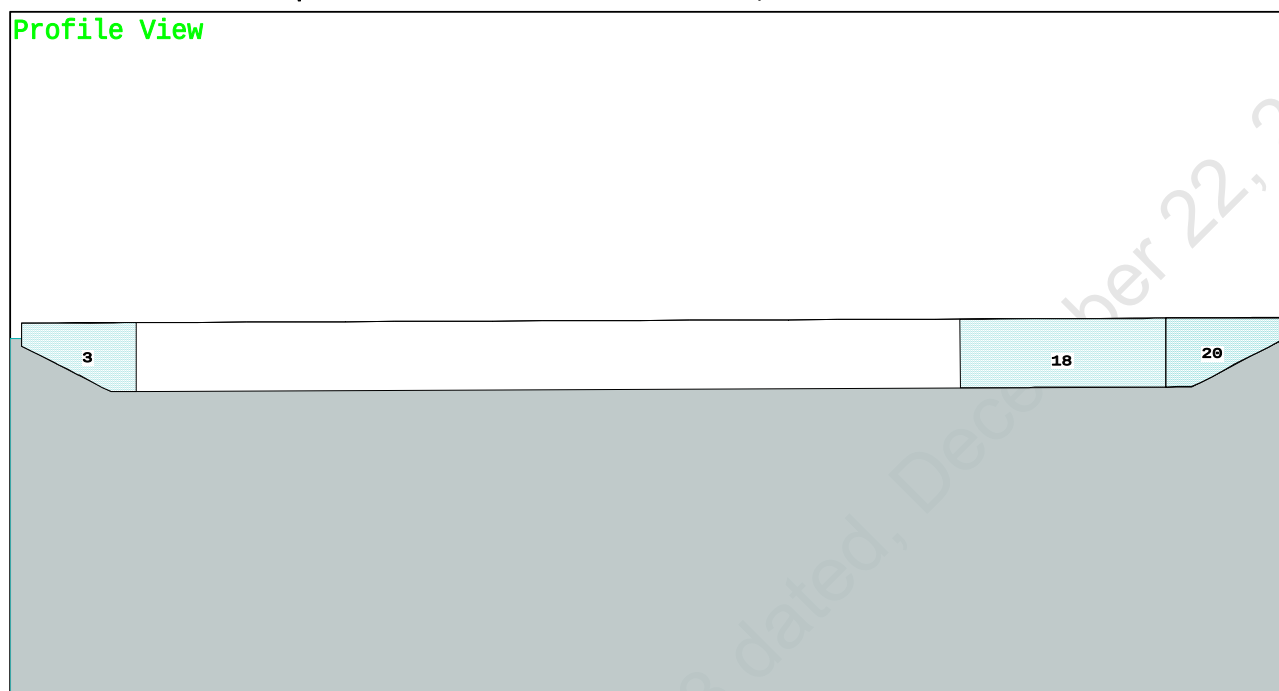
GHS 17.22

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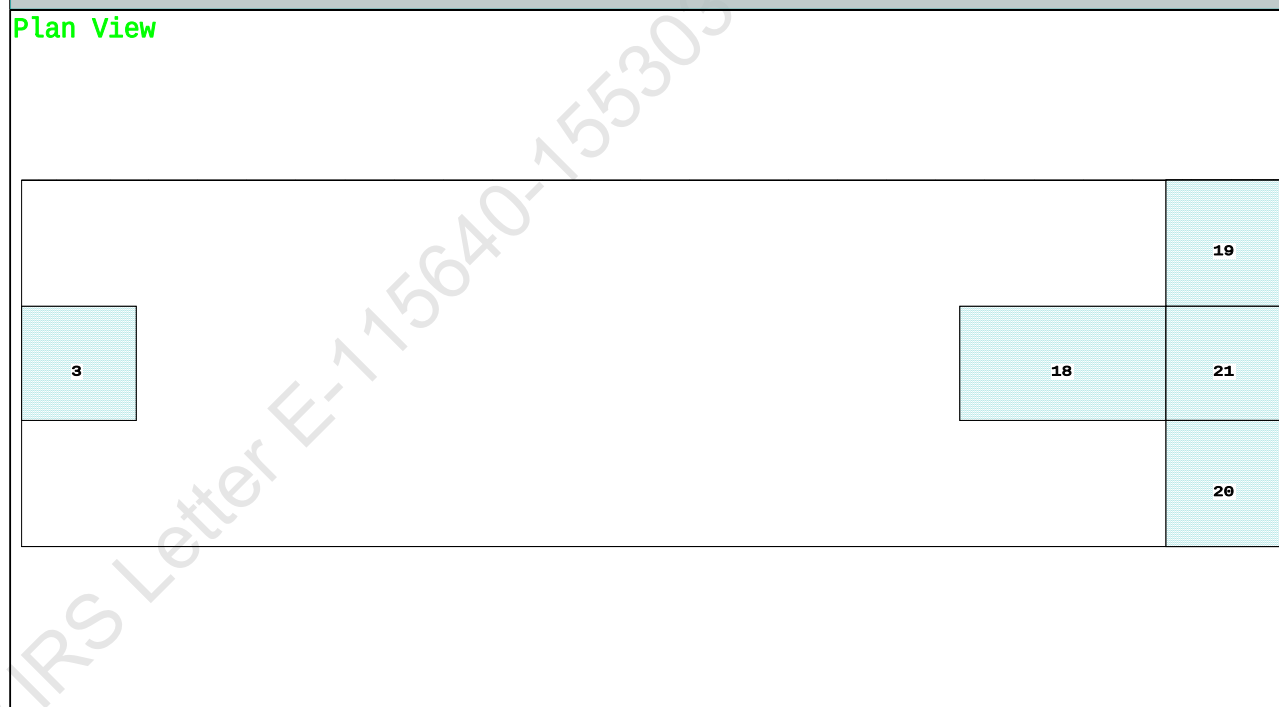
Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Aft

Condition Graphic - Draft: 2.092 @ 55.000f, 2.331 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

3 N01_WBTK.C 18 N05_WBTK.C 19 N06_WBTK.P 20 N06_WBTK.S 21 N06_WBTK.C

17-12-20 15:26:31 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.092 @ 55.00f, 2.212 @ 27.50f, 2.331 @ 0.00

Trim: Aft 0.239/55.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,486.06	23.200f	0.000	5.714	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,855.24	26.984f	0.000	4.913	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.20	26.968f	0.000	1.132	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	880.5	27.491f	0.000	122.49	10.209
			MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
			9.03		40.04	118.70	6.428
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

360.2 Cu.M. (18%) WB

0.0 Cu.M. (0%) VOID

250.00 MT Crane

36.60 MT Crane Load

776.11 MT Deck Cargo

DRAFT AT FP (M) = 2.092
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.331
 DRAFT AT FWD MARK (M) = 2.110
 DRAFT AT AFT MARK (M) = 2.314

HYDROSTATIC PROPERTIES

Trim: Aft 0.239/55.000, No Heel, VCG = 4.913

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm trim-----	GML-----	GMT-----		
2.212	1,855.20	26.968f	1.132	9.03	27.491f	40.04	118.70	6.428

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 338.1 m.-MT

17-12-20 15:26:31 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.092 @ 55.00f, 2.212 @ 27.50f, 2.331 @ 0.00

Trim: Aft 0.239/55.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	44.6	0.408	1.484	0.05140	3.40
CRANE.C	36.2	2.963	4.039	0.05140	7.51
DECKCARGO.C	70.6	1.820	2.896	0.05140	10.50
Total wind heeling moment to starboard-->					21.41
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.984f TCG = 0.000 VCG = 4.913

Free Surface Adjustment: 0.182

Adjusted CG: LCG = 26.985f TCG = 0.000 VCG = 5.095

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. 50% DeckImm Area	Angle
2.319	0.25a	0.00	1,855.2	0.000	-0.011	0.0000 2.460
2.319	0.25a	0.10s	1,855.2	0.000	0.000	-0.0000 2.361
2.318	0.25a	2.46s	1,855.2	0.000	0.268	0.0055 -0.000
2.311	0.25a	5.10s	1,855.2	0.000	0.564	0.0247 -2.639
2.412	0.35a	10.10s	1,855.2	0.000	0.846	0.0904 -7.639
2.444	0.37a	10.95s	1,855.2	0.000	0.850	0.1030 -8.489
2.634	0.45a	15.10s	1,855.2	0.000	0.784	0.1635 -12.639
2.903	0.55a	20.10s	1,855.2	0.000	0.590	0.2249 -17.639
3.189	0.68a	25.10s	1,855.2	0.000	0.323	0.2653 -22.639
3.461	0.82a	30.10s	1,855.2	0.000	0.011	0.2801 -27.639
3.470	0.83a	30.26s	1,855.2	0.000	0.000	0.2802 -27.801
3.709	0.96a	35.10s	1,855.2	0.000	-0.323	0.2667 -32.639
3.929	1.09a	40.10s	1,855.2	0.000	-0.666	0.2236 -37.639
4.119	1.21a	45.10s	1,855.2	0.000	-1.010	0.1505 -42.639
4.278	1.33a	50.10s	1,855.2	0.000	-1.351	0.0474 -47.639
4.402	1.43a	55.10s	1,855.2	0.000	-1.685	-0.0851 -52.639
4.491	1.51a	60.10s	1,855.2	0.000	-2.007	-0.2463 -57.639
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.						

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 338.1 m.-MT was applied to artificially modify the CG.

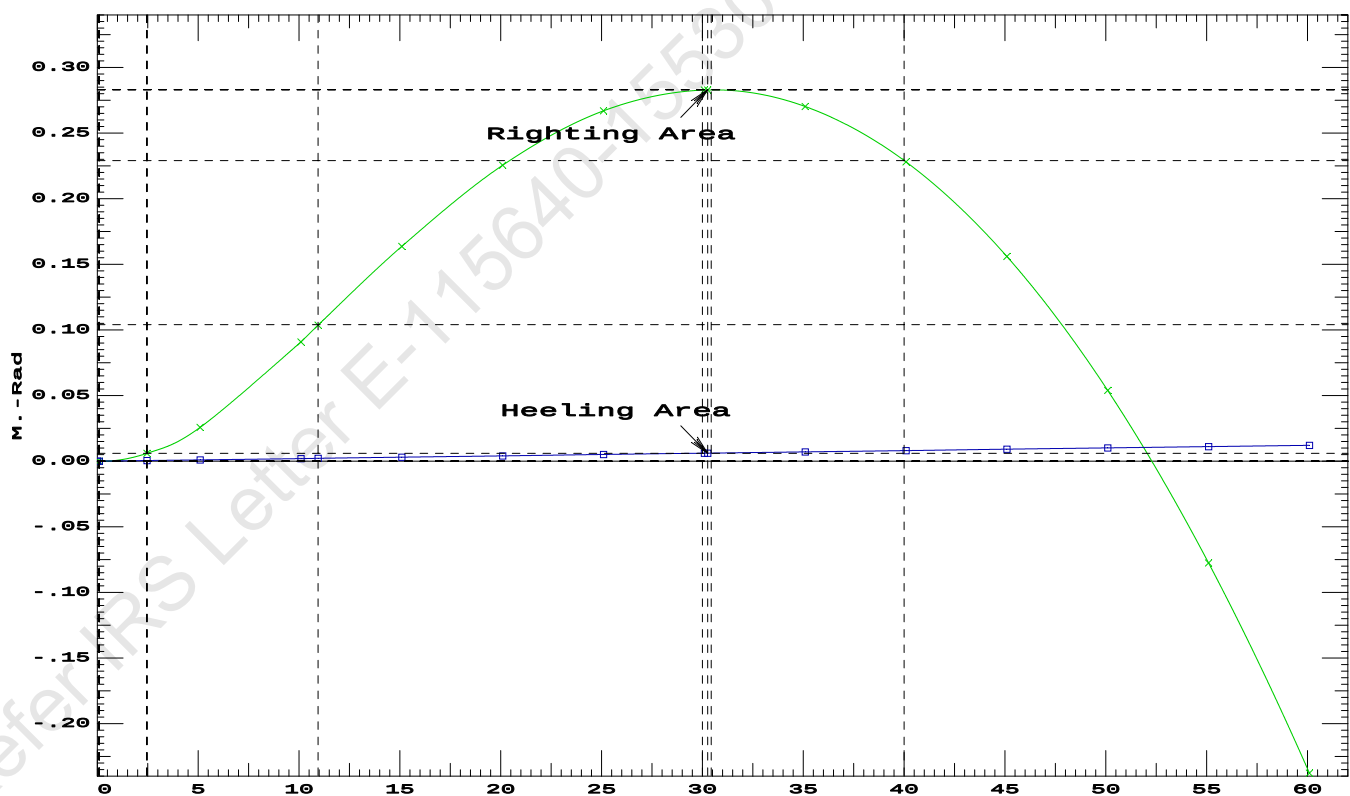
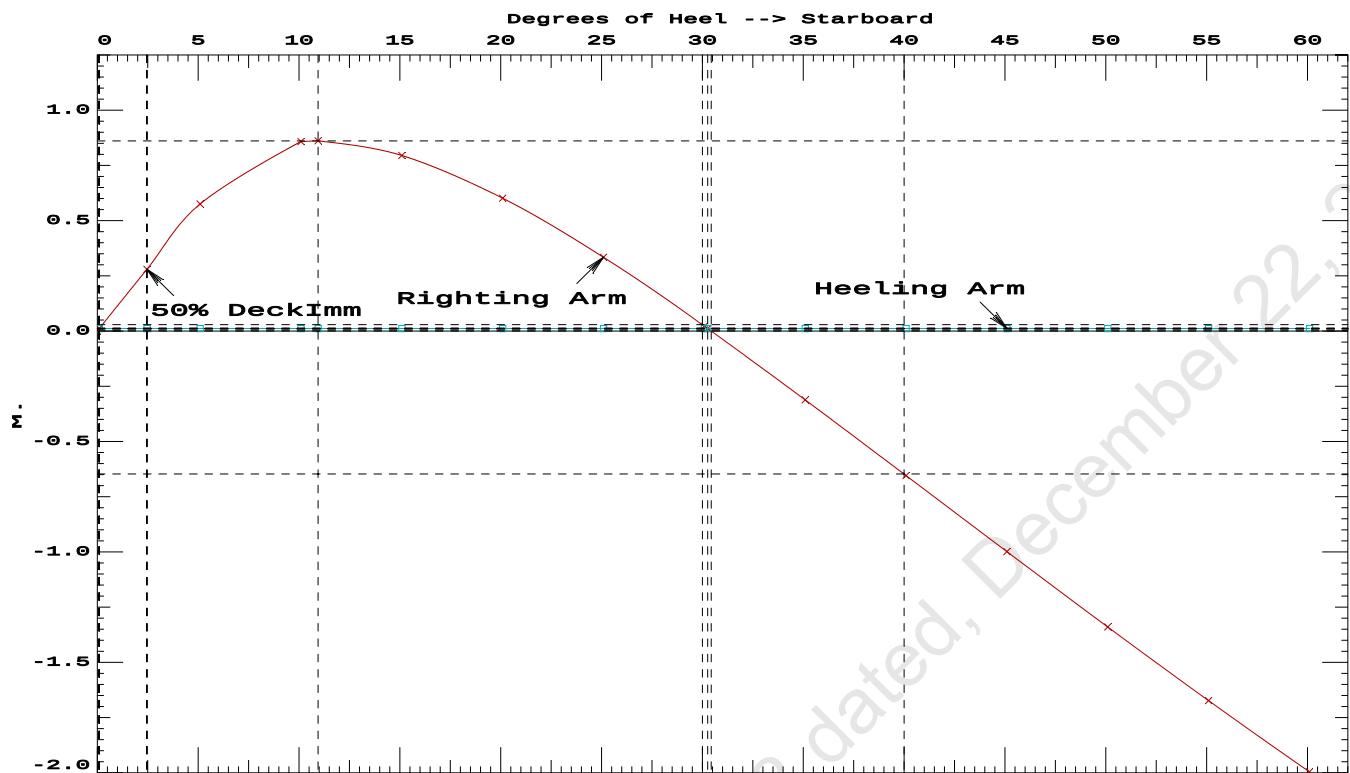
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.41 (constant)

LIM	IS CODE 2008 PART B CHAPTER-2 SEC.2	Min/Max	Attained
(1)	Residual Area from abs 0 deg to MaxRA	> 0.0800 m.-Rad	0.1030 P
(2)	Angle from Equilibrium to RAZero or Flood	> 20.00 deg	30.16 P
(3)	Angle from Equilibrium to 50% Dk Imm Angle	> 0.00 deg	2.36 P

17-12-20 15:26:31 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Aft

17-12-20 15:26:31 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo****Crane Lifting 36.60T @ 18m Radius Aft****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 26.984f TCG = 0.000 VCG = 4.913

Free Surface Adjustment: 0.182

Adjusted CG: LCG = 26.985f TCG = 0.000 VCG = 5.095

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area
2.319	0.25a	0.00s	1,855.2	0.000	0.000	0.0000
2.311	0.25a	5.00s	1,855.2	0.000	0.564	0.0246
2.409	0.35a	10.00s	1,855.2	0.000	0.856	0.0908
2.445	0.37a	10.96s	1,855.2	0.000	0.861	0.1052
2.629	0.45a	15.00s	1,855.2	0.000	0.797	0.1649
2.897	0.55a	20.00s	1,855.2	0.000	0.605	0.2276
3.184	0.68a	25.00s	1,855.2	0.000	0.339	0.2693
3.456	0.82a	30.00s	1,855.2	0.000	0.027	0.2856
3.478	0.83a	30.42s	1,855.2	0.000	0.000	0.2857
3.704	0.96a	35.00s	1,855.2	0.000	-0.306	0.2736
3.925	1.09a	40.00s	1,855.2	0.000	-0.648	0.2321
4.116	1.21a	45.00s	1,855.2	0.000	-0.993	0.1605
4.275	1.32a	50.00s	1,855.2	0.000	-1.334	0.0589
4.400	1.42a	55.00s	1,855.2	0.000	-1.668	-0.0721
4.490	1.51a	60.00s	1,855.2	0.000	-1.990	-0.2318

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 338.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

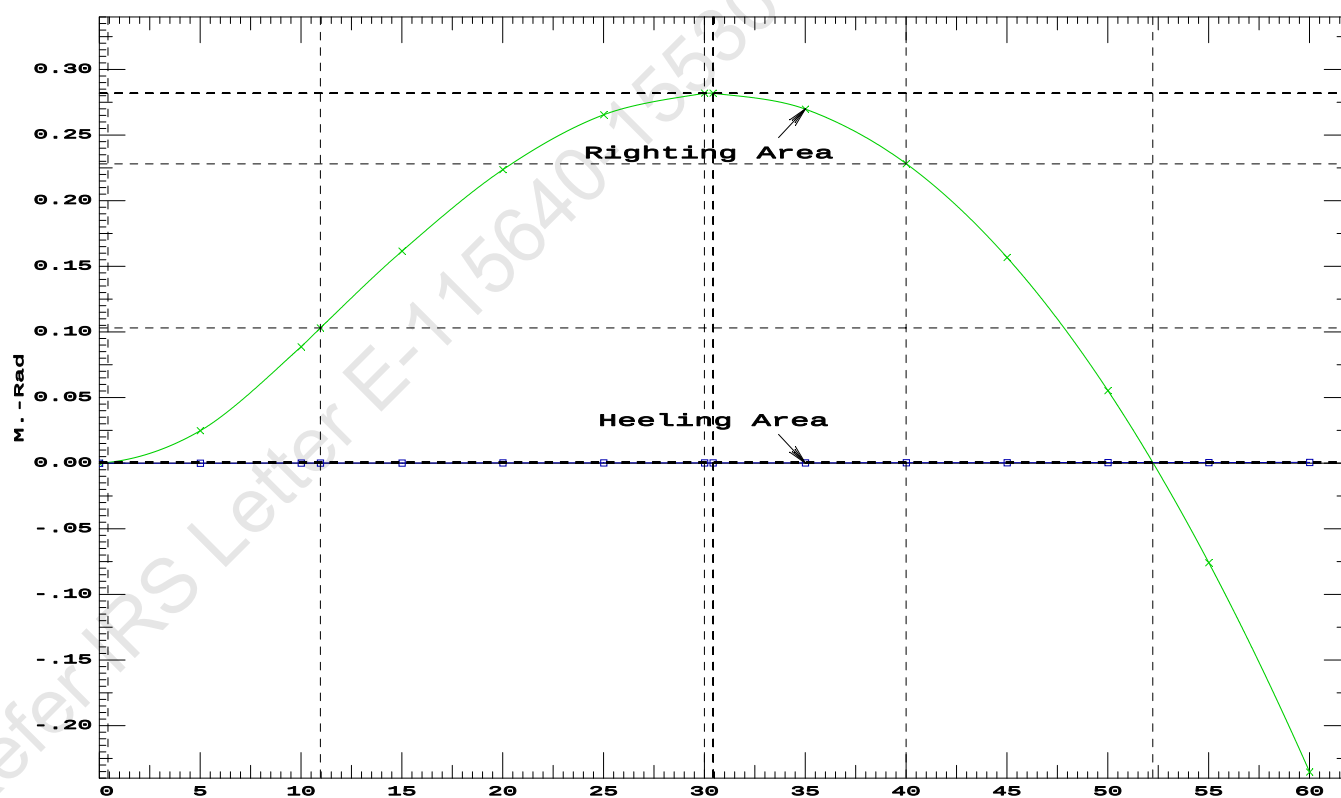
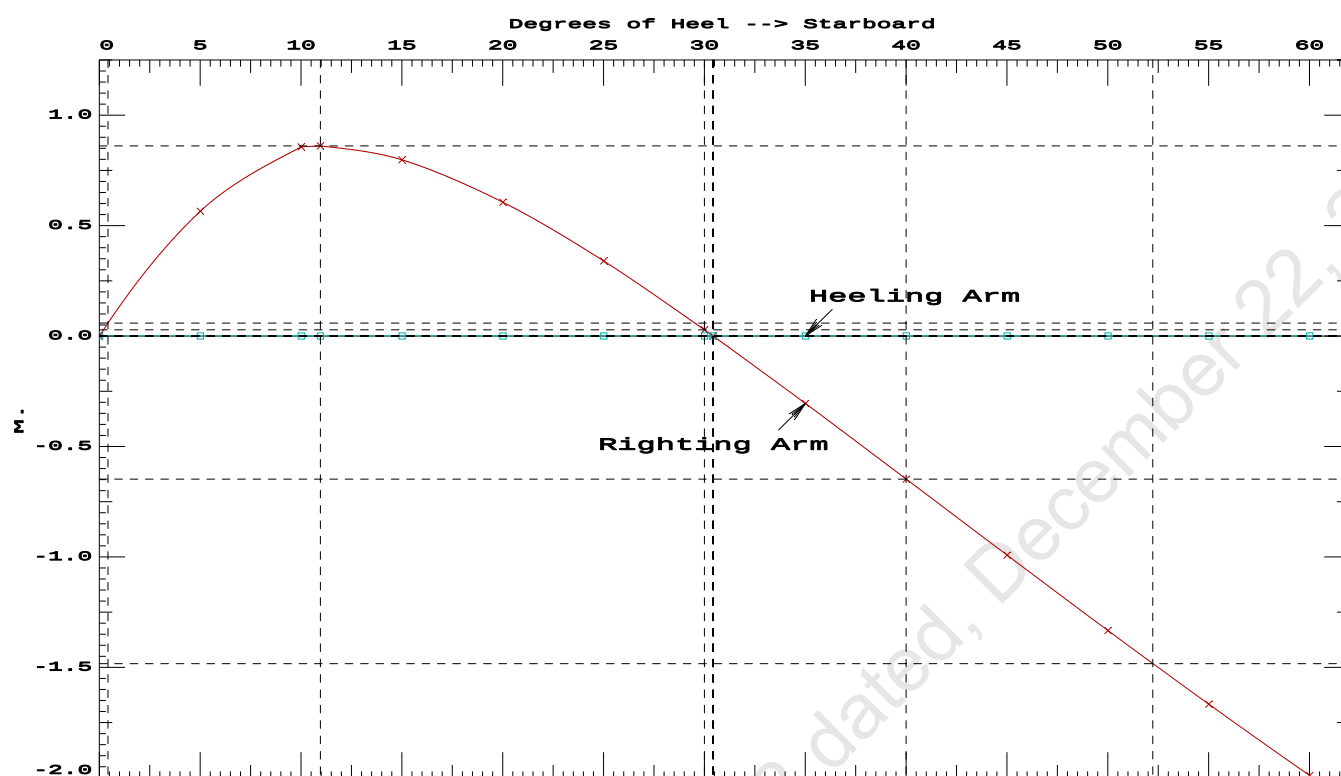
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	159651 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	999.446 P

17-12-20 15:26:31 Cybermarine Technologies PTE. Ltd.

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PONTON

Condition 7 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Aft

17-12-20 15:26:37 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 7 : Lightship + Load at Aft + Ballast + Deck Cargo**Crane Lifting 36.60T @ 18m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.092 @ 55.00f, 2.212 @ 27.50f, 2.331 @ 0.00

Trim: Aft 0.239/55.000, Heel: Stbd 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,486.06	23.200f	0.000	5.714	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,855.24	26.984f	0.000	4.913	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.24	26.968f	0.001	1.132	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		880.5	27.491f	0.000	122.48	10.209
			MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
			9.03		40.04	118.70	6.428
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

17-12-20 15:26:37 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.225 @ 55.00f, 2.171 @ 27.50f, 2.118 @ 0.00

Trim: Fwd 0.107/55.000, Heel: 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,449.46	24.050f	0.000	3.915	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.073f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,818.64	27.738f	0.000	3.462	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,818.64	27.743f	0.001	1.110	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	880.6	27.502f	0.000	124.96	10.416
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.03		40.54	122.61	8.063
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

17-12-20 15:26:37 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.050f TCG = 0.000 VCG = 3.915

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.106	0.11f	0.00s	1,818.6	0.000	0.000	0.0000
2.106	0.11f	0.07p	1,818.6	0.000	0.011	0.0000
2.094	0.11f	5.00p	1,818.6	0.000	0.723	0.0315
2.114	0.15f	10.00p	1,818.7	0.000	1.208	0.1175
2.253	0.19f	15.00p	1,818.6	0.000	1.313	0.2302
2.267	0.20f	15.40p	1,818.6	0.000	1.314	0.2395
2.439	0.24f	20.00p	1,818.7	0.000	1.268	0.3439
2.620	0.30f	25.00p	1,818.6	0.000	1.134	0.4494
2.780	0.36f	30.00p	1,818.6	0.000	0.946	0.5405

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1449.46 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

17-12-20 15:26:37 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.050f TCG = 0.000 VCG = 3.915

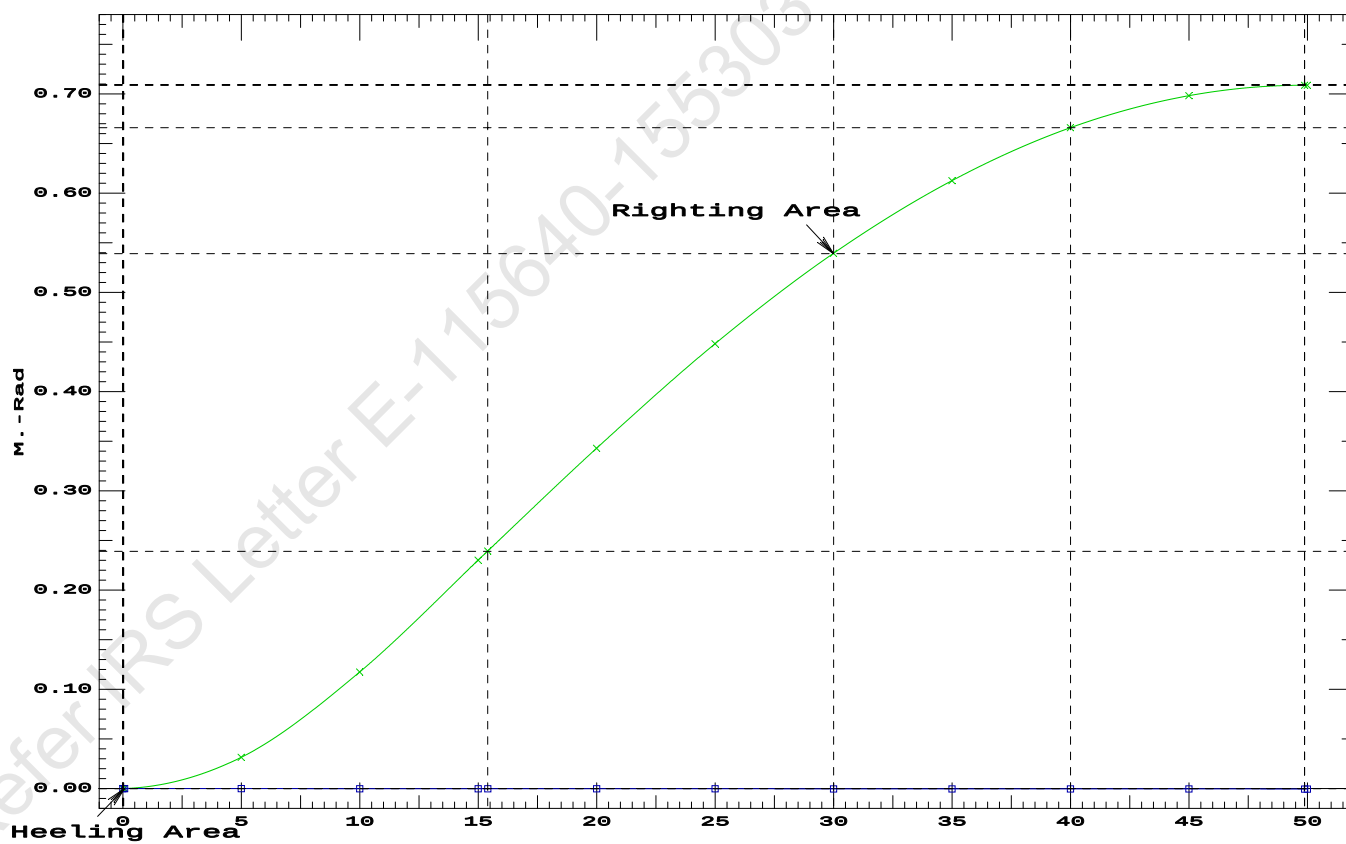
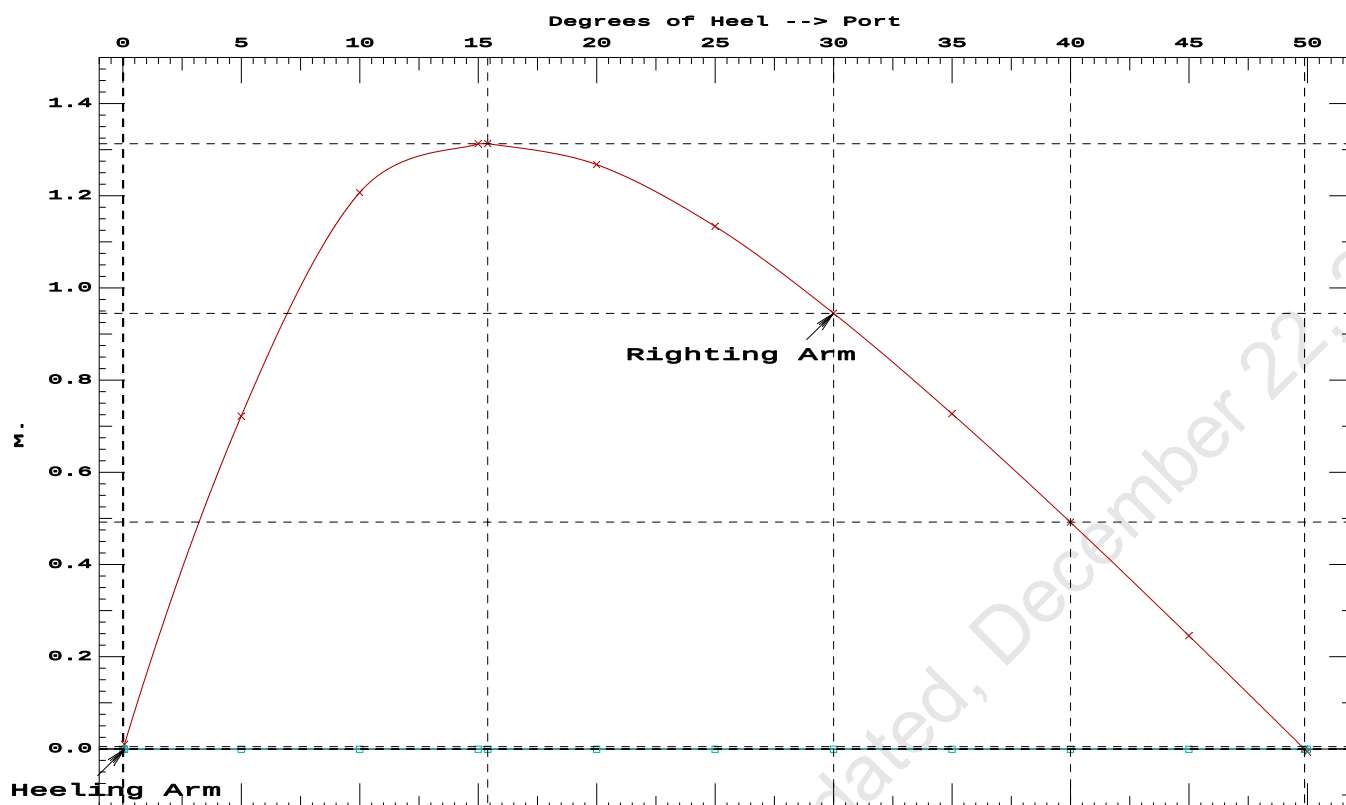
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.106	0.11f	0.00s	1,818.6	0.000	0.000	0.0000
2.106	0.11f	0.07p	1,818.6	0.000	0.011	0.0000
2.094	0.11f	5.00p	1,818.6	0.000	0.723	0.0315
2.114	0.15f	10.00p	1,818.7	0.000	1.208	0.1175
2.253	0.19f	15.00p	1,818.6	0.000	1.313	0.2302
2.267	0.20f	15.40p	1,818.6	0.000	1.314	0.2395
2.439	0.24f	20.00p	1,818.7	0.000	1.268	0.3439
2.620	0.30f	25.00p	1,818.6	0.000	1.134	0.4494
2.780	0.36f	30.00p	1,818.6	0.000	0.946	0.5405
2.918	0.41f	35.00p	1,818.6	0.000	0.728	0.6138
3.033	0.47f	40.00p	1,818.6	0.000	0.492	0.6671
3.125	0.52f	45.00p	1,818.6	0.000	0.246	0.6994
3.191	0.57f	49.88p	1,818.6	0.000	0.000	0.7100
3.193	0.57f	50.00p	1,818.6	0.000	-0.006	0.7100

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1449.46 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 49.81 P



17-12-20 15:26:40 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Stbd

Condition Graphic - Draft: 2.175 @ 55.000f, 2.249 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

3 N01_WBTK.C 18 N05_WBTK.C 19 N06_WBTK.P 20 N06_WBTK.S 21 N06_WBTK.C

17-12-20 15:26:40 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.175 @ 55.00f, 2.212 @ 27.50f, 2.249 @ 0.00

Trim: Aft 0.074/55.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,486.06	23.643f	0.000	5.714	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,855.24	27.339f	0.000	4.913	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.20	27.334f	0.000	1.131	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	880.6	27.496f	0.000	122.51	10.212
			MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
			9.03		40.05	118.73	6.429
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

360.2 Cu.M. (18%) WB

0.0 Cu.M. (0%) VOID

250.00 MT Crane

36.60 MT Crane Load

776.11 MT Deck Cargo

DRAFT AT FP (M) = 2.175
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.249
 DRAFT AT FWD MARK (M) = 2.180
 DRAFT AT AFT MARK (M) = 2.244

HYDROSTATIC PROPERTIES

Trim: Aft 0.074/55.000, No Heel, VCG = 4.913

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm trim-----	GML-----	GMT-----		
2.212	1,855.20	27.334f	1.131	9.03	27.496f	40.05	118.73	6.429

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 338.1 m.-MT

17-12-20 15:26:40 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.175 @ 55.00f, 2.212 @ 27.50f, 2.249 @ 0.00

Trim: Aft 0.074/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	44.5	0.405	1.480	0.05140	3.39
CRANE.C	36.2	3.023	4.098	0.05140	7.62
DECKCARGO.C	70.6	1.806	2.881	0.05140	10.45
Total wind heeling moment to starboard----->					21.46
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.339f TCG = 0.000 VCG = 4.913

Free Surface Adjustment: 0.182

Adjusted CG: LCG = 27.339f TCG = 0.000 VCG = 5.095

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle
2.237	0.08a	0.00	1,855.2	0.000 -0.011 0.0000 2.758
2.237	0.08a	0.10s	1,855.2	0.000 0.000 -0.0000 2.659
2.234	0.08a	2.76s	1,855.2	0.000 0.302 0.0070 0.000
2.227	0.08a	5.10s	1,855.2	0.000 0.564 0.0247 -2.341
2.295	0.11a	10.10s	1,855.2	0.000 0.851 0.0908 -7.341
2.321	0.12a	10.95s	1,855.2	0.000 0.854 0.1035 -8.192
2.484	0.14a	15.10s	1,855.3	0.000 0.788 0.1643 -12.341
2.718	0.17a	20.10s	1,855.2	0.000 0.595 0.2262 -17.341
2.964	0.21a	25.10s	1,855.2	0.000 0.329 0.2670 -22.341
3.191	0.26a	30.10s	1,855.2	0.000 0.017 0.2824 -27.341
3.202	0.26a	30.36s	1,855.2	0.000 0.000 0.2825 -27.600
3.393	0.30a	35.10s	1,855.2	0.000 -0.317 0.2695 -32.341
3.570	0.34a	40.10s	1,855.2	0.000 -0.660 0.2270 -37.341
3.720	0.38a	45.10s	1,855.2	0.000 -1.005 0.1543 -42.341
3.841	0.41a	50.10s	1,855.2	0.000 -1.346 0.0518 -47.341
3.933	0.45a	55.10s	1,855.2	0.000 -1.680 -0.0804 -52.341
3.994	0.47a	60.10s	1,855.2	0.000 -2.004 -0.2412 -57.341
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 338.1 m.-MT was applied to artificially modify the CG.

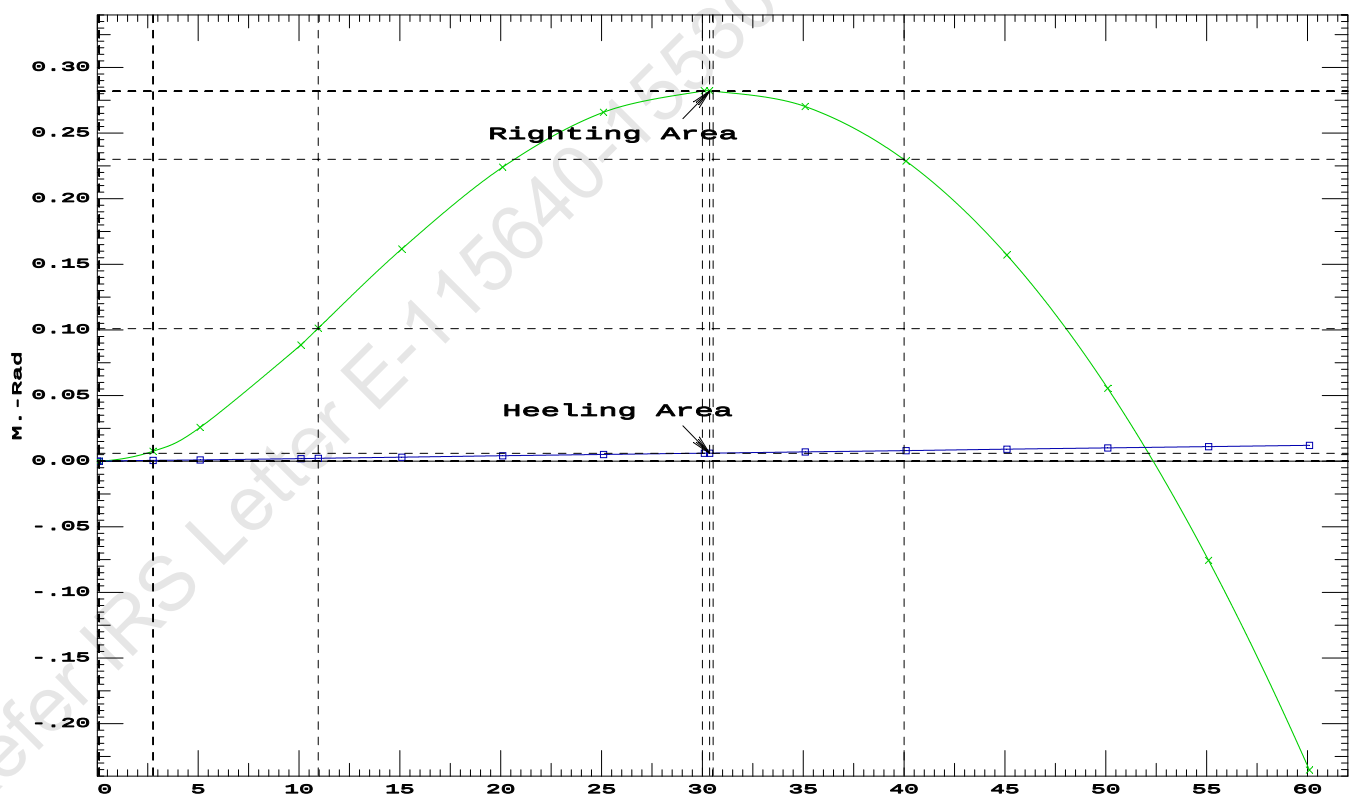
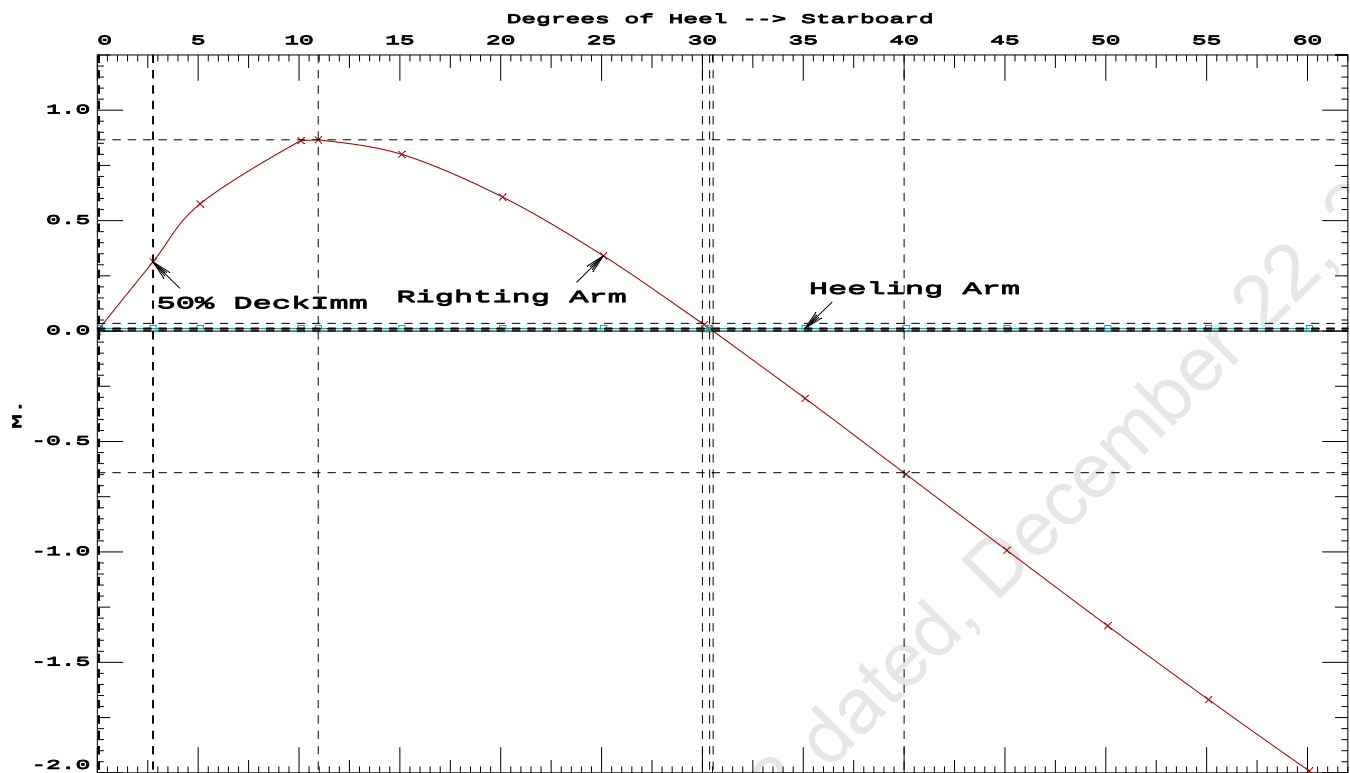
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.46 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1035 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	30.26 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.66 P

17-12-20 15:26:40 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Stbd

17-12-20 15:26:40 Cybermarine Technologies PTE. Ltd.

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PONTOON**Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo****Crane Lifting 36.60T @ 18m Radius Stbd****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 27.339f TCG = 0.000 VCG = 4.913

Free Surface Adjustment: 0.182

Adjusted CG: LCG = 27.339f TCG = 0.010p VCG = 5.095

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms		Res. Area
				in Trim--	in Heel---	
2.234	0.08a	3.04s	1,855.2	0.000	0.000	0.0000
2.246	0.10a	8.04s	1,855.3	0.000	0.469	0.0205
2.328	0.12a	11.16s	1,855.2	0.000	0.520	0.0482
2.397	0.13a	13.04s	1,855.2	0.000	0.501	0.0650
2.619	0.16a	18.04s	1,855.2	0.000	0.350	0.1030
2.865	0.20a	23.04s	1,855.2	0.000	0.111	0.1238
2.960	0.21a	25.00s	1,855.2	0.000	0.000	0.1257
3.100	0.24a	28.04s	1,855.2	0.000	-0.186	0.1208
3.313	0.28a	33.04s	1,855.2	0.000	-0.514	0.0906
3.500	0.32a	38.04s	1,855.2	0.000	-0.854	0.0310
3.662	0.36a	43.04s	1,855.2	0.000	-1.200	-0.0586
3.795	0.40a	48.04s	1,855.2	0.000	-1.544	-0.1783
3.899	0.43a	53.04s	1,855.2	0.000	-1.882	-0.3278
3.973	0.46a	58.04s	1,855.2	0.000	-2.211	-0.5065
4.015	0.49a	63.04s	1,855.2	0.000	-2.527	-0.7133

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 338.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

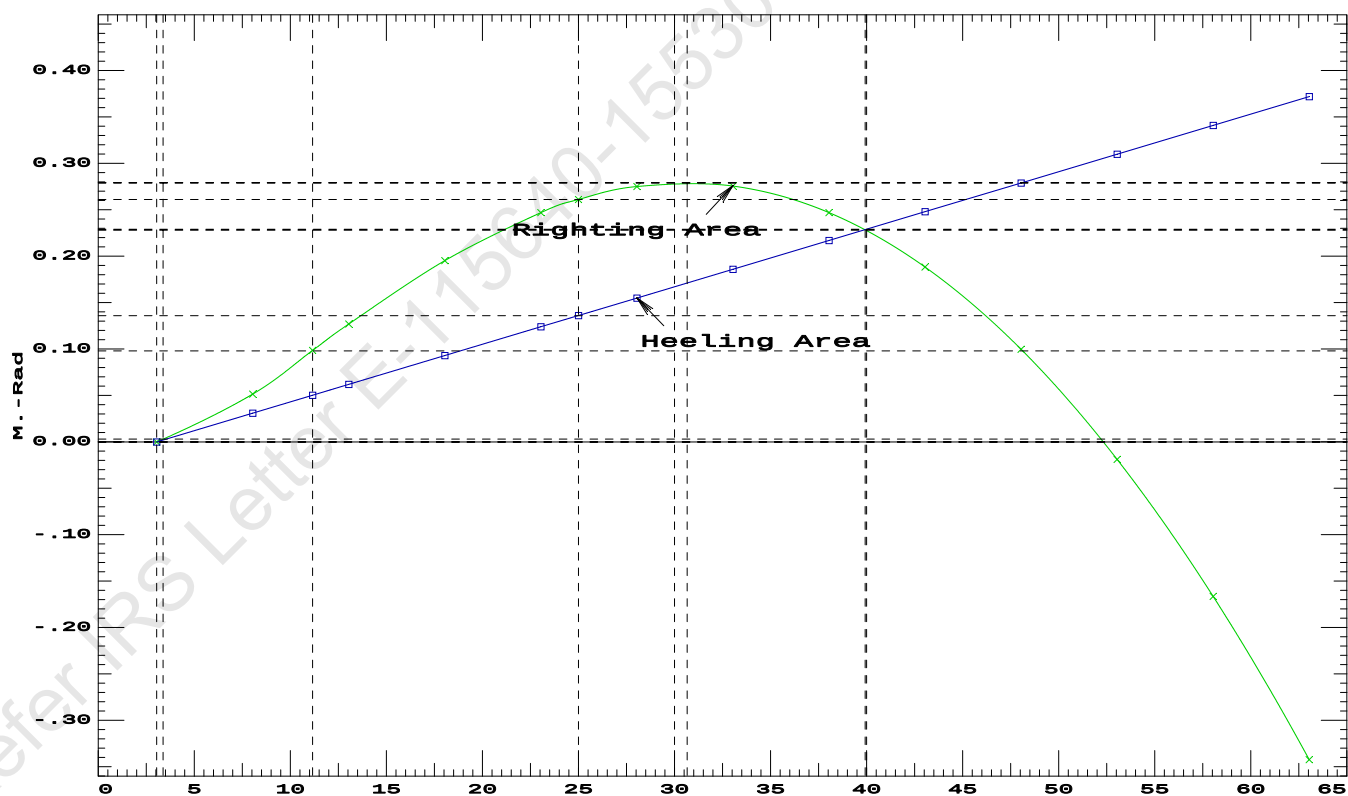
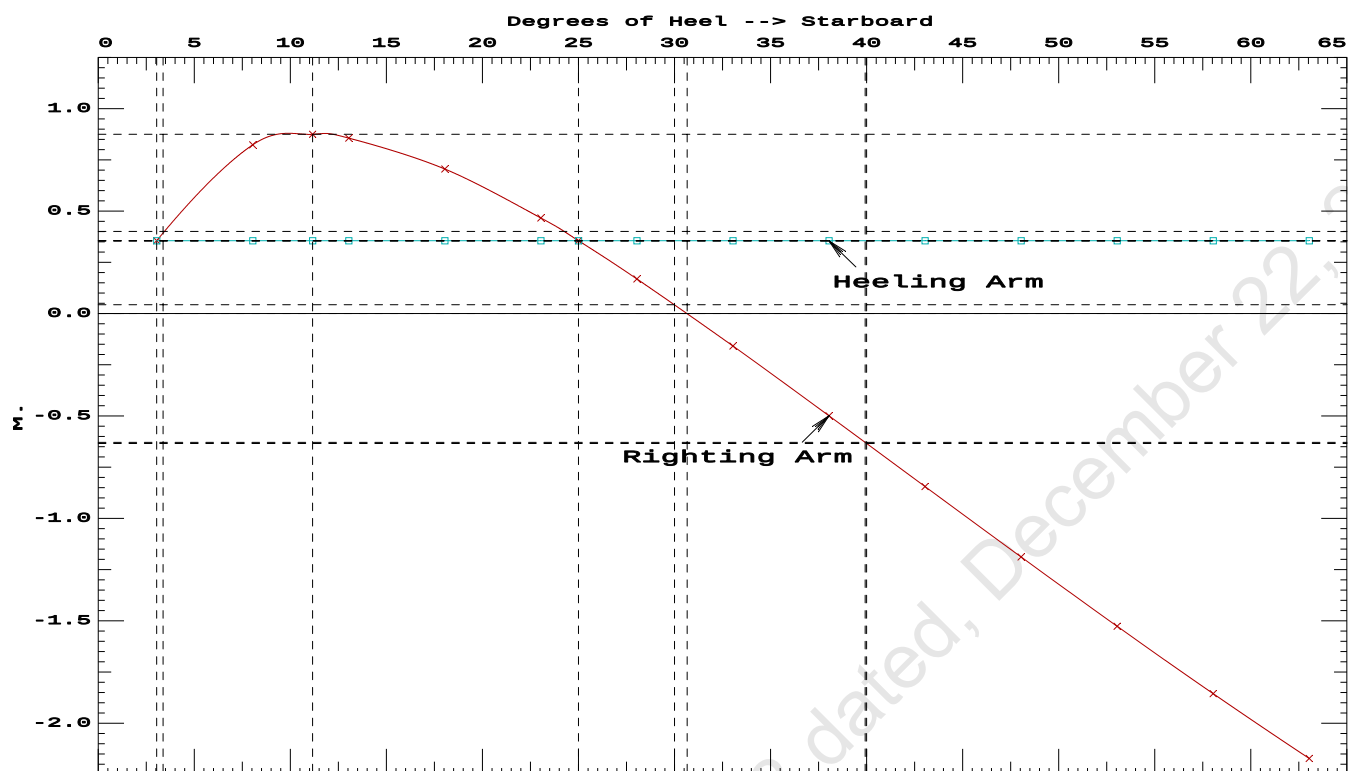
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.04 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	146.4 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1.924 P

17-12-20 15:26:40 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

Condition 8 : Lightship + Load at Stbd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Stbd

17-12-20 15:26:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 8 : Lightship + Load at Stbd + Ballast + Deck Cargo**Crane Lifting 36.60T @ 18m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.174 @ 55.00f, 2.211 @ 27.50f, 2.249 @ 0.00

Trim: Aft 0.075/55.000, Heel: Stbd 3.04 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,486.06	23.643f	0.000	5.714	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,855.24	27.339f	0.000	4.913	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.24	27.334f	0.556s	1.145	

	Righting Arms:			0.000	0.355s		
	External Arms:			0.000	0.355s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		878.9	27.487f	0.027s	121.50	10.180
			MT/cm				
			9.01				
Distances in METERS.-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.05

17-12-20 15:26:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.225 @ 55.00f, 2.171 @ 27.50f, 2.118 @ 0.00

Trim: Fwd 0.107/55.000, Heel: 0.00 deg.

Part	Weight(MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	423.35	27.500f	0.000	3.000			
Crane	250.00	7.500f	0.000	5.200			
Deck Cargo	776.11	27.500f	0.000	4.000			
Total Fixed	1,449.46	24.050f	0.000	3.915			
Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM	
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.073f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks			369.18	42.218f	0.000	1.685	338.08
Total Weight			1,818.64	27.738f	0.000	3.462	
			Displ(MT)	LCB	TCB	VCB	
HULL	1.025		1,818.64	27.743f	0.001	1.110	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	880.6	27.502f	0.000	124.96	10.416	
		MT/cm	m	MT/cm	GML	GMT	
		9.03		40.54	122.61	8.063	
Distances in METERS						Moments in m.-MT	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 4517.05; t3: 30.00; t: 15.00
limmarg: 1866.70; t3: 15.00; t: 7.50
limmarg: 425.83; t3: 7.00; t: 3.75
limmarg: -2.44; t3: 3.00; t: 1.88
limmarg: 169.51; t3: 5.00; t: 0.94
limmarg: 73.06; t3: 4.00; t: 0.47
limmarg: 73.06; t3: 4.00; t: 0.23
limmarg: 73.06; t3: 4.00; t: 0.12
limmarg: 73.06; t3: 4.00; t: 0.06
limmarg: 73.06; t3: 4.00; t: 0.03
limmarg: 73.06; t3: 4.00; t: 0.01

Theta3 is 4.00 degrees, as demonstrated below.

17-12-20 15:26:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.050f TCG = 0.000 VCG = 3.915

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.101	0.11f	3.05s	1,818.6	0.000	-0.441	0.0000
2.106	0.11f	0.00s	1,818.6	0.000	0.000	-0.0117
2.098	0.11f	4.00p	1,818.6	0.000	0.580	0.0086
2.095	0.11f	5.00p	1,818.6	0.000	0.723	0.0199
2.114	0.15f	10.00p	1,818.7	0.000	1.208	0.1054
2.253	0.19f	15.00p	1,818.6	0.000	1.313	0.2182
2.267	0.20f	15.40p	1,818.6	0.000	1.314	0.2274
2.439	0.24f	20.00p	1,818.7	0.000	1.268	0.3318
2.620	0.30f	25.00p	1,818.6	0.000	1.134	0.4373
2.780	0.36f	30.00p	1,818.6	0.000	0.946	0.5285

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1449.46 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -3 deg to abs 4 > 1.000 1.731 P

17-12-20 15:26:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.050f TCG = 0.000 VCG = 3.915

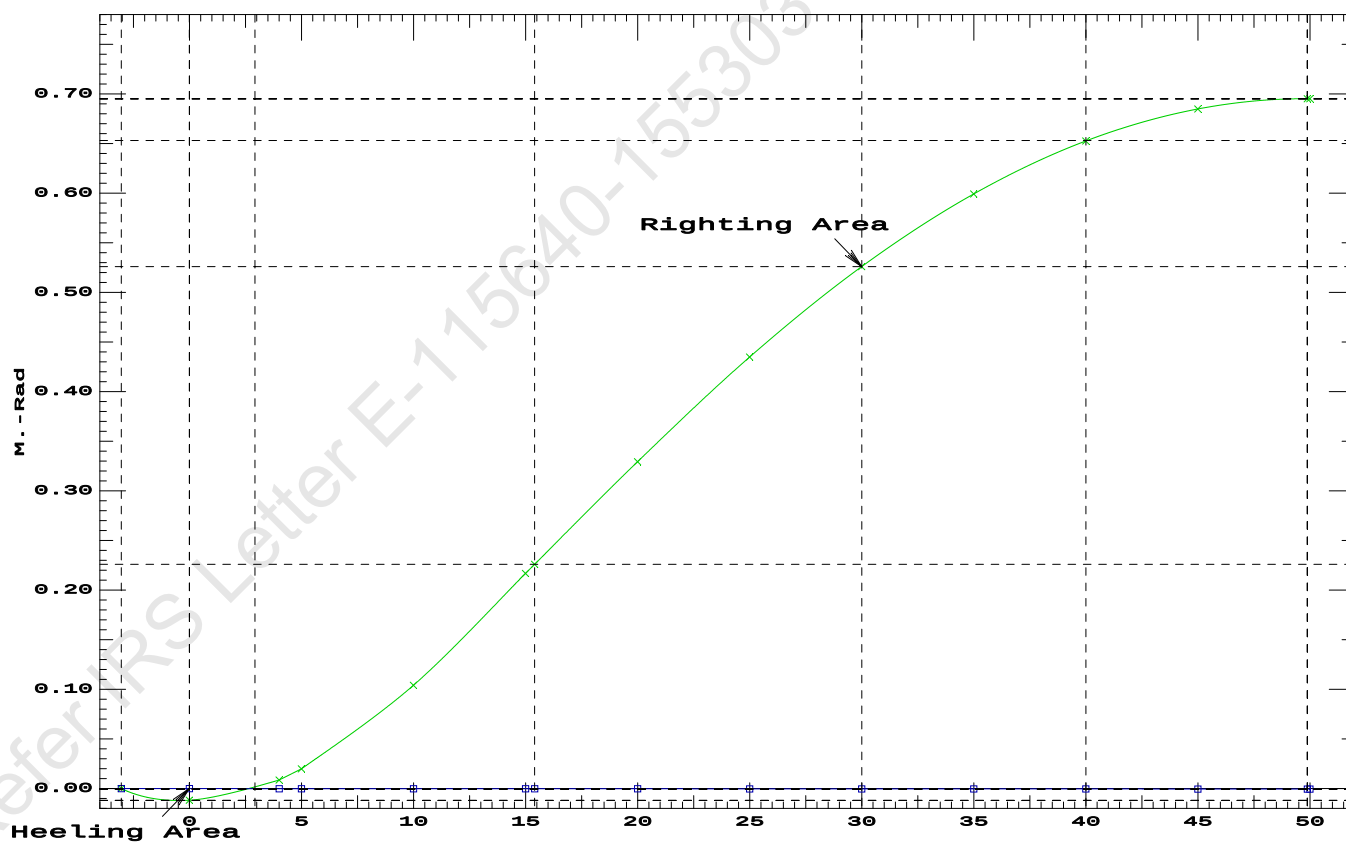
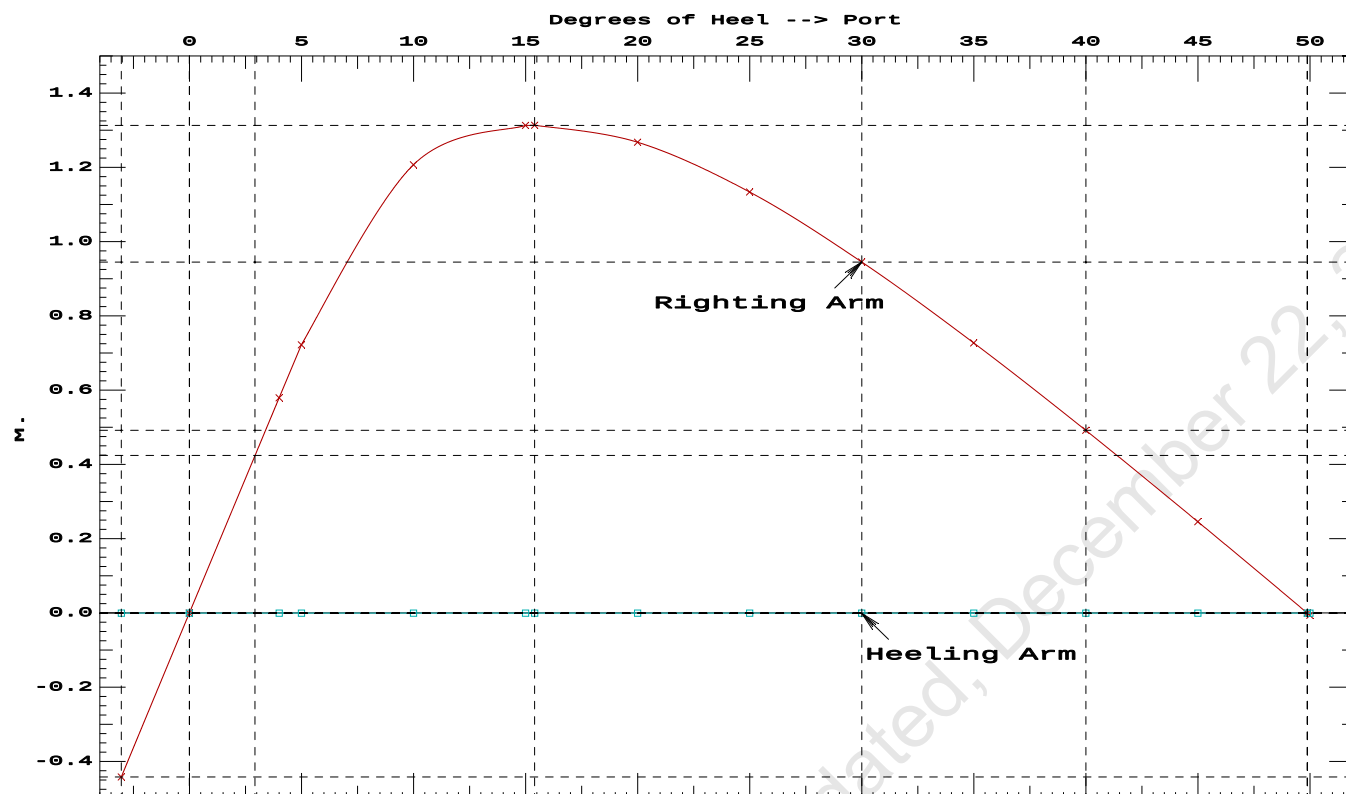
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.101	0.11f	3.05s	1,818.6	0.000	-0.441	0.0000
2.106	0.11f	0.00s	1,818.6	0.000	0.000	-0.0117
2.098	0.11f	4.00p	1,818.6	0.000	0.580	0.0086
2.095	0.11f	5.00p	1,818.6	0.000	0.723	0.0199
2.114	0.15f	10.00p	1,818.7	0.000	1.208	0.1054
2.253	0.19f	15.00p	1,818.6	0.000	1.313	0.2182
2.267	0.20f	15.40p	1,818.6	0.000	1.314	0.2274
2.439	0.24f	20.00p	1,818.7	0.000	1.268	0.3318
2.620	0.30f	25.00p	1,818.6	0.000	1.134	0.4373
2.780	0.36f	30.00p	1,818.6	0.000	0.946	0.5285
2.918	0.41f	35.00p	1,818.6	0.000	0.728	0.6017
3.033	0.47f	40.00p	1,818.6	0.000	0.492	0.6551
3.125	0.52f	45.00p	1,818.6	0.000	0.246	0.6874
3.191	0.57f	49.88p	1,818.6	0.000	0.000	0.6979
3.193	0.57f	50.00p	1,818.6	0.000	-0.006	0.6979

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1449.46 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -3 deg to RAZero > 1.000 60.614 P
(2) Angle from abs 4 deg to RAZero > 20.00 deg 45.88 P



17-12-20 15:26:53 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONT00N

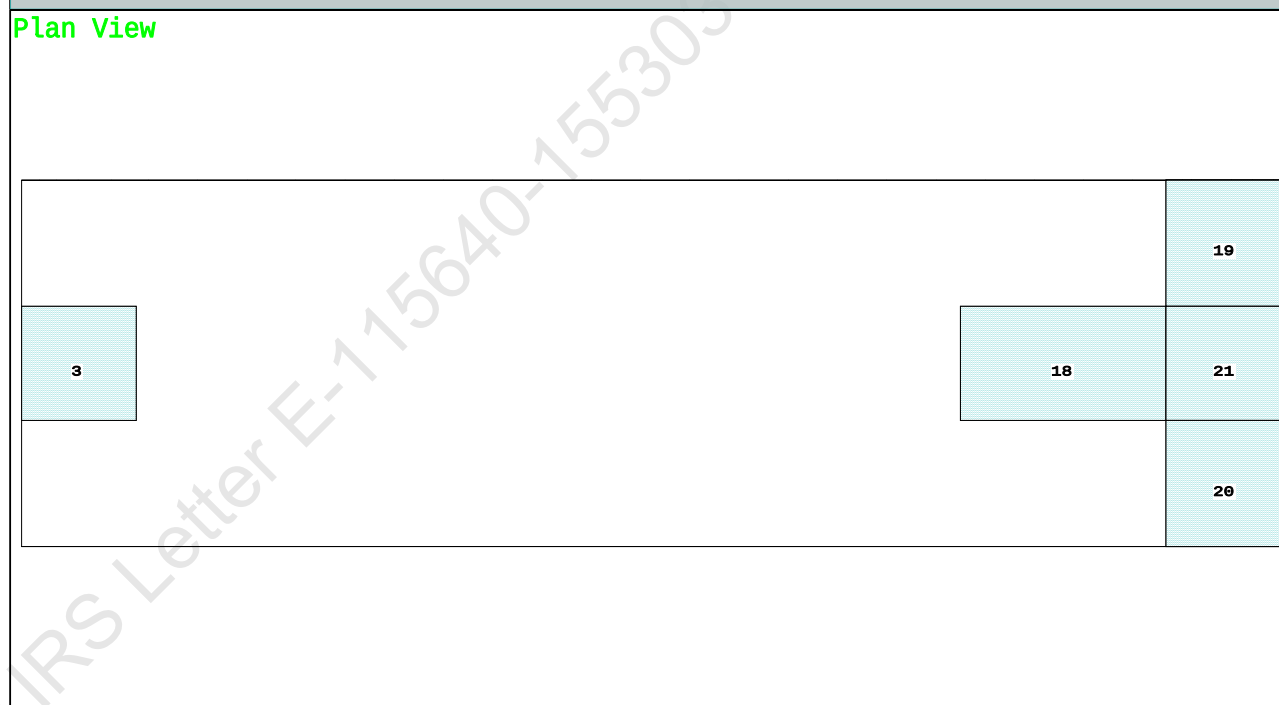
Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 2.257 @ 55.000f, 2.167 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

3 N01_WBTK.C 18 N05_WBTK.C 19 N06_WBTK.P 20 N06_WBTK.S 21 N06_WBTK.C

17-12-20 15:26:53 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.257 @ 55.00f, 2.212 @ 27.50f, 2.167 @ 0.00

Trim: Fwd 0.090/55.000, Heel: zero

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,486.06	24.086f	0.000	5.714	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.074f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.074f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,855.24	27.694f	0.000	4.913	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.20	27.700f	0.000	1.131	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	880.7	27.502f	0.000	122.53	10.214
			MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
			9.03		40.06	118.75	6.432
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

360.2 Cu.M. (18%) WB

0.0 Cu.M. (0%) VOID

250.00 MT Crane

36.60 MT Crane Load

776.11 MT Deck Cargo

DRAFT AT FP (M) = 2.257
 DRAFT AT MS (M) = 2.212
 DRAFT AT AP (M) = 2.167
 DRAFT AT FWD MARK (M) = 2.250
 DRAFT AT AFT MARK (M) = 2.173

HYDROSTATIC PROPERTIES

Trim: Fwd 0.090/55.000, No Heel, VCG = 4.913

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm trim-----	GML-----	GMT-----		
2.212	1,855.20	27.700f	1.131	9.03	27.502f	40.06	118.75	6.432

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
 Trim is per 55.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 338.1 m.-MT

17-12-20 15:26:53 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.257 @ 55.00f, 2.212 @ 27.50f, 2.167 @ 0.00

Trim: Fwd 0.090/55.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	44.5	0.405	1.480	0.05140	3.39
CRANE.C	36.2	3.083	4.158	0.05140	7.73
DECKCARGO.C	70.6	1.793	2.868	0.05140	10.40
Total wind heeling moment to starboard----->					21.52
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 27.694f TCG = 0.000 VCG = 4.913

Free Surface Adjustment: 0.182

Adjusted CG: LCG = 27.694f TCG = 0.000 VCG = 5.095

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight(MT)---	in Trim--in Heel---	Area---Angle
2.155	0.09f	0.00	1,855.2	0.000 -0.011 0.0000 2.729
2.155	0.09f	0.10s	1,855.2	0.000 0.000 -0.0000 2.630
2.152	0.09f	2.73s	1,855.3	0.000 0.298 0.0068 0.000
2.144	0.10f	5.10s	1,855.3	0.000 0.564 0.0247 -2.370
2.179	0.13f	10.10s	1,855.3	0.000 0.850 0.0908 -7.370
2.200	0.14f	10.97s	1,855.2	0.000 0.854 0.1037 -8.240
2.335	0.17f	15.10s	1,855.3	0.000 0.788 0.1642 -12.370
2.537	0.20f	20.10s	1,855.2	0.000 0.595 0.2261 -17.370
2.739	0.25f	25.10s	1,855.2	0.000 0.329 0.2669 -22.370
2.920	0.31f	30.10s	1,855.2	0.000 0.017 0.2823 -27.370
2.928	0.31f	30.35s	1,855.2	0.000 0.000 0.2823 -27.624
3.077	0.36f	35.10s	1,855.2	0.000 -0.317 0.2693 -32.370
3.210	0.41f	40.10s	1,855.2	0.000 -0.660 0.2267 -37.370
3.319	0.45f	45.10s	1,855.2	0.000 -1.005 0.1541 -42.370
3.403	0.50f	50.10s	1,855.2	0.000 -1.346 0.0515 -47.370
3.462	0.54f	55.10s	1,855.2	0.000 -1.681 -0.0807 -52.370
3.495	0.57f	60.10s	1,855.2	0.000 -2.004 -0.2415 -57.370
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 338.1 m.-MT was applied to artificially modify the CG.

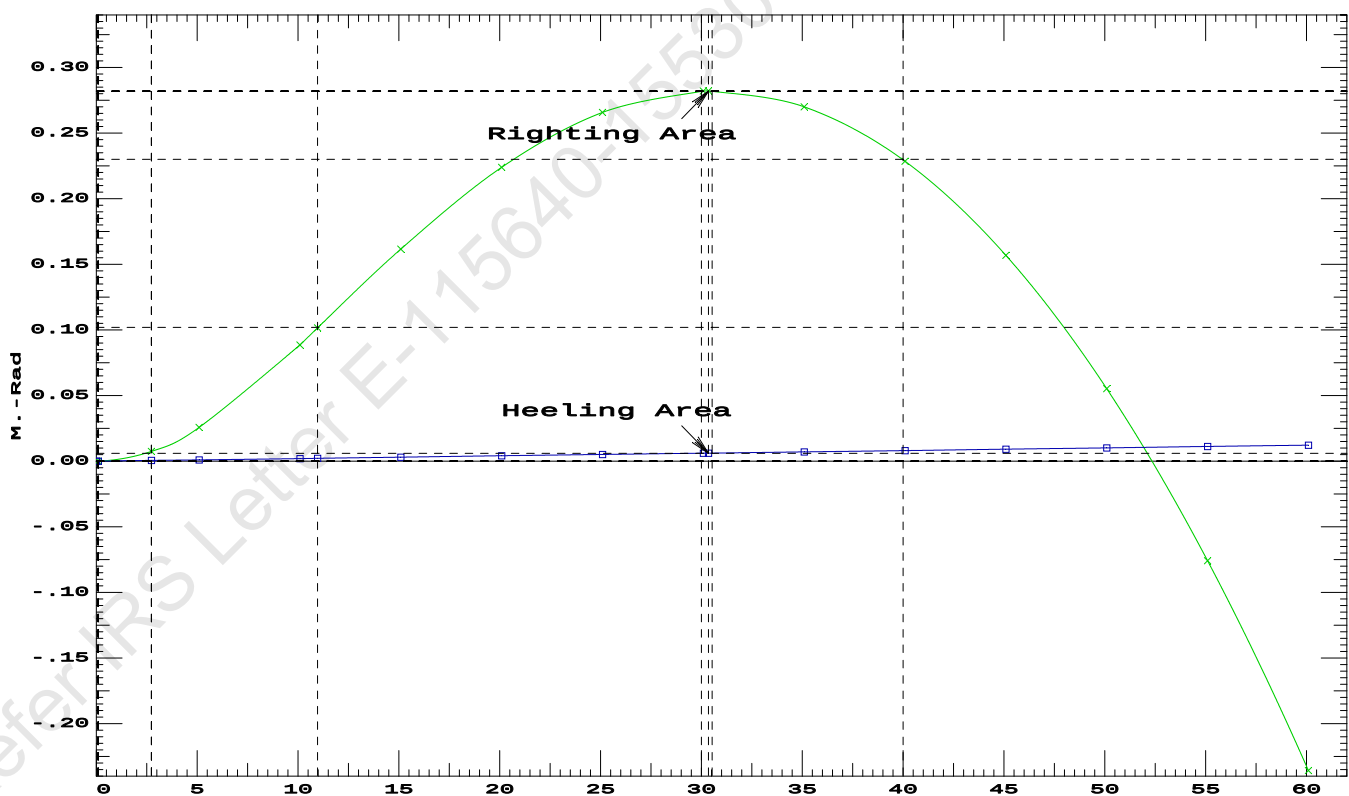
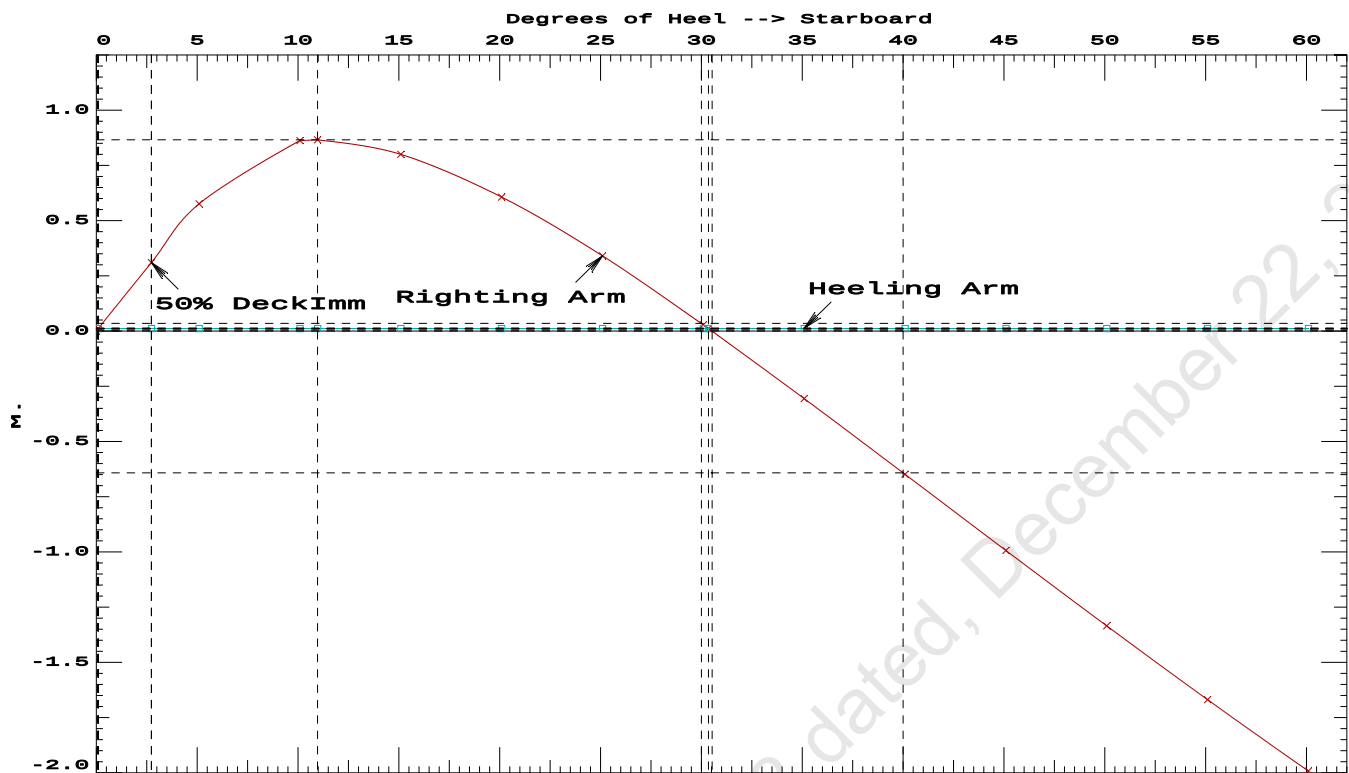
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.52 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1037 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	30.25 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.63 P

17-12-20 15:26:53 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Fwd

17-12-20 15:26:53 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTOON**Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo****Crane Lifting 36.60T @ 18m Radius Fwd****RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Total CG: LCG = 27.694f TCG = 0.000 VCG = 4.913

Free Surface Adjustment: 0.182

Adjusted CG: LCG = 27.694f TCG = 0.000 VCG = 5.095

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms		Res. Area
				in Trim--	in Heel---	
2.155	0.09f	0.00s	1,855.2	0.000	0.000	0.0000
2.144	0.09f	5.00s	1,855.3	0.000	0.565	0.0246
2.177	0.13f	10.00s	1,855.3	0.000	0.860	0.0912
2.200	0.14f	10.98s	1,855.2	0.000	0.865	0.1059
2.331	0.17f	15.00s	1,855.3	0.000	0.802	0.1657
2.533	0.20f	20.00s	1,855.2	0.000	0.610	0.2288
2.736	0.25f	25.00s	1,855.2	0.000	0.345	0.2710
2.916	0.31f	30.00s	1,855.2	0.000	0.034	0.2878
2.934	0.31f	30.52s	1,855.2	0.000	0.000	0.2880
3.074	0.36f	35.00s	1,855.2	0.000	-0.300	0.2763
3.208	0.41f	40.00s	1,855.2	0.000	-0.643	0.2353
3.318	0.45f	45.00s	1,855.2	0.000	-0.988	0.1642
3.402	0.50f	50.00s	1,855.2	0.000	-1.329	0.0631
3.461	0.53f	55.00s	1,855.2	0.000	-1.663	-0.0676
3.494	0.57f	60.00s	1,855.2	0.000	-1.987	-0.2269

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 338.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

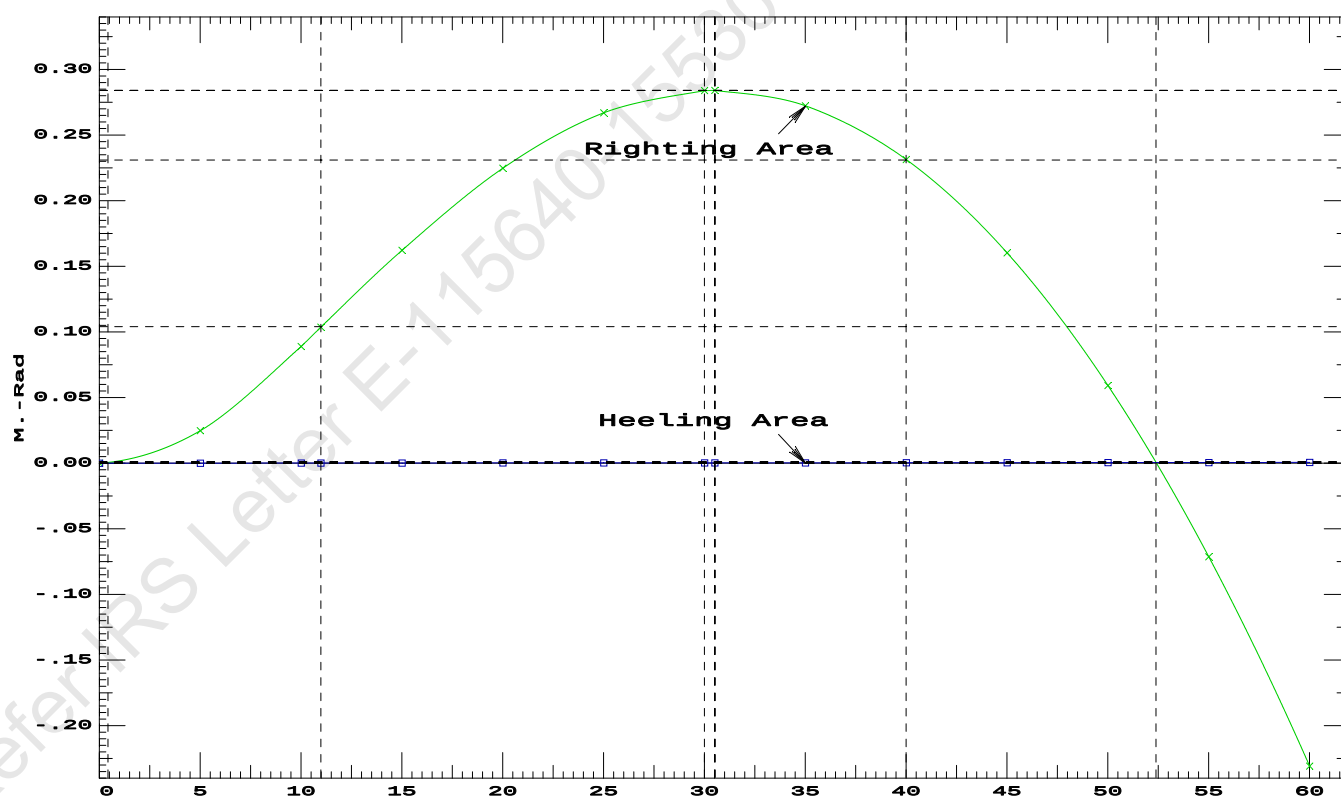
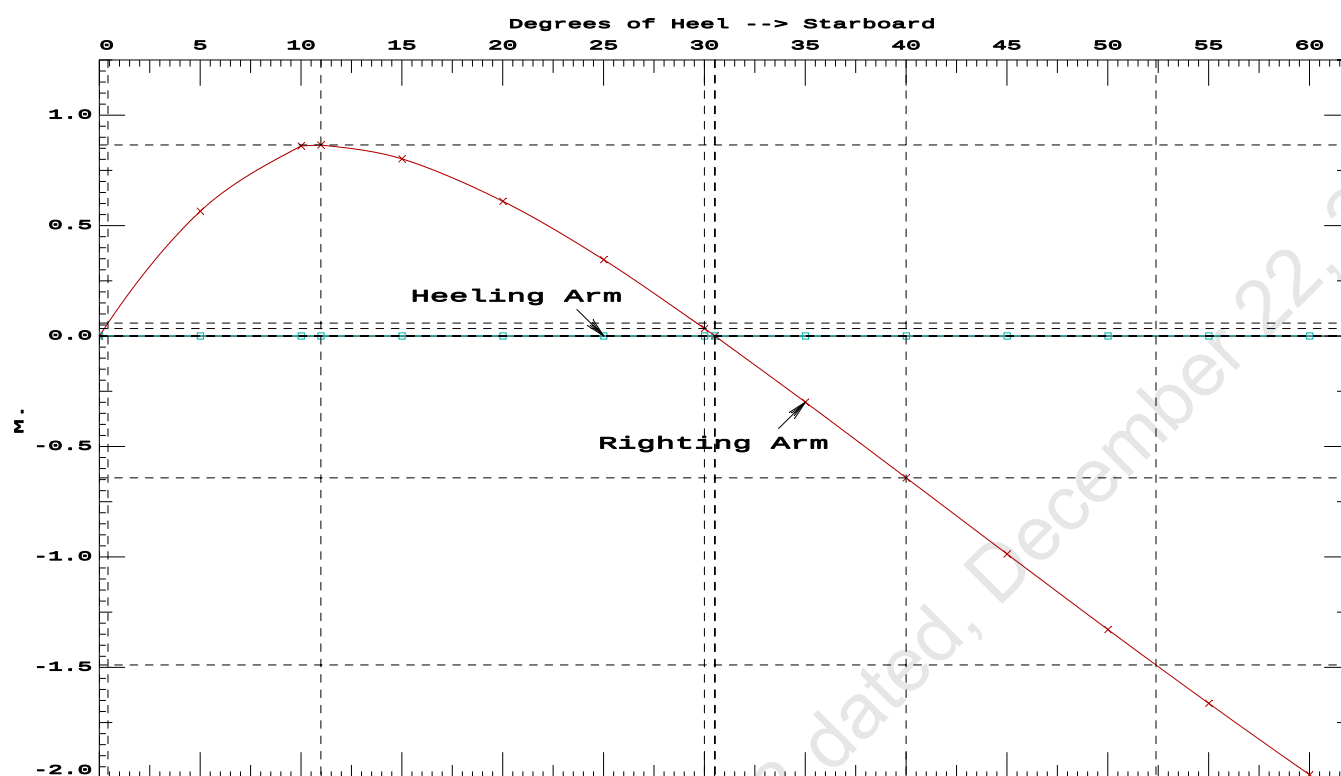
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	160454 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1004.20 P

17-12-20 15:26:53 Cybermarine Technologies PTE. Ltd.

GHS 17.22

PONTON

Condition 9 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 36.60T @ 18m Radius Fwd

17-12-20 15:27:00 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

Condition 9 : Lightship + Load at Fwd + Ballast + Deck Cargo**Crane Lifting 36.60T @ 18m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.257 @ 55.00f, 2.212 @ 27.50f, 2.167 @ 0.00

Trim: Fwd 0.090/55.000, Heel: Stbd 0.00 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			423.35	27.500f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Deck Cargo			776.11	27.500f	0.000	4.000	
Total Fixed----->			1,486.06	24.086f	0.000	5.714	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.073f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks----->			369.18	42.218f	0.000	1.685	338.08
Total Weight----->			1,855.24	27.694f	0.000	4.913	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,855.24	27.701f	0.001	1.131	

	Righting Arms:			0.001	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.001	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		880.7	27.502f	0.000	122.53	10.214
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			9.03		40.06	118.75	6.432
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

17-12-20 15:27:00 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.225 @ 55.00f, 2.171 @ 27.50f, 2.118 @ 0.00

Trim: Fwd 0.107/55.000, Heel: 0.00 deg.

Part	Weight(MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	423.35	27.500f	0.000	3.000			
Crane	250.00	7.500f	0.000	5.200			
Deck Cargo	776.11	27.500f	0.000	4.000			
Total Fixed	1,449.46	24.050f	0.000	3.915			
Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM	
N01_WBTK.C	1.000	1.025	55.90	2.922f	0.000	1.789	52.32*
N05_WBTK.C	1.000	1.025	135.60	45.500f	0.000	1.500	94.17*
N06_WBTK.P	1.000	1.025	60.88	52.073f	5.223p	1.797	69.64*
N06_WBTK.S	1.000	1.025	60.88	52.073f	5.223s	1.797	69.64*
N06_WBTK.C	1.000	1.025	55.92	52.078f	0.000	1.788	52.32*
Total Tanks			369.18	42.218f	0.000	1.685	338.08
Total Weight			1,818.64	27.738f	0.000	3.462	
			Displ(MT)	LCB	TCB	VCB	
HULL	1.025		1,818.64	27.743f	0.001	1.110	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	880.6	27.502f	0.000	124.96	10.416	
		MT/cm	m	MT/cm	GML	GMT	
		9.03		40.54	122.61	8.063	
Distances in METERS						Moments in m.-MT	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

17-12-20 15:27:00 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.050f TCG = 0.000 VCG = 3.915

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
2.106	0.11f	0.00s	1,818.6	0.000	0.000	0.0000
2.106	0.11f	0.07p	1,818.6	0.000	0.011	0.0000
2.094	0.11f	5.00p	1,818.6	0.000	0.723	0.0315
2.114	0.15f	10.00p	1,818.7	0.000	1.208	0.1175
2.253	0.19f	15.00p	1,818.6	0.000	1.313	0.2302
2.267	0.20f	15.40p	1,818.6	0.000	1.314	0.2395
2.439	0.24f	20.00p	1,818.7	0.000	1.268	0.3439
2.620	0.30f	25.00p	1,818.6	0.000	1.134	0.4494
2.780	0.36f	30.00p	1,818.6	0.000	0.946	0.5405

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1449.46 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

17-12-20 15:27:00 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONT00N

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 24.050f TCG = 0.000 VCG = 3.915

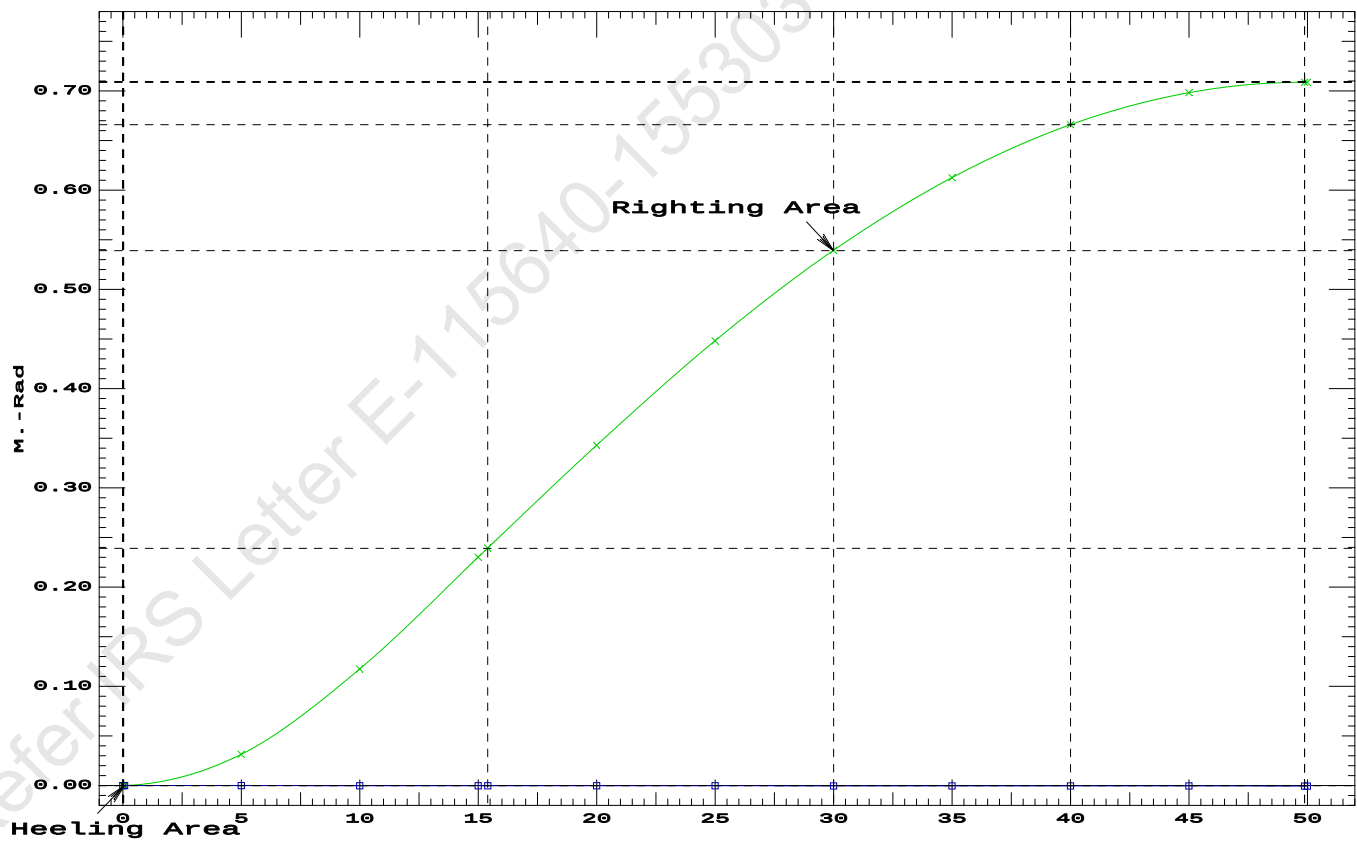
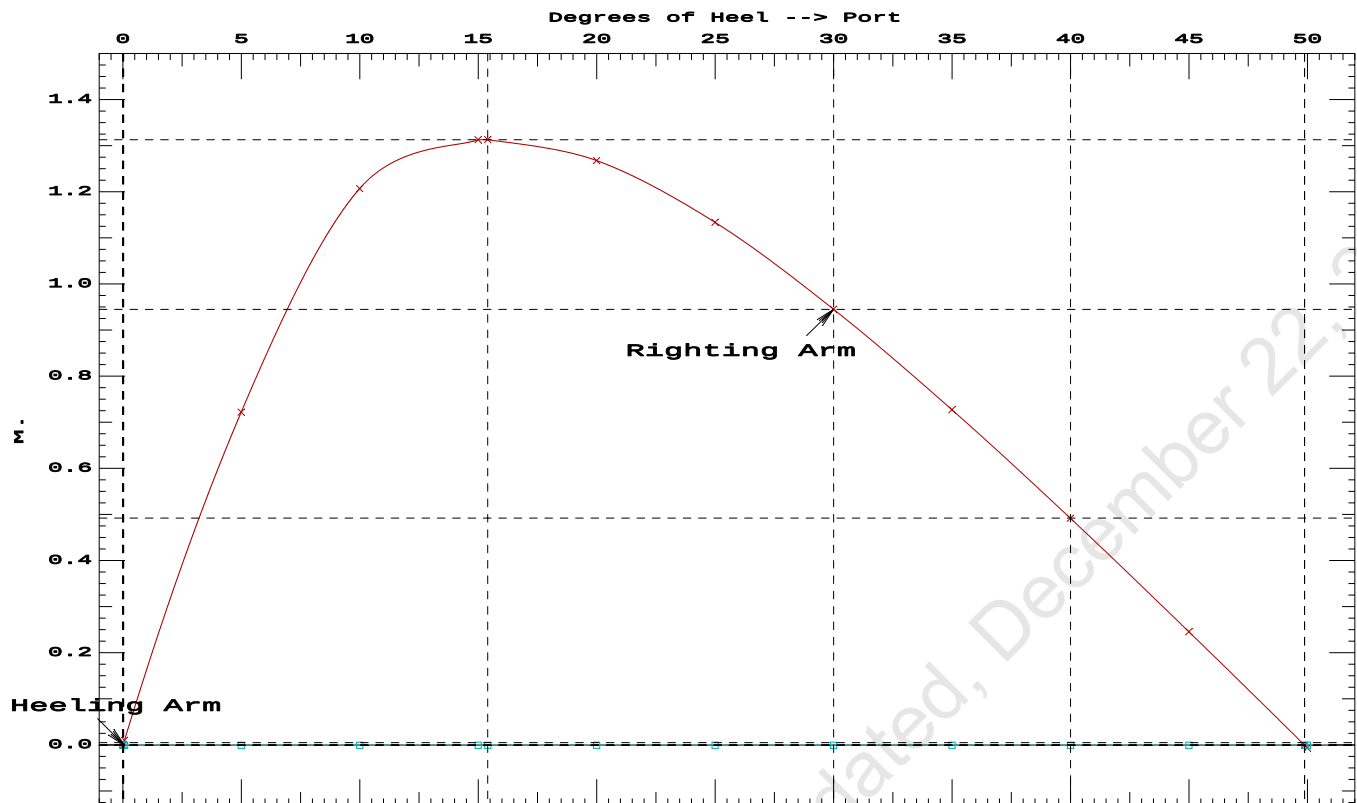
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.106	0.11f	0.00s	1,818.6	0.000	0.000	0.0000
2.106	0.11f	0.07p	1,818.6	0.000	0.011	0.0000
2.094	0.11f	5.00p	1,818.6	0.000	0.723	0.0315
2.114	0.15f	10.00p	1,818.7	0.000	1.208	0.1175
2.253	0.19f	15.00p	1,818.6	0.000	1.313	0.2302
2.267	0.20f	15.40p	1,818.6	0.000	1.314	0.2395
2.439	0.24f	20.00p	1,818.7	0.000	1.268	0.3439
2.620	0.30f	25.00p	1,818.6	0.000	1.134	0.4494
2.780	0.36f	30.00p	1,818.6	0.000	0.946	0.5405
2.918	0.41f	35.00p	1,818.6	0.000	0.728	0.6138
3.033	0.47f	40.00p	1,818.6	0.000	0.492	0.6671
3.125	0.52f	45.00p	1,818.6	0.000	0.246	0.6994
3.191	0.57f	49.88p	1,818.6	0.000	0.000	0.7100
3.193	0.57f	50.00p	1,818.6	0.000	-0.006	0.7100

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1449.46 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 49.81 P



LIGHTSHIP PARTICULARS

Lightship Weight	=	423.35	T
VCG	=	3.000	M above Baseline
LCG	=	27.500	M fwd of AP
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)



MEMORANDUM OF FREEBOARDS

30th JUNE 2020

GMS MARINE SERVICES

Yard No: GMS- 005

Type B+ freeboard

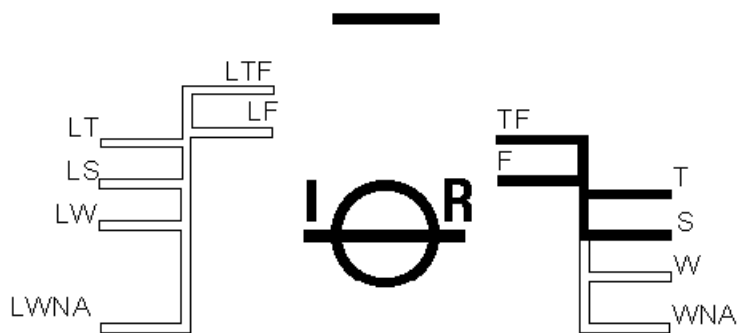
Freeboard Length : 52.800 m

Summer WL Extreme Draught : 2.12 m
- loadline amidships

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of
* Wood/ * steel Deck * at side / * continued to side. (*Delete words not applicable).

Freeboards from Deck Line		Seasonal, Fresh Water, or Timber allowances		
Tropical Fresh Water	713	mm.	97	mm. above summer
Fresh Water	759	mm.	51	mm. above summer
Tropical	764	mm.	46	mm. above summer
Summer	810	mm.	Upper edge of line through centre	
Winter	Not	mm.	Assigned	mm. below summer
Winter North Atlantic	Not	mm.	Assigned	mm. below summer
Timber Tropical fresh water	—	mm.	—	mm. above timber summer
Timber fresh water	—	mm.	—	mm. above timber summer
Timber Tropical	—	mm.	—	mm. above timber summer
Timber Summer	—	mm.	—	mm. above centre of ring
Timber Winter	—	mm.	—	mm. below timber summer
Timber Winter North Atlantic	—	mm.	—	mm. below timber summer

Valid only for unmanned operation while under tow along Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.



400 mm Aft of Fr. 58



IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244

52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-115327-154861

17-December-2020

Company : Cybermarine Knowledge System pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002414
Project name	GMS Marine Services-GMS-005
Project Manager	Rishi Doshi
Project In-Charge	Amit Waje

Document Details	
Plan No.	CM-19-1021-0206
Plan Title	0206-Draft Survey Report
Rev.No.	0
Upload.No.	1
Submitted Date	11-December-2020
Review Status	Approved
Parent Plan Title	

The plan has been examined for compliance with:
ISCODE 2008

Notes

1. The review status Approved indicates:

The following Lightship particulars are approved for GMS yard No. 005 based on lightship survey conducted on 04-12-2020

Lightship Weight : 423.35 T

Lightship L.C.G : 27.5 m forward of transom (Fr.0)

Lightship V.C.G : 3.00 m above baseline (on Main deck)

Lightship T.C.G : 0.000 m off centerline

Stability Booklet prepared based on the approved Lightship particulars shall be submitted for approval. Copy of the approved lightship survey report should form part of the stability booklet.

Thanking you,
Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Renganathan S

Sr. Principal Surveyor

This is an electronically published document and does not require signature.

LATERAL PLANE STATUS
USK draft: 0.500 @ 55.00f, 0.500 @ 27.50f, 0.500 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	24.4	27.578f	-0.250	134.0	27.486f	1.289
CRANE.C				36.2	7.500f	4.762
DECKCARGO.C				70.6	32.000f	3.512
Total Lateral Plane->	24.4	27.578f	-0.250	240.7	25.807f	2.462
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 0.600 @ 55.00f, 0.600 @ 27.50f, 0.600 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	29.3	27.578f	-0.300	129.1	27.482f	1.236
CRANE.C				36.2	7.500f	4.662
DECKCARGO.C				70.6	32.000f	3.412
Total Lateral Plane->	29.3	27.578f	-0.300	235.8	25.770f	2.412
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 0.700 @ 55.00f, 0.700 @ 27.50f, 0.700 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	34.2	27.578f	-0.350	124.2	27.479f	1.182
CRANE.C				36.2	7.500f	4.562
DECKCARGO.C				70.6	32.000f	3.312
Total Lateral Plane->	34.2	27.578f	-0.350	231.0	25.732f	2.362
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 0.800 @ 55.00f, 0.800 @ 27.50f, 0.800 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	39.1	27.532f	-0.397	119.3	27.490f	1.129
CRANE.C				36.2	7.500f	4.462
DECKCARGO.C				70.6	32.000f	3.212
Total Lateral Plane->	39.1	27.532f	-0.397	226.0	25.699f	2.312
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 0.900 @ 55.00f, 0.900 @ 27.50f, 0.900 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	44.1	27.473f	-0.442	114.3	27.511f	1.076
CRANE.C				36.2	7.500f	4.362
DECKCARGO.C				70.6	32.000f	3.112
Total Lateral Plane->	44.1	27.473f	-0.442	221.0	25.670f	2.264
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 1.000 @ 55.00f, 1.000 @ 27.50f, 1.000 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	49.2	27.491f	-0.491	109.2	27.504f	1.024
CRANE.C				36.2	7.500f	4.262
DECKCARGO.C				70.6	32.000f	3.012
Total Lateral Plane->	49.2	27.491f	-0.491	215.9	25.623f	2.216
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 1.100 @ 55.00f, 1.100 @ 27.50f, 1.100 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	54.4	27.507f	-0.540	104.0	27.496f	0.972
CRANE.C				36.2	7.500f	4.162
DECKCARGO.C				70.6	32.000f	2.912
Total Lateral Plane->	54.4	27.507f	-0.540	210.8	25.573f	2.169
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 1.200 @ 55.00f, 1.200 @ 27.50f, 1.200 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	59.5	27.519f	-0.589	98.9	27.488f	0.920
CRANE.C				36.2	7.500f	4.062
DECKCARGO.C				70.6	32.000f	2.812
Total Lateral Plane->	59.5	27.519f	-0.589	205.6	25.521f	2.122
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 1.300 @ 55.00f, 1.300 @ 27.50f, 1.300 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	64.7	27.500f	-0.637	93.7	27.500f	0.868
CRANE.C				36.2	7.500f	3.962
DECKCARGO.C				70.6	32.000f	2.712
Total Lateral Plane->	64.7	27.500f	-0.637	200.4	25.476f	2.075
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 1.400 @ 55.00f, 1.400 @ 27.50f, 1.400 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	70.0	27.491f	-0.685	88.4	27.507f	0.816
CRANE.C				36.2	7.500f	3.862
DECKCARGO.C				70.6	32.000f	2.612
Total Lateral Plane->	70.0	27.491f	-0.685	195.2	25.425f	2.030
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.500 @ 55.00f, 1.500 @ 27.50f, 1.500 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	75.3	27.505f	-0.733	83.1	27.495f	0.765
CRANE.C				36.2	7.500f	3.762
DECKCARGO.C				70.6	32.000f	2.512
Total Lateral Plane->	75.3	27.505f	-0.733	189.9	25.361f	1.985
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.600 @ 55.00f, 1.600 @ 27.50f, 1.600 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	80.6	27.517f	-0.782	77.8	27.482f	0.714
CRANE.C				36.2	7.500f	3.662
DECKCARGO.C				70.6	32.000f	2.412
Total Lateral Plane->	80.6	27.517f	-0.782	184.6	25.294f	1.941
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.700 @ 55.00f, 1.700 @ 27.50f, 1.700 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	85.9	27.506f	-0.830	72.5	27.493f	0.663
CRANE.C				36.2	7.500f	3.562
DECKCARGO.C				70.6	32.000f	2.312
Total Lateral Plane->	85.9	27.506f	-0.830	179.2	25.233f	1.897
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.800 @ 55.00f, 1.800 @ 27.50f, 1.800 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	91.3	27.494f	-0.878	67.1	27.508f	0.612
CRANE.C				36.2	7.500f	3.462
DECKCARGO.C				70.6	32.000f	2.212
Total Lateral Plane->	91.3	27.494f	-0.878	173.8	25.169f	1.855
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.900 @ 55.00f, 1.900 @ 27.50f, 1.900 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	96.8	27.494f	-0.925	61.6	27.510f	0.561
CRANE.C				36.2	7.500f	3.362
DECKCARGO.C				70.6	32.000f	2.112
Total Lateral Plane->	96.8	27.494f	-0.925	168.4	25.094f	1.813
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 2.000 @ 55.00f, 2.000 @ 27.50f, 2.000 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	102.2	27.500f	-0.973	56.2	27.500f	0.511
CRANE.C				36.2	7.500f	3.262
DECKCARGO.C				70.6	32.000f	2.012
Total Lateral Plane->	102.2	27.500f	-0.973	162.9	25.010f	1.772
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 2.100 @ 55.00f, 2.100 @ 27.50f, 2.100 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	107.7	27.500f	-1.021	50.7	27.500f	0.461
CRANE.C				36.2	7.500f	3.162
DECKCARGO.C				70.6	32.000f	1.912
Total Lateral Plane->	107.7	27.500f	-1.021	157.4	24.923f	1.732
Distances in METERS.-----						

LATERAL PLANE STATUS
USK draft: 2.212 @ 55.00f, 2.212 @ 27.50f, 2.212 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	113.9	27.500f	-1.075	44.5	27.500f	0.405
CRANE.C				36.2	7.500f	3.050
DECKCARGO.C				70.6	32.000f	1.800
Total Lateral Plane->	113.9	27.500f	-1.075	151.3	24.818f	1.688
Distances in METERS.-----						

Refer IRS Letter E-115640-15533 dated 22, 2020